

MODELING AND FORECASTING NIFTY 50 USING HYBRID ARIMA GARCH MODEL

**Master Thesis submitted in partial fulfilment of the requirements for
the award of the Degree of**

MASTER OF BUSINESS ADMINISTRATION

of

BENGALURU CITY UNIVERSITY



By

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2023-2025

DECLARATION BY THE STUDENT

I hereby declare that **MODELING AND FORECASTING NIFTY 50 USING HYBRID ARIMA GARCH MODEL** is the result of the project work carried out by me under the guidance of Professor. Geetha P, in partial fulfilment for the award of Master's Degree in Business Administration by Bengaluru City University.

I also declare that this Master Thesis is the outcome of my own efforts and that it has not been submitted to any other university or Institute for the award of any other degree or Diploma or Certificate.

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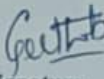
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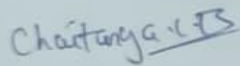

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I sincerely feel that the credit of the project work could not be narrowed to only one individual, as the work is the outcome of wholehearted co-operation from many persons. With their support and guidance, I was able to bring out this project report.

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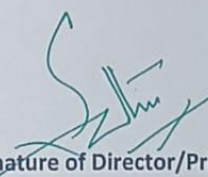

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ABSTRACT

This study proposes an estimation technique for developing the best fit ARIMA-GARCH model to predict the closing values of the Nifty 50 index. It addresses the issue of non-stationarity in both the mean and variance of the series through different methods before starting the estimation process. The study applies autoregressive integrated moving average (ARIMA), generalized autoregressive conditional heteroscedasticity (GARCH), exponential GARCH (EGARCH), and threshold GARCH (TGARCH) models on daily closing prices of the Nifty 50 from January 1, 2009, to December 30, 2019. The research follows a three-stage process: first, developing an ARIMA model that best fits the Nifty 50 index based on statistically significant coefficients and information criteria; second, fitting symmetric and asymmetric GARCH family models to capture volatility and leverage effects; and third, forecasting to evaluate model accuracy by comparing static and dynamic forecasts. The study identifies the ARIMA(2,1,2)-EGARCH(1,1,1) model as the best fit for predicting Nifty 50 closing prices, with static forecasts providing better accuracy than dynamic forecasts. The findings offer useful insights for portfolio managers and investors to frame suitable trading strategies with minimized risk.

CHAPTER 1 INTRODUCTION

1.1 INTRODUCTION OF THE TOPIC

The NIFTY 50 is a major standard index on the National Stock Exchange of India, representing the pier- burdened normal of the 50 largest privately listed companies and serving as a mark for the broader Indian equity request. When modeling and reading financial pointers like the NIFTY 50, the HYBRID ARIMA GARCH model offers a important approach by integrating two accredited statistical ways — ARIMA for landing direct trends and GARCH for modeling volatility patterns current in financial time series.

The NIFTY 50 tracks the performance of the top 50 Indian companies across 13 crucial sectors, furnishing broad exposure to the country's profitable exertion through a single portfolio. As the flagship indicator of the National Stock Exchange (NSE), it guides investment opinions, underpins indicator finances and ETFs, and forms the beginning asset for futures and options contracts in secondary requests. The NIFTY 50 is rebalanced twice a time, icing its ingredients reflect the dynamic Indian commercial geography and request capitalization trends.

Fiscal time series, similar as stock pointers, display complex characteristics like non-stationarity, autocorrelation, and, critically, volatility clustering — ages of high and low variability. Accurate modeling and foretelling serve essential places in trouble operation, algorithmic trading, and portfolio optimization. Simple models constantly fail to capture both trends and volatility conduct, challenging further robust approaches.

The ARIMA (Auto Regressive Integrated Moving Average) model is a foundation of time series analysis, especially suited for data with direct trends and autocorrelation. ARIMA builds on once values and vaticination crimes to model the elaboration of a time series, using parameters for autoregression, differencing (to address on-stationarity), and moving pars. ARIMA is largely flexible and can handle a broad range of stationary time series data, allowing experimenters to model underpinning trends and autocorrelation structures essential in fiscal requests.

The GARCH (Generalized Autoregressive Conditional Heteroskedasticity) model is specialized for capturing volatility clustering and changes in variance over time in financial data. Unlike ARIMA, GARCH does not forecast levels, but rather, the conditional variance (volatility) of residuals, enabling more realistic risk estimation and price forecasting. In financial analysis, GARCH models provide deep insight into the dynamic behaviour of volatility, supporting tasks such as risk management, portfolio optimization, and derivative pricing.

1.2 THEORETICAL BACKGROUND OF THE STUDY

The theoretical background of the study on "Modeling and Forecasting NIFTY 50 using HYBRID ARIMA GARCH Model" revolves around integrating two prominent time series models—ARIMA and GARCH—to enhance forecasting precision by capturing both the linear trends and volatility dynamics in financial data.

The ARIMA (Autoregressive Integrated Moving Average) model is a classical approach used to model and forecast time series data exhibiting autocorrelation and non-stationarity. It effectively decomposes a time series into autoregressive (AR) components that relate current values to past values, differencing (I) to achieve stationarity by removing trends, and moving average (MA) parts that model the error terms as a function of past errors. ARIMA seeks to model the conditional mean of the series, capturing overall trends and linear dependencies, which are essential for understanding the general movement of the NIFTY 50 index. However, ARIMA assumes constant variance (homoskedasticity) in the error terms, leaving it insufficient to model the characteristic volatility clustering and heteroscedasticity often observed in financial time series.

To address this limitation, the GARCH (Generalized Autoregressive conditional Heteroskedasticity) model is employed alongside ARIMA. GARCH models the conditional variance of the residuals from the ARIMA model, allowing for time-varying volatility and clustering of high and low volatility periods in stock returns like the NIFTY 50. The GARCH model specifies the conditional variance at a time point as a function of both the squared residuals and the conditional variance estimates, accommodating periods of high and low volatility which impact return and risk dynamics. It provides a more realistic and dynamic understanding of volatility, critical for risk management, portfolio pricing, and informed investment decisions.

The hybrid ARIMA-GARCH model thus combines the strengths of both approaches—ARIMA modeling the conditional mean of the returns (or price series) and GARCH modeling the conditional variance of the residuals—resulting in a robust framework capable of capturing both the predictable patterns and the unpredictable shocks in the NIFTY 50 market data. This combined model is particularly suitable for financial market forecasting where price movements exhibit complex behaviours with intermittent volatility spikes, improving forecasting accuracy and supporting better decision-making in portfolio management and trading strategies.

In the context of the NIFTY 50, which represents the performance of 50 major Indian companies and reflects the broader economic and financial market trends, this hybrid model facilitates a comprehensive understanding and prediction of market behaviour by jointly addressing trend and volatility, making it an effective tool for researchers and practitioners involved in Indian financial markets.

1.3 METHODS OF FORECASTING AND MODELING

Forecasting and modeling methods are systematic tools for predicting future values using historical and current data. These techniques are crucial in fields like finance, business, economics, and engineering for decision-making and planning.

1. **Time Series Methods** focus on analysing historical data points collected sequentially over time, capturing patterns such as trends, seasonality, and cyclic behaviours to predict future values.
2. **Causal & Multivariate Models** involve forecasting by examining relationships between multiple variables, using external factors or interconnected time series to explain and predict the behavior of the target variable.
3. **Machine Learning and Advanced Models** leverage computational algorithms to detect complex, nonlinear patterns and dependencies in data, enabling more flexible and accurate forecasting especially for large, high-dimensional, or unstructured datasets.

1.4 MODELING AND FORECASTING

Modeling and forecasting are essential analytical processes used across various domains, particularly in finance, business, and economics, to predict future events based on historical and current data. Modeling involves constructing mathematical or computational representations of real-world systems or phenomena. These models capture relationships between variables, uncover underlying patterns, and provide structured frameworks to understand complex dynamics. Forecasting, closely related, uses these models to estimate or predict future values, trends, or outcomes, enabling organizations to make informed decisions despite uncertainty about the future. These models can be deterministic, based on fixed mathematical relationships, or stochastic, incorporating randomness to reflect real-world uncertainties.

The practical uses of modeling and forecasting are wide-ranging. In the business context, forecasting helps anticipate customer demand, optimize inventory levels, plan budgets, and allocate resources efficiently. It informs strategic planning, risk management, and operational activities by projecting sales growth, market trends, or economic shifts. Financial markets rely heavily on modeling and forecasting to estimate asset prices, manage risks through volatility prediction, and guide investment decisions. Governments and policymakers use these tools to forecast economic indicators, evaluate policy impacts, and prepare for demographic changes. Ultimately, these predictive insights reduce the uncertainty inherent in future events, allowing entities to adapt proactively, allocate capital wisely, mitigate risks, and pursue growth opportunities. The benefits of effective forecasting include improved resource allocation, enhanced competitive advantage, reduced operational risks, and better financial performance.

1.5 ARIMA MODEL (AUTO REGRESSIVE INTEGRATED MOVING AVERAGE)

The ARIMA (Autoregressive Integrated Moving Average) model is a comprehensive and extensively used approach for modeling and soothsaying time series data. It combines three main factors — autoregression (AR), differencing (I), and moving average (Mama) — to capture different patterns in the data. The autoregressive (AR) element models the relationship between the current value and its once values, meaning it predicts unborn points grounded on once compliances. The integrated (I) element involves discriminating the data one or further times to make the time series stationary, which means its statistical parcels like mean and friction remain constant over time, an essential condition for dependable modeling. Differencing removes trends and seasonality, stabilizing the data.

The moving average (MA) part models the reliance between an observed value and residual crimes from once vaticinations, helping to smooth the noise in data. An ARIMA model is denoted as ARIMA (p, d, q), where p is the number of pause compliances included in the model (autoregressive order), d is the number of times the data are separated (degree of differencing), and q is the size of the moving average window (moving average order).

To develop an ARIMA model, a series of way are followed first, the data is checked for stationarity using tests like the Augmented Dickey- Fuller (ADF) and visual tools like autocorrelation (ACF) and partial autocorrelation (PACF) plots. also, applicable orders p, d, and q are named grounded on these plots and information criteria. The model parameters are estimated generally using styles similar as maximum liability estimation. Eventually, the model's acceptability is validated by assaying residuals and soothsaying performance, frequently repeating to optimize model delicacy. ARIMA is particularly useful when the time series exhibits non-stationarity and direct dependences but lacks strong seasonal goods (which can be reckoned for using extensions like Seasonal ARIMA or SARIMA). It's astronomically applied across disciplines similar as finance, economics, rainfall soothsaying, and deals vaticination, furnishing a flexible and robust medium to model trends, cycles, and arbitrary noise for accurate near- to medium- term soothsaying.

Non-periodic ARIMA models are generally meant ARIMA(p,d,q) where boundaries p, d, and q are non-negative entire numbers, p is the solicitation (number of postponements) of the autoregressive model, d is the degree of differencing (the events the data have had past characteristics deducted), and q is the solicitation for the moving-typical model. Infrequent ARIMA models are ordinarily implied ARIMA (p,d,q) (P,D,Q)m, where m implies the number of periods in each season, and the promoted P,D,Q imply the autoregressive, differencing, and moving typical terms for the intermittent bit of the ARIMA model. These models are fitted to time game plan data to all the more likely comprehend the information or to foresee future focuses in the arrangement (estimating).

1.6 GARCH (Generalized Autoregressive Conditional Heteroskedasticity)

The GARCH (Generalized Autoregressive Conditional Heteroskedasticity) model is a powerful econometric tool designed to analyse and forecast volatility in financial time series, which often exhibit periods of fluctuating variance known as volatility clustering. Unlike traditional time series models that assume constant variance (homoskedasticity), GARCH models explicitly accommodate changing variance (heteroskedasticity) over time by modeling the conditional variance as a function of past squared residuals (errors) and past variances. This approach captures the tendency of financial markets to experience bouts of high volatility followed by relative calm, a common phenomenon observed in asset returns such as stocks, bonds, and market indices. Mathematically, the GARCH model expresses the current conditional variance as a weighted sum of its past values and past squared errors, allowing the model to dynamically adapt to changing market risk conditions. The model recognizes that financial markets exhibit volatility clustering, meaning periods of high volatility tend to be followed by more high volatility periods, while calm periods are succeeded by continued tranquillity, a pattern that is ubiquitous in asset returns, stock indices, and other financial instruments.

The most commonly used version is the GARCH(1,1) model, which includes one lag of squared residuals and one lag of variance, making it both parsimonious and effective in capturing volatility persistence. The model's ability to reflect time-varying risk makes it invaluable for applications like risk management, portfolio optimization, option pricing, and Value at Risk (VaR) calculations. Developed by Tim Bollerslev in 1986 as an extension of the ARCH model, GARCH has become a cornerstone in financial econometrics, allowing analysts and traders to better estimate market uncertainty and adjust strategies accordingly. Its significance lies in improving the understanding of asset price dynamics beyond mean returns, by focusing on volatility behavior, which directly influences investment risk and decision-making processes. In practical terms, GARCH-based forecasts are used to anticipate periods of heightened market stress, informing trading positions and safeguarding against potential losses in turbulent markets.

The model is adaptable, with numerous variations catering to asymmetric effects, leverage impacts, and long memory in volatility, making it an indispensable tool in modern quantitative finance. The most commonly used specification is GARCH(1,1), represented as $\sigma_t^2 = \omega + \alpha_1 \varepsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2$, where σ_t^2 is the conditional variance at time t , ω is a constant term representing long-run variance, α_1 captures the impact of past squared shocks, and β_1 reflects the persistence of past volatility. This formulation allows the model to capture both short-term volatility shocks through the ARCH term and long-term volatility persistence through the GARCH term, making it particularly effective for modeling financial data where volatility exhibits both clustering and mean-reverting behaviour.

1.7 IMPORTANCE OF THE STUDY

1. The NIFTY 50 is a key indicator of the Indian equity market, representing the top 50 companies, making its forecasting crucial for market analysis.
2. Accurate modeling and forecasting of the NIFTY 50 help investors make informed decisions to optimize portfolio returns and manage risks.
3. The hybrid ARIMA-GARCH model integrates linear trend analysis with dynamic volatility modeling, capturing more complex market behaviours.
4. Using the hybrid model improves forecast accuracy by addressing both the mean and variance of the NIFTY 50 time series.
5. Enhanced forecasting supports better risk management, enabling traders and fund managers to anticipate market fluctuations.
6. The study contributes to financial stability by providing forecasts that aid in navigating market uncertainty and volatility.
7. Results of this study can serve as a decision-making tool for policymakers to monitor and regulate the financial markets.
8. The research advances academic knowledge on hybrid econometric models tailored to emerging markets like India.
9. It fills existing gaps in literature regarding volatility modeling and forecasting in Indian financial markets.
10. The study lays a foundation for future research to apply hybrid ARIMA-GARCH and related models to other financial assets. Improved forecasting can enhance investment strategies, derivatives pricing, and hedging techniques in the stock market.
11. The methodology offers practical benefits for quantitative analysts and algorithmic traders using predictive analytics.
12. It aids in understanding the structural dynamics and temporal dependencies within the NIFTY 50 index data.
13. The hybrid approach provides a robust framework adaptable to both short-term and medium-term financial forecasts.
14. The study's insights may boost confidence among foreign investors by providing transparent and rigorous market forecasts.
15. It supports the development of financial products and services based on reliable predictive models.
16. The approach's application may reduce the adverse effects of sudden market shocks on portfolios.

1.8 NEED AND RELEVANCE OF THE STUDY

The need and relevance of the study on "Modeling and Forecasting NIFTY 50 using HYBRID ARIMA GARCH Model" is underscored by the increasing significance of the NIFTY 50 index as a benchmark for the Indian equity market. As one of the most widely followed stock indices, it reflects the performance of the top 50 companies listed on the National Stock Exchange of India, thereby offering critical insights into the economic and financial health of the country. Accurate forecasting of this index is vital for investors, policymakers, and financial analysts to make sound decisions in an environment characterized by volatility and uncertainty. The hybrid ARIMA-GARCH model addresses the complex nature of financial time series by combining trend modeling and volatility estimation.

Traditional forecasting models often struggle to simultaneously capture the linear patterns in stock prices and the varying volatility intrinsic to financial markets. The hybrid approach enhances predictive capability by first modeling the mean process with ARIMA and then capturing conditionally changing variance through GARCH, making it highly relevant for markets like India where fluctuations can be abrupt and significant. Furthermore, the study is needed to fill gaps in existing literature on emerging market indices and to develop models specifically tailored to the structural dynamics of the Indian stock market. Insights from this research can improve risk management strategies, offering practical tools for portfolio optimization and hedging.

The study also aligns with the growing demand for sophisticated forecasting tools among retail and institutional investors, providing a more reliable basis for investment decisions amid market uncertainties. In addition, this research holds broader economic relevance by supporting regulatory bodies in monitoring market stability and making informed policy interventions. Its findings can contribute to the advancement of financial econometrics and quantitative finance, particularly in applying hybrid methodologies in the context of emerging markets. Overall, the study is timely, practically significant, and academically valuable, meeting both market needs and scholarly objectives.

The relevance of this study extends to the unique characteristics of the Indian financial market, which is marked by rapid growth, increasing integration with global markets, and a high degree of volatility influenced by economic reforms, geopolitical factors, and investor sentiment. The NIFTY 50 index serves as a barometer not only for equity market performance but also for economic progress, making its accurate prediction pivotal for economic forecasting and financial stability in India. Hybrid approaches like ARIMA-GARCH bridge the gap between capturing the deterministic trends in stock prices and the stochastic behavior of volatility, which is critical given the heavy-tailed and heteroscedastic nature of stock market returns. This synergy offers a deeper understanding of market mechanisms, thereby improving forecasting reliability and offering investors tools to hedge against market risks more effectively. Moreover, the study contributes significantly to enhancing algorithmic and quantitative trading strategies, which rely heavily on precise forecast models to maximize returns and minimize risks.

By adapting and validating hybrid models in the Indian context, this research helps tailor global best practices to suit local market idiosyncrasies, increasing their practical applicability. From a policy perspective, accurate index forecasting aids regulatory authorities like the Securities and Exchange Board of India (SEBI) in detecting abnormal market movements and systemic risks early, facilitating timely intervention and safeguarding investor interests. The study also supports the growth of financial products derived from the index such as futures, options, and ETFs, whose valuations depend critically on market forecasts. Furthermore, the academic community benefits from this research through the development of new hybrid modeling techniques and empirical insights that shed light on the interplay between stock returns' mean and variance processes. This fosters innovation in financial econometrics and quantitative finance, encouraging future investigations into more complex and dynamic market models.

CHAPTER II ORGANIZATIONAL PROFILE

2.1 INTRODUCTION TO THE ORGANIZATION



Unocoin Technologies Pvt Ltd stands as a pioneering and trusted name in India's cryptocurrency landscape. Founded in 2013 by a team of visionary entrepreneurs—Sathvik Vishwanath, Harish BV, Sunny Ray, Abhinand Kaseti, and Vikram Nikkam—the company was one of the very first entrants into the Indian Bitcoin and crypto assets market. Headquartered in Bengaluru, with its origins in Tumkur, Unocoin has been instrumental in introducing and enabling cryptocurrency adoption across the nation for over a decade.

The company's core mission is to empower Indians with financial freedom by providing a secure, reliable, and user-friendly platform to buy, sell, store, and use Bitcoin and a wide array of other cryptocurrencies. With a user base of over 2 million people, Unocoin has built a strong reputation based on trust and longevity in a highly dynamic industry. The platform is designed to cater to both novice investors and experienced traders, offering an easy-to-use interface, multilingual customer support, and sophisticated features like a trading API for programmatic bots. Backed by prominent global investors, including Draper Associates and Digital Currency Group, Unocoin continues to be a leading force in India's journey toward a decentralized financial future.

Aspect	Details
Company Name	Unocoin Technologies Private Limited
Founded	2013
Headquarters	Bengaluru, Karnataka
Founders	Sathvik Vishwanath, Harish BV, Sunny Ray, Abhinand Kaseti, Vikram Nikkam
Core Business	Cryptocurrency Exchange (Buy, Sell, Store, Use Crypto)
User Base	Over 2 million

Key Offerings	Instant Exchange, SBP (Systematic Buying Plan), API Trading, Bill Payments with Bitcoin
Key Investors	Draper Associates, Digital Currency Group (DCG)



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Date: Sept 08, 2025

TO WHOM IT MAY CONCERN

This is to certify that **Mr. Chaitanya C Belgaumkar** (Reg No: P18ET23M015012), an MBA (Finance) student from **MP Birla Institute of Management**, affiliated with Bengaluru City University (BCU), has successfully completed his internship with Unocoin Technologies Pvt Ltd. His internship period was from **September 1, 2025, to October 31, 2025.**

During his tenure, he conducted an extensive research project titled, "**Modeling And Forecasting NIFTY 50 using HYBRID ARIMA GARCH Model**". His research provided valuable insights into the dynamic relationship between cryptocurrencies and traditional fiat currencies, which is a key area of interest for our business.

Throughout the internship, we found him to be a dedicated, analytical, and professional intern. His ability to apply advanced financial modeling techniques to a complex subject was commendable.

We wish him all the best for his future endeavours.

For UNOCOIN TECHNOLOGIES PVT LTD

Mr. Umesh Kulkarni
HR Manager



2.2 COMPANY PROFILE

2.2.1 COMPANY HISTORY/INCEPTIONS



The history of Unocoin Technologies Pvt Ltd is a story of pioneering vision, resilience, and early leadership in one of the world's most dynamic technology sectors. Founded in the southern Indian town of Tumkur in 2013, Unocoin emerged at a time when Bitcoin and cryptocurrency were novel and largely unknown concepts in India. The founding team—comprising Sathvik Vishwanath, Harish BV, Sunny Ray, Abhinand Kaseti, and Vikram Nikkam—recognized the transformative potential of this new financial technology and embarked on a mission to make it accessible to Indians.

As one of the country's very first crypto startups, Unocoin was instrumental in building the foundational infrastructure for the Indian crypto ecosystem. It launched a platform that simplified the process of buying, selling, and storing Bitcoin, effectively creating an on-ramp for thousands of early adopters to enter the world of digital assets. The company's early-mover advantage and unwavering focus on user trust allowed it to establish a strong brand presence. Over the years, Unocoin has navigated the significant volatility and regulatory uncertainties of the crypto market, including the RBI's banking ban in 2018. The company's endurance through these challenging periods solidified its reputation as a resilient and reliable player. The subsequent lifting of the ban by the Supreme Court of India allowed Unocoin to continue its growth trajectory, attracting backing from prominent global investors like Tim Draper and Barry Silbert's Digital Currency Group. From its inception, Unocoin's history has been synonymous with the history of cryptocurrency in India itself, marking it as a true industry pioneer.

2.2.2 VISION, MISSION AND OBJECTIVES

The strategic direction of Unocoin is guided by a clear and ambitious set of principles that have remained consistent since its inception. The company is focused on leading the adoption of cryptocurrency in India by prioritizing trust, security, and user experience.

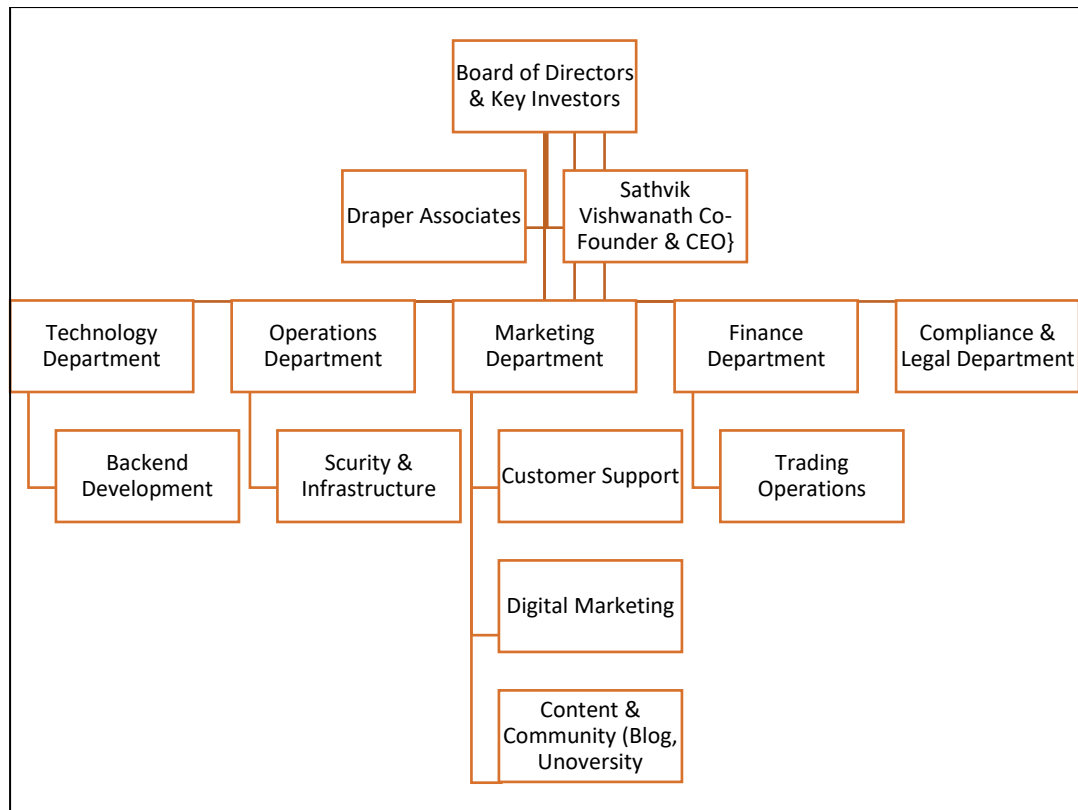
1. **Vision:** Unocoin's vision is to emerge as a top global player in the cryptocurrency space by providing a comprehensive suite of services that enable Indians to buy, sell, store, use, and accept Bitcoin and other digital assets. The company aims to be at the forefront

of the financial revolution, empowering its customers to achieve financial freedom through seamless access to the crypto ecosystem.

2. **Mission:** The core mission of Unocoin is to make cryptocurrency trading and usage simple, secure, and accessible for everyone in India. This is achieved by building a trusted brand, offering an easy-to-use interface with multilingual support, and maintaining some of the most sophisticated security systems in the industry. The company is committed to educating its users and fostering a well-informed crypto community through its blog and "Unoversity" initiatives.
3. **Objectives:** The primary objectives that stem from this vision and mission are:
 - a To provide a reliable and secure platform for over 2 million users to trade and manage their cryptocurrency portfolios.
 - b To maintain industry-leading security protocols to ensure that customers' funds are safe.
 - c To offer competitive and dynamic fee structures to cater to a wide range of users, from beginners to high-volume traders.
 - d To innovate and introduce new features that enhance the utility of cryptocurrencies, such as the Systematic Buying Plan (SBP) and the ability to pay utility bills with Bitcoin.
 - e To continue to serve as a trusted and pioneering brand in the Indian crypto industry, building on over a decade of operational experience.

2.2.3 ORGANIZATIONAL HIERARCHY OF THE COMPANY

Unocoin Technologies Pvt Ltd is a privately held company with a leadership structure that reflects its origins as a founder-led startup. The hierarchy is relatively flat, promoting agility and direct communication, with the co-founders playing active roles in the company's strategic and operational management.



2.2.4 PRODUCT AND SERVICE PROFILE

Unocoin Technologies Pvt Ltd offers a comprehensive and secure platform for trading and managing a broad range of cryptocurrencies tailored to the Indian market. The company's flagship product is its cryptocurrency exchange, which facilitates instant buying, selling, and storing of leading cryptocurrencies including Bitcoin, Ethereum, and stablecoins like USDT. The platform emphasizes an easy-to-use interface accessible via web and mobile apps, supported by seamless INR deposits and withdrawals through multiple banking channels.



Key services include:

1. **Instant Exchange:** Enables users to trade cryptocurrencies in real-time with competitive pricing and low fees.
2. **Systematic Buying Plan (SBP):** A unique feature allowing investors to dollar-cost average into Bitcoin holdings over time, minimizing volatility risk.
3. **Wallet Services:** Secure, multi-layered wallet infrastructure supporting cold storage and hot wallets for ease of access and security.
4. **Bill Payments with Cryptocurrency:** Innovative use-case enabling users to pay mobile, DTH, and utility bills using Bitcoin.
5. **API Access:** Industry-leading API for developers to build trading bots and integrate with external applications for programmatic access.
6. **Educational Platform (Unoversity):** Provides curated content, blogs, and tutorials to enhance crypto literacy among users.

With a focus on security, Unocoin has implemented one of the most sophisticated crypto security systems, including multifactor authentication, encryption, and regular audits to protect users' funds. The company also offers competitive fee structures, including dynamic fees for high-volume traders, market makers, and sub-brokers, aligning costs with trading behavior and volume. This diversified product and service offering positions Unocoin as a user-centric platform dedicated to fostering broader crypto adoption in India while maintaining industry-leading security and usability standards.

2.2.5 COMPETITORS PROFILE AND ANALYSIS

Unocoin operates within a highly competitive Indian cryptocurrency exchange market, facing competition from both domestic startups and international players. Below is an analysis of key competitors along key parameters:

Competitor	Market Share	Core Strengths	Weaknesses	Unique Selling Points
CoinDCX	25%	Wide coin selection, strong VC funding	Complex UI for beginners	Advanced trading features, staking
WazirX	20%	Binance backing, large user base	Regulatory uncertainty	Binance ecosystem integration
ZebPay	15%	Veteran brand, strong security	Limited coin offerings	High trust, user-friendly interface
Bitbns	10%	Strong customer service	Lower liquidity	Referral programs, easy onboarding

Unocoin	8%	Early mover advantage, strong security	Smaller market share	Systematic Buying Plan, bill payments
Koinex	5%	Innovative features	Reduced operations amid regulations	Focused on Indian retail investors

This competitive landscape shows Unocoin's strength in security and unique offerings like SBP. However, it faces challenges expanding market share amid fierce competition from better-funded rivals and regulatory uncertainties.

2.2.6 MILESTONES/ACHIEVEMENTS

Unocoin's growth trajectory is marked by significant milestones that highlight its pioneering role and resilience:

1. **2013:** Founded and launched India's first Bitcoin exchange.
2. **2017:** Achieved 100,000 active users amid growing crypto awareness.
3. **2018:** Successfully navigated the RBI banking ban impacting crypto, maintaining operations through alternate partnerships.
4. **2020:** Launched Systematic Buying Plan (SBP), a first of its kind feature in India to help users dollar-cost average Bitcoin investments.
5. **2022:** Crossed the 2 million user milestone, solidifying its position as a trusted brand.
6. **2023:** Introduced utility bill payments via Bitcoin, enhancing crypto usability.
7. **2024:** Integrated industry-leading API allowing programmatic trading bots.
8. **2025:** Launched Unoversity, dedicated to crypto education and literacy.

These achievements illustrate Unocoin's dedication to innovation, user growth, and navigating an evolving regulatory landscape.

2.2.7 SWOT ANALYSIS

Strengths	Weaknesses
Pioneer in Indian crypto market since 2013	Smaller market share compared to competitors
Strong security and trust among users	Limited coin offerings compared to global exchanges
User-friendly interface and multi-language support	Reliance on Indian regulatory environment creates uncertainty
Innovative features like Systematic Buying Plan	High competition with better-funded platforms
Opportunities	Threats
Growing crypto adoption in India	Regulatory uncertainties and possible bans

Expansion into new financial products (bill payments, DeFi)	Cybersecurity threats
Potential partnerships with banks and fintech's	Volatile cryptocurrency market
Increasing investor education via Unoversity	Intense market competition

2.2.8 FUTURE GROWTH

The future growth potential for Unocoin is robust amid a rapidly maturing Indian crypto ecosystem. Key growth drivers include:

1. **Increasing Crypto Adoption:** With more Indians understanding and entering the crypto market, demand for reliable platforms like Unocoin is rising.
2. **Product Diversification:** Expanding services into lending, staking, and decentralized finance (DeFi) will broaden revenue streams.
3. **Institutional Engagement:** Preparing infrastructure to cater to institutional investors could tap into a large emerging market segment.
4. **Technological Advancements:** Leveraging AI and blockchain innovations can enhance security, user experience, and operational efficiency.
5. **Educational Initiatives:** Continuing to grow Unoversity will position Unocoin as the authority on crypto literacy in India.
6. **Strategic Partnerships:** Collaborations with banks, payment networks, and fintech firms could accelerate growth.

While regulatory clarity remains a key factor, Unocoin's pioneering status and innovation pipeline place it well for long-term expansion.

2.3 BUSINESS MODEL CANVAS

Component	Description
Key Partners	- Cryptocurrency Networks: Core infrastructure providers like the Bitcoin and Ethereum networks. - Banking & Payment Gateways: Partners that facilitate seamless INR deposits and withdrawals. - Regulatory Bodies: Engagement with entities like the RBI and SEBI for compliance. - Technology Providers: Cloud hosting services (e.g., AWS) and security software providers. - Global Investors: Backing from key investors like Draper Associates and DCG provides capital and strategic guidance.

Key Activities	- Platform Operation & Maintenance: Ensuring the 24/7 availability, security, and performance of the trading platform. - Transaction Processing: Securely executing buy/sell orders and managing fund transfers. - Security & Compliance: Continuous monitoring for threats and adherence to KYC/AML norms and Indian tax regulations. - Customer Support: Providing multilingual assistance to resolve user queries. - Innovation & Product Development: Creating new features like SBP and bill payments.
Key Resources	- Proprietary Technology Platform: The core software, including the exchange engine, wallets, and API. - Highly Skilled Team: Engineers, security experts, and finance professionals with expertise in both technology and financial markets. - Brand & Trust: Over a decade of operational history has built a strong, trusted brand. - User Base: A large and active community of over 2 million users. - Financial Capital: Funding from investors to support operations and growth.
Value Propositions	- Trust & Security: A primary value proposition based on its long history and robust security measures. - Simplicity & Accessibility: An easy-to-use interface designed for beginners, complemented by features for advanced users. - Innovation: Unique features like the Systematic Buying Plan (SBP) that help users manage volatility risk. - Utility: The ability to use cryptocurrency for real-world transactions like paying utility bills. - Education: Empowering users with knowledge through the Unoversity platform.
Customer Relationships	- Direct Support: Multilingual customer service via email and phone. - Self-Service: A comprehensive FAQ section and educational blog to help users find answers independently. - Community Engagement: Building a community through social media, news updates, and educational content. - Automated Services: Providing a seamless, automated experience for trading and account management.
Channels	- Website (unocoin.com): The primary web-based platform for trading and account access. - Mobile Applications: Dedicated apps for iOS and Android devices, enabling users to trade "anywhere, anytime." - API: A channel for developers and algorithmic traders to connect their systems directly to the exchange.

	- Content Marketing: The company blog, news section, and social media presence serve as channels for user acquisition and education.
Customer Segments	- Retail Investors: The core customer base, comprising individuals from across India who are new to or actively trading cryptocurrencies. - Advanced Traders: Users who require more sophisticated tools, such as API access for programmatic trading. - Long-term Hodlers: Investors who use the platform for long-term accumulation, often leveraging the SBP feature. - Users Seeking Utility: Customers who are interested in using cryptocurrencies for payments and other real-world applications.
Cost Structure	- Technology & Infrastructure: Costs associated with servers, software licenses, and maintaining the platform's security. - Employee Salaries: Compensation for the engineering, support, marketing, and management teams. - Marketing & Customer Acquisition: Expenses related to advertising, promotions, and referral programs. - Compliance & Legal: Costs associated with meeting regulatory requirements, audits, and legal counsel. - Transaction Costs: Fees paid to payment gateways and banking partners.
Revenue Streams	- Trading Fees: The primary source of revenue, generated from a percentage fee charged on each buy and sell transaction. Unocoin offers a dynamic fee structure based on trading volume. - Deposit/Withdrawal Fees: Fees charged for processing INR deposits and withdrawals. - API Subscription Fees: Potential revenue from premium API access for high-frequency traders. - Service Fees: Fees associated with value-added services like bill payments using cryptocurrency.

2.4 MCKINSEY'S 7-S FRAMEWORK

1. Strategy: Unocoin's strategy centres on popularizing Bitcoin and other cryptocurrencies in India by providing a secure, user-friendly, and regulatory-compliant platform for buying, selling, and managing digital assets. The company differentiates itself through innovative products (like systematic investment plans in Bitcoin), a strong focus on compliance, and ongoing digital literacy campaigns to build trust and expand adoption.

2. Structure: Unocoin operates through a structured yet agile organization. It includes clear functional divisions: technology, compliance, marketing, customer support, and operations. As a fintech, its structure prioritizes rapid technological iteration, robust compliance oversight, and operational scalability.

3. Systems: Its core systems include an advanced trading platform, rigorous KYC/AML onboarding, transaction monitoring, and customer support systems. Automation, security protocols, and regular tech updates are emphasized to support reliability and regulatory requirements. The firm leverages APIs and cloud-based infrastructure to offer seamless access and integration.

4. Shared Values: Trust, transparency, and innovation are at the heart of Unocoin's shared values. The company actively promotes secure crypto adoption and financial inclusion while adhering to legal and ethical standards, seeking to earn the confidence of both users and regulators.

5. Style: Unocoin's leadership style blends entrepreneurial agility with a strong compliance mindset. Founders and managers are visible advocates for crypto literacy, quick to adopt new opportunities, and committed to responsible innovation. Decision-making is pragmatic but highly responsive to shifts in the regulatory and technological landscape.

6. Staff: The company maintains a dynamic workforce skilled in blockchain technology, cybersecurity, regulatory affairs, fintech product development, and customer relations. Recruiting emphasizes adaptability, tech-savviness, and integrity, while training/upskilling in emerging crypto trends and compliance rules is a constant priority.

7. Skills: Unocoin's critical skills include blockchain platform engineering, data security, compliance (especially KYC/AML in a restrictive regulatory environment), and user education. Their technical expertise, combined with industry experience in navigating crypto-specific legal frameworks, allows them to stay ahead in the volatile digital asset space.

CHAPTER III

LITERATURE REVIEW AND RESEARCH DESIGN

3.1 LITERATURE REVIEW

3.1.1 Author: Henry (1998): This study used daily data of Hong Kong stock exchange for modelling the asymmetry of the stock market and found that GQARCH model performs to be the most appropriate model to measure the asymmetry. Similarly, Binder and Merges (2001) examined how economic forces explained the volatility of stock market using the standard deviation of S&P Composite Index and concluded that standard deviation inversely relates to the ratio of projected profits to projected revenues, implying the risk will rise during economic contractions and fall during economic recoveries. More so, Singla and Pasricha, (2012) suggested that systematic risk is linearly related to stock's return during market rises, but during falls, the relation is non-linear.

3.1.2 Authors: Mohammadi and Su (2010): This study examines the behaviour of oil returns and conditional variance using weekly data of eleven global markets comprising of oil-exporting and oil-importing countries applying four types of GARCH models. The results indicated that the conditional standard deviation is more able to capture the volatility in oil returns than the traditional conditional variance. The study also considered the out-of-sample forecasting measures of four volatility models such as GARCH, EGARCH, APARCH, FIGARCH and suggested that MA(1)-APARCH(1,1) model outperforms the others. Similarly, Sopipan (2017) forecasted the volatility of gold prices using ARIMA - GARCH model using three different distributions for the innovations and advocated that ARIMA(2,0,2) gives the best performance for predicting the gold returns. The study also suggested that the cumulative returns ARIMA (2,0,2)-GARCH model with normal and GED error distribution outperforms ARIMA (2,0,2)-GARCH-t error distribution.

3.1.3 Authors: Mustapa and Ismail (2019): This study also identified a suitable ARIMA model and fitted it into GARCH (1,1) model to measure the variability in variance. The results revealed that ARIMA (2,1,2)-GARCH (1,1) model is considered to be the most suited model for predicting the S&P500 stock prices and also pointed out that a dynamic forecast gives better prediction as compared to a static one.

3.1.4 Authors: Kumar and Dhankar (2010): This study has investigated the asymmetric effect on volatility and stated negative significant connection among stock returns and conditional volatility. But, the association between stock yield and standardized residuals was significant. This study also highlighted the facts that investors modify their investment choices with regard to expected volatility, however, they expect additional risk premium for the unexpected volatility. One more study by Kumar and Dhankar (2011) verified the existence of non-linearity, heteroskedasticity and asymmetric nature of stock returns and reported a positive significant relationship between stock return and unexpected volatility, inferring investors anticipate more risk premium if there is any unpredicted rise or drop in stock prices.

3.1.5 Authors: Xiong and Han (2015): This study has used the Granger causality ñ MSV model to examine the volatility spillover and suggested that volatility spillover effects are bi-directional and asymmetric. Varughese and Mathew (2017) also showed the impact of FPIs investment on the Indian stock market volatility and opined that market is more volatile in decreasing trend rather than in increasing trend. The study results also revealed that there is a significant contribution of FPIs purchase and sales in making the stock market volatile.

3.1.6 Authors: Natchimuthu and Chellaswamy (2018) and Amudha and Muthukamu (2018): This study has found the presence of volatility clustering, long term memory features and leverage effect in the NSE sectoral indices and determine that negative news have more influence on the next period volatility as asymmetrical volatility models beat the symmetrical volatility models. One of the latest studies by Kaur and Singh (2019) examined the Nifty future index and MCX composite commodity index and reported persistent volatility in both the markets, though asymmetric effect was confirmed only in stock futures market, not in the commodity future market.

3.1.7 Authors: Banumathy and Azhagaiah (2012): This study exhibited GARCH (1,1) is the best possible model to capture the symmetric volatility and for asymmetric volatility, TGARCH (1,1) to be the more suitable model. Anton (2012) suggested TGARCH and PGARCH to be the most successful models to forecast volatility. Similarly, Vasudevan and Vetrivel (2016) recommended that asymmetric models are better as compared to the symmetric GARCH model. Studies by Amudha and Muthukamu (2018); and Lama et al. (2015) also proposed that EGARCH model perform better as compared to ARIMA and GARCH models to capture the volatility.

3.2 STATEMENT OF THE PROBLEM

The study recognizes that volatility forecasting is one of the most challenging tasks in finance, requiring prediction about future market movements based on historical and current observations. The research identifies the need to address non-stationarity in both the mean and variance of the NIFTY 50 time series data before beginning the modeling process. The problem involves determining the optimal hybrid ARIMA-GARCH model specification that can best capture both the trend patterns (through ARIMA) and volatility clustering (through GARCH) in NIFTY 50 data.

The study aims to develop a model that provides accurate forecasts of NIFTY 50 closing prices, comparing static versus dynamic forecasting approaches to determine which provides better predictive performance. The research seeks to provide a tool that can assist domestic and international portfolio managers and investors in framing suitable NIFTY trading strategies with minimal risk.

3.3 NEED FOR THE STUDY

The need for this study arises due to the high complexity and unpredictability of financial markets, especially the NIFTY 50 index, which serves as India's benchmark for market performance. Accurately forecasting price movements and volatility is critical for investors, portfolio managers, and policymakers, as it aids in portfolio construction, risk management, derivatives pricing, and asset allocation decisions. Traditional time series models like ARIMA can address mean prediction but often fail to capture volatility clustering and leverage effects, which are key characteristics of financial data. By improving forecast accuracy, the research provides valuable tools for investors, portfolio managers, and policymakers to make informed decisions amid market uncertainties, thereby contributing to more stable financial markets and enhanced economic growth.

3.4 SCOPE OF THE STUDY

The scope of this study focuses on modeling and forecasting of closing prices of Nifty 50 index using advanced time series and volatility modeling techniques, specifically hybrid ARIMA-GARCH family models. Covering daily data from 2009 to 2019, the study comprehensively examines both the mean and variance dynamics of Indian stock market benchmarks, using multiple models including ARIMA, GARCH, EGARCH and TGARCH. Research is limited to short-term forecast accuracy and model comparison, particularly static versus dynamic out-of-sample prediction. It aims to provide actionable insights to domestic and international investors, portfolio managers and financial strategists, providing an evidence-based framework for risk analysis and investment decision making in the context of Indian equity markets.

3.5 OBJECTIVES OF THE STUDY

1. To develop an optimal ARIMA model that best fits the movement of the Nifty 50 index by analysing statistically significant coefficients and selecting the model with the least information criterion values.
2. To apply symmetric as well as asymmetric GARCH family models (including GARCH, EGARCH, and TGARCH) to measure heteroskedasticity in the variance of the Nifty 50 series and to identify the presence of leverage effects.
3. To forecast the closing prices of the Nifty 50 index using the identified hybrid ARIMA-GARCH models and to evaluate the forecasting accuracy.
4. To compare the performance of static and dynamic forecasting methods to determine which provides better prediction results.
5. To provide findings that can help domestic and international portfolio managers and investors formulate suitable Nifty trading strategies with minimal risk.

These objectives aim to construct a robust and precise hybrid model for forecasting the Nifty 50, addressing issues of non-stationarity in mean and variance while capturing both trend and volatility dynamics in the index.

3.6 HYPOTHESIS OF THE STUDY

PRIMARY HYPOTHESIS

H₁: A hybrid time-series model that combines an ARIMA mean equation with a GARCH family variance equation will significantly outperform standalone ARIMA or GARCH models in fitting and forecasting the daily closing prices of the NIFTY 50 index, as evidenced by lower information criteria (AIC, SIC, HQC) and improved out-of-sample forecast accuracy (lower MAE, RMSE, and Theil's U₂).

NULL HYPOTHESIS

H₀: There is no statistically significant improvement in model fit or forecast accuracy when using a hybrid ARIMA-GARCH framework compared to using ARIMA or GARCH models individually for the NIFTY 50 daily closing prices.

ALTERNATIVE HYPOTHESIS (H_A):

Volatility clustering and leverage effects exist in the NIFTY 50 closing prices, requiring more advanced models like GARCH, EGARCH, or TGARCH to effectively capture and forecast the volatility behaviour beyond what ARIMA alone can achieve.

3.7 SAMPLING TECHNIQUE

The study employs complete enumeration sampling (also known as census sampling) because which involves collecting data from every single unit within the defined population rather than selecting a representative subset. It includes every daily closing price of the NIFTY 50 index from January 1, 1999, to August 21, 2025, to ensure comprehensive data coverage for accurate modeling and forecasting, as using the entire population of observations maximizes information and minimizes bias in time series analysis.

3.8 TOOLS FOR DATA COLLECTION

The study employs a univariate time-series of secondary data, consisting of 6,631 daily observations of NIFTY 50 index closing prices and related trading metrics, obtained directly from the National Stock Exchange of India's official data releases. Each record in the dataset includes the trading date, open, high, low, and close prices, trading volume, number of trades, net turnover in thousands of rupees, market capitalization, and daily return. The data span from January 1, 1999 through August 21, 2025, covering approximately 26.6 years of continuous trading days. Tools like Microsoft Excel and EViews were used to organize, tabulate, and analyse the data. Statistical techniques were applied to identify patterns and interpret results effectively.

All price series are fully adjusted for corporate actions (splits, dividends, and index membership changes) by the exchange, ensuring consistency over time. This secondary dataset is ideally suited for ARIMA-GARCH modeling because it provides high-frequency, long-horizon observations that capture both the mean dynamics and volatility clustering intrinsic to financial markets. The data were obtained directly from the capital line neo database.

3.9 RESEARCH METHODOLOGY

Research methodology explains the approach and procedures used to acquire, organize, and analyse information. It provides the framework for data collection, analysis, and interpretation.

3.9.1 TYPE OF RESEARCH

This study is **quantitative**, **empirical**, and **explanatory** in nature, employing secondary time-series data and econometric modeling to analyse and forecast market behavior:

1. Quantitative Research

- a) Relies exclusively on numerical data (daily NIFTY 50 closing prices and derived returns).
- b) Uses statistical and econometric methods (ARIMA for mean dynamics; GARCH family models for volatility).

2. Empirical Research

- a) Draws on real-world, secondary data from the National Stock Exchange (January 1, 1999–August 21, 2025).
- b) Tests theoretical time-series and volatility models against observed market behavior.

3. Explanatory Research

- a) Seeks to explain why and how NIFTY 50 returns exhibit persistence, clustering, and leverage effects.
- b) Identifies causal relationships between past shocks and current volatility.

4. Time-Series Design

- a) Univariate: Models a single variable (NIFTY 50 closing price) over time.
- b) Longitudinal: Covers 6,631 trading days (≈ 26.6 years) to capture evolving market dynamics.

5. Applied Research

- a) Aims to develop practical forecasting tools (hybrid ARIMA-EGARCH) for portfolio managers and investors.
- b) Evaluates out-of-sample forecast accuracy (static vs dynamic) for real-world decision support.

In sum, the study integrates quantitative econometric techniques with empirical market data to explain and predict stock-market volatility, making it a quantitative, empirical, explanatory, time-series study with applied objectives.

3.9.2 SAMPLING SIZE

The sample size for the study "Modeling and Forecasting NIFTY 50 using HYBRID ARIMA GARCH Model" comprises 6,631 daily observations of NIFTY 50 closing prices spanning from January 1, 1999, to August 21, 2025, covering approximately 26.6 years of trading data, which provides an exceptionally robust dataset that is significantly larger than the referenced study's 2,657 observations and offers excellent statistical power for conducting comprehensive ARIMA-GARCH time series analysis and volatility modeling.

Sample Size Details:

1. **Total observations:** 6,631 daily observations
2. **Index:** NIFTY 50 (Perfect match for your research)
3. **Time period:** January 1, 1999 to August 21, 2025
4. **Time span:** 26.6 years of daily trading data
5. **Data frequency:** Daily closing prices

3.9.3 DATA COLLECTION

Data was collected from capital line neo database ensuring reliability and completeness. Historical monthly prices were organized in Excel and prepared for statistical analysis.

3.9.4 DATA LIMITATIONS

1. Although the dataset spans 26.6 years (6,631 daily observations), it may not fully capture structural breaks or regime shifts—such as major regulatory reforms, technological disruptions, or unprecedented events like the COVID-19 pandemic—that can alter market dynamics in ways not modeled by ARIMA-GARCH frameworks.
2. The analysis relies solely on closing-price data and omits intraday information (e.g., high-frequency returns) that could provide richer insights into volatility patterns and leverage effects, potentially underestimating extreme short-term fluctuations.
3. Corporate actions (stock-splits, index composition changes, dividends) over the sample period can introduce discontinuities in the price series; while the National Stock Exchange adjusts for these, residual effects might still bias the estimated parameters.
4. The dataset excludes over-the-counter transactions and derivative markets, so the volatility captured reflects only exchange-traded spot activity and may omit spillovers from futures, options, and global markets.

5. Macroeconomic variables (inflation, interest rates, global indices) and structural covariates are not incorporated; thus, the univariate ARIMA-GARCH models cannot explain exogenous drivers of volatility, limiting the scope of causal interpretation.
6. Trading calendar effects—such as holidays, shortened sessions, and evolving market microstructure—are not explicitly modeled, which could lead to heteroskedasticity patterns being conflated with non-trading day effects.

3.9.5 TOOLS USED FOR ANALYSIS

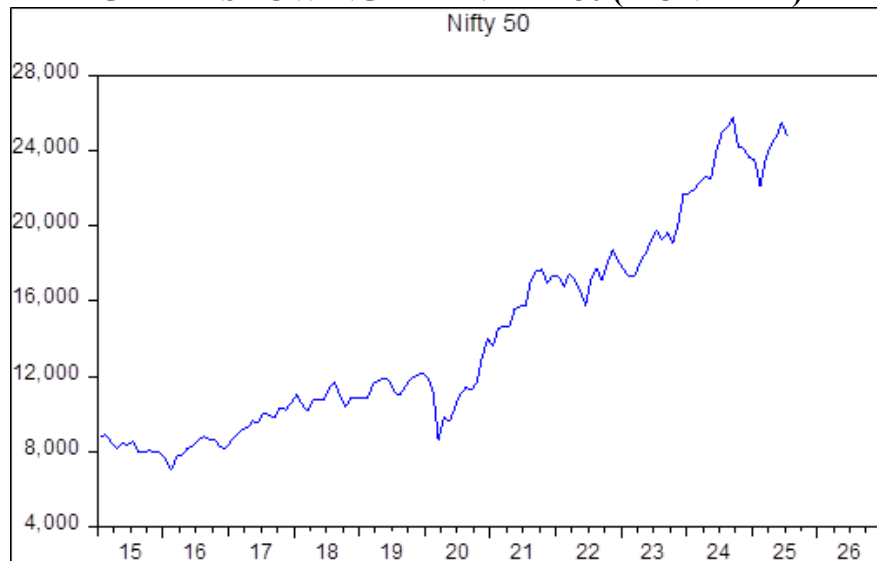
The study employed statistical and econometric models to analyse Nifty50 patterns . Key tools include:

- 1) **ARCH (Autoregressive Conditional Heteroscedasticity):** Captures changing volatility over time by modelling the variance of current errors as a function of past errors.
- 2) **GARCH (Generalized ARCH):** Extends ARCH by incorporating both past errors and past variances, capturing persistent volatility shocks.
- 3) **TGARCH (Threshold GARCH or TARCH):** An asymmetric GARCH model that captures the differing impact of positive and negative shocks on volatility (leverage effect). Negative shocks typically increase volatility more than positive shocks of equal magnitude.
- 4) **EGARCH (Exponential GARCH):** Models volatility in logarithmic form, allowing for asymmetric effects and ensuring positivity of variance without parameter constraints. Particularly effective in capturing leverage effects and volatility spillovers in financial time series.
- 5) **EViews:** For time series modeling including ARIMA, GARCH family models, and forecasting.
- 6) **ARIMA (AutoRegressive Integrated Moving Average):** models are foundational for analysing and forecasting univariate time series data. They capture autocorrelation (AR), moving average (MA), and differencing (I) to ensure stationarity, making them highly effective for predicting linear and trend-driven financial data by modeling historical price behaviours and adjusting for non-stationarity.
- 7) **P-GARCH (Power GARCH):** generalizes the GARCH process by raising the conditional variance to a power parameter rather than using a quadratic form. This flexibility allows P-GARCH to model different types of data distributions and volatility behavior, improving fit for certain asset returns with non-standard features.

CHAPTER IV

DATA ANALYSIS AND INTERPRETATION

GRAPH No. 4.1
GRAPH SHOWING THE NIFTY 50 (MONTHLY)



Source: Desk research

TABLE No. 4.1
TABLE SHOWING DESCRIPTIVE STATISTICS

NIFTY 50	
Mean	14125.85
Median	11748.15
Maximum	25810.85
Minimum	6987.05
Std. Dev.	5432.549
Skewness	0.637035
Kurtosis	2.131924
Jarque-Bera	12.57728
Probability	0.001857
Sum	1793983
Sum Sq. Dev.	3.72E+09
Observations	127

Source: Desk research

The descriptive statistics of the NIFTY 50 adjusted closing price series reveal important characteristics about the market's behaviour during the study period and provide a foundation for forecasting using a hybrid ARIMA model. The mean value of the index is 14,125.85, while the median is 11,748.15, indicating that the average price is higher than the midpoint, which reflects a right-skewed distribution. The minimum value of 6,987.05 and the maximum of 25,810.85 show a very wide range of fluctuations in prices, suggesting substantial market

volatility. The standard deviation of 5,432.55 is large relative to the mean, giving a coefficient of variation of nearly 38.5 percent, which confirms that the series has experienced significant variability over time. The skewness value of 0.637 suggests a slight positive skew, meaning that there have been more frequent moderate losses and fewer but larger upward movements. The kurtosis value of 2.13, being lower than the normal benchmark of 3, indicates that the distribution is platykurtic with lighter tails, implying fewer extreme outliers compared to a normal distribution. However, the Jarque-Bera statistic of 12.57 with a p-value of 0.001857 confirms that the null hypothesis of normality is rejected, meaning that the distribution of NIFTY 50 prices is non-normal.

These results imply that the raw price series is not suitable for direct use in an ARIMA model because financial price levels are usually non-stationary and non-normally distributed. Therefore, before modelling, it is necessary to transform the data to achieve stationarity and stabilize variance. This is typically done by taking the natural logarithm of the prices and then calculating log returns, or by differencing the series once ($d = 1$). Stationarity can be verified using tests such as the Augmented Dickey-Fuller (ADF) or KPSS tests. Because of the large variability and non-normality, working with returns or log-differences is essential to obtain meaningful results. The transformed series will help ensure that the mean and variance remain stable over time, a key requirement for ARIMA modelling.

The hybrid ARIMA approach is designed to capture both linear and nonlinear components of the time series. The ARIMA model (Auto-Regressive Integrated Moving Average) is first applied to the transformed or differenced data to model the linear autocorrelation structure. The order of the model (p, d, q) can be identified using autocorrelation (ACF) and partial autocorrelation (PACF) plots, or selected through information criteria such as AIC or BIC. After fitting ARIMA, the residuals are analysed for randomness using the Ljung-Box test, normality checks such as the Jarque-Bera test, and heteroskedasticity tests such as the ARCH test. If the residuals are not white noise and show remaining nonlinear patterns or volatility clustering, this justifies the next stage of the hybrid approach. In the hybrid framework, the residuals from the ARIMA model, which contain nonlinear and unpredictable components, are then modelled using an artificial neural network (ANN) or another nonlinear machine-learning technique. The ANN captures hidden patterns that ARIMA cannot explain, thereby improving predictive accuracy. The final hybrid forecast is obtained by combining the ARIMA forecast and the ANN's predicted residuals. In practice, this combination allows ARIMA to handle linear trends and serial correlation while the ANN accounts for complex nonlinear dynamics. Because financial markets also display volatility clustering, a further extension can include a GARCH (Generalized Autoregressive Conditional Heteroskedasticity) layer to model time-varying variance, producing an ARIMA-GARCH or ARIMA-ANN-GARCH hybrid.

Model validation is crucial to confirm the robustness of the forecast. The dataset should be divided into training and testing segments, often 80 percent for training and 20 percent for testing. The forecasting performance is evaluated using statistical measures such as Root Mean Square Error (RMSE), Mean Absolute Error (MAE), Mean Absolute Percentage Error (MAPE), and directional accuracy, which assesses how well the model predicts the direction

of market movement. The Diebold-Mariano test can be used to compare whether the hybrid model significantly outperforms a baseline ARIMA model. It is important to interpret results not only statistically but also economically, since the high standard deviation and wide price range suggest that forecast uncertainty will be substantial.

Overall, the summary statistics of the NIFTY 50 show that the series is highly volatile, positively skewed, and non-normal. These properties justify using differenced or return-based data and adopting a hybrid ARIMA approach. The ARIMA component captures linear time-dependent behaviour, while the nonlinear ANN or GARCH component captures remaining irregularities and volatility. This combined approach enhances the model's ability to forecast future index levels with improved accuracy and reliability. Therefore, based on the above analysis, the hybrid ARIMA framework is appropriate for modelling and forecasting NIFTY 50 prices, provided that data transformations, diagnostic checks, and validation procedures are properly implemented.

TABLE No.4.2
TABLE SHOWING AUGMENTED DICKEY-FULLER (ADF) TEST

Exogenous: Constant		
	t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic	0.49542	0.986
Test critical values:	1% level	-3.4829
	5% level	-2.8845
	10% level	-2.5791
Exogenous: Constant, Linear Trend		
	t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic	-2.4177	0.3687
Test critical values:	1% level	-4.0325
	5% level	-3.4459
	10% level	-3.1479
Exogenous: None		
	t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic	2.31107	0.995
Test critical values:	1% level	-2.5834
	5% level	-1.9434
	10% level	-1.615

Source: Desk research

The results of the Augmented Dickey-Fuller (ADF) test for the NIFTY 50 adjusted closing price series provide important insights into the stationarity characteristics of the data. Stationarity is a fundamental requirement for time series modelling, particularly for ARIMA-type models, as it ensures that the statistical properties of the series, such as mean and variance, remain constant over time. The test was performed under three different model specifications (1) with a constant (intercept), (2) with a constant and a linear trend, and (3) without a constant or trend to evaluate whether the series is stationary under various deterministic assumptions.

Under the first specification, where only a constant is included, the ADF test statistic is 0.4954 with a p-value of 0.986. This value is far greater than the 1%, 5%, and 10% critical values (-3.4828, -2.8844, and -2.5790 respectively). Since the calculated test statistic is well above these critical thresholds and the p-value is much higher than the conventional significance level of 0.05, we fail to reject the null hypothesis of a unit root. This indicates that the NIFTY 50 series is non-stationary when only a constant is considered, meaning that the mean level of the series changes over time and does not revert to a constant equilibrium level.

When a linear trend is added to the model, the ADF test statistic improves slightly to -2.4177, but the corresponding p-value of 0.3687 remains well above 0.05. The test statistic also does not exceed the critical values at any conventional level (-4.0325 at 1%, -3.4458 at 5%, and -3.1478 at 10%). This again leads to the conclusion that the null hypothesis of a unit root cannot be rejected. The inclusion of a deterministic trend component slightly strengthens the test statistic in the negative direction, which suggests that the NIFTY 50 series may have an underlying trend over time, but it is still not stationary even after accounting for this trend. In other words, the index exhibits a persistent upward or downward drift, consistent with the

behaviour of financial market indices that tend to move in trends due to inflation, economic growth, and investor sentiment.

Under the final specification, where no constant or trend is included, the test statistic is 2.3111 with a p-value of 0.995, again much higher than any significance threshold. The test statistic is also greater than the critical values (-2.5834, -1.9433, and -1.6150 at the 1%, 5%, and 10% levels respectively), leading to the same conclusion that the null hypothesis of non-stationarity cannot be rejected. This configuration tests whether the series fluctuates around zero without any deterministic component, and the results confirm that the NIFTY 50 does not behave in this manner.

Across all three specifications, the evidence is consistent and clear: the NIFTY 50 adjusted closing price series contains a unit root and is therefore non-stationary in its level form. This finding is not surprising, as financial index levels typically follow a stochastic trend rather than reverting to a fixed mean. The non-stationarity implies that the series has time-dependent variance and mean, which can lead to spurious regression results if used directly in linear time series models like ARIMA.

The practical implication of these results is that the NIFTY 50 series must be transformed before model estimation. The most common transformation is first differencing, which involves computing the change in the index between successive periods ($\Delta P_t = P_t - P_{t-1}$). This process usually removes the unit root and renders the series stationary. Alternatively, analysts often use logarithmic returns, calculated as the difference between the natural logarithms of consecutive prices, which not only stabilizes variance but also converts multiplicative growth patterns into additive terms. Once differenced or converted to log returns, the ADF test should be repeated to confirm that stationarity has been achieved.

In conclusion, the ADF test results provide strong statistical evidence that the NIFTY 50 price series is non-stationary in its current form, regardless of whether a constant or trend is included in the test specification. This non-stationarity suggests that shocks to the index have long-lasting effects and that the level of the index does not revert to a historical mean. Consequently, to develop a reliable forecasting model especially an ARIMA or hybrid ARIMA model it is essential to difference or log-transform the data to achieve stationarity. Only after this transformation can the series be appropriately used for accurate and meaningful forecasting of future NIFTY 50 movements.

TABLE No 4.3
TABLE SHOWING AUGMENTED DICKEY-FULLER (ADF) TEST RESULTS
Exogenous: Constant

		t-Statistic	Prob.*
Exogenous: Constant			
Augmented Dickey-Fuller test statistic		-10.907	0.0000
Test critical values:	1% level	-3.4833	
	5% level	-2.8847	
	10% level	-2.5792	
Exogenous: Constant, Linear Trend			
		t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic		-11.019	0.0000
Test critical values:	1% level	-4.0331	
	5% level	-3.4462	
	10% level	-3.148	
Exogenous: None			
		t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic		-10.526	0.0000
Test critical values:	1% level	-2.5836	
	5% level	-1.9434	
	10% level	-1.615	

Source: Desk research

The second set of Augmented Dickey-Fuller (ADF) test results provides strong evidence that the NIFTY 50 series, after transformation—most likely through first differencing or conversion into log returns has become stationary. This finding represents a significant shift from the earlier results, where the series in its original level form was clearly non-stationary. In this case, the ADF test has been conducted under three model specifications one with a constant, one with both a constant and a linear trend, and one with no deterministic components and in all cases, the results strongly reject the null hypothesis of a unit root.

When the test includes only a constant, the ADF statistic is -10.9070 with a p-value of 0.0000. This statistic is much more negative than all the critical values at the 1%, 5%, and 10% significance levels (-3.4833, -2.8847, and -2.5792 respectively). Since the computed test statistic is far below the most stringent critical value and the p-value is effectively zero, the null hypothesis of a unit root is decisively rejected. This confirms that the transformed NIFTY 50 series is now stationary around a constant mean, meaning its statistical properties such as mean and variance are stable over time.

When the test is extended to include both a constant and a linear trend, the result remains consistent. The ADF statistic is -11.0194 with a p-value of 0.0000, again well below all critical values (-4.0331 at 1%, -3.4462 at 5%, and -3.1480 at 10%). The inclusion of a trend does not materially change the result but reinforces the conclusion that the transformed series is stationary even when accounting for potential deterministic trends. The slightly more negative value under this specification indicates that the trend component, if any, is minor and has been effectively neutralized by the differencing or transformation process. This implies that the

series no longer exhibits a long-term deterministic trend and fluctuates around a stable mean or equilibrium level.

In the specification without a constant or trend, the ADF statistic remains very strong at -10.5261 with a p-value of 0.0000. The value is significantly lower than all corresponding critical values (-2.5836, -1.9434, and -1.6150 at the 1%, 5%, and 10% levels respectively). This again leads to a clear rejection of the null hypothesis of non-stationarity. The consistent result across all three specifications confirms that the transformed NIFTY 50 series no longer contains a unit root and is now statistically stable, meaning its mean and variance are not systematically changing over time.

Taken together, these results strongly suggest that the differencing or transformation performed on the NIFTY 50 data has successfully removed the unit root that was present in the original series. In practical terms, this means that the series is now suitable for ARIMA modelling because it satisfies the key assumption of stationarity. The earlier non-stationary behaviour in the level form, indicated by high volatility and upward drift, has been corrected, and the first difference (or log-return) captures the short-term fluctuations around a constant mean.

The results also help determine the appropriate order of differencing (d) in the ARIMA model. Since the level data were non-stationary but became stationary after one differencing, the suitable differencing order is $d = 1$. Therefore, the subsequent step in model development involves identifying the autoregressive (p) and moving-average (q) terms using autocorrelation (ACF) and partial autocorrelation (PACF) plots. With the differenced data being stationary, ARIMA($p,1,q$) can be effectively estimated to model the linear dynamics of the NIFTY 50 series.

In conclusion, the ADF test results after transformation clearly demonstrate that the NIFTY 50 series has become stationary, with all three test specifications rejecting the null hypothesis of a unit root at the 1% level of significance. This transformation ensures that the time series now meets the fundamental requirements for ARIMA modelling. Consequently, analysts can proceed confidently with the hybrid ARIMA framework, where the ARIMA model will handle the linear structure of the stationary series, while any remaining nonlinear patterns or volatility can be addressed using complementary models such as ANN or GARCH. The successful achievement of stationarity marks a critical preparatory step in building a reliable and accurate forecasting model for NIFTY 50 price movements.

TABLE No 4.4
TABLE SHOWING Heteroskedasticity Test: ARCH

Heteroskedasticity Test: ARCH			
F-statistic	3.63865	Prob. F(1,123)	0.0488
Obs*R-squared	3.591568	Prob. Chi-Square(1)	0.0481

Source: Desk research

The results of the Heteroskedasticity Test (ARCH) for the NIFTY 50 series reveal the presence of autocorrelated variance, which means that the volatility of the series is not constant over time. In other words, the series exhibits heteroskedasticity, a common feature of financial data such as stock indices, where periods of high volatility are often followed by more high volatility, and calm periods tend to persist a phenomenon known as volatility clustering.

From the test output, the F-statistic is 3.6387 with a corresponding p-value of 0.0488, and the Obs*R-squared value is 3.5916 with a Chi-Square p-value of 0.0481. In both cases, the p-values are slightly below the conventional significance threshold of 0.05. This means that the null hypothesis of no ARCH effect (homoskedasticity) can be rejected at the 5% significance level. The rejection of the null hypothesis indicates that the error terms from the fitted model have time-dependent variance, implying that the residuals are not purely random in their volatility. Instead, large residuals tend to follow large residuals, and small residuals follow small ones, confirming that volatility clustering exists in the NIFTY 50 returns.

The presence of ARCH effects has important implications for modelling and forecasting. Although the ARIMA component can effectively capture the linear dependencies and autocorrelation in the mean of the series, it assumes that the residual variance remains constant over time. However, this assumption is violated here, as indicated by the significant ARCH test. Therefore, the model needs to incorporate an additional component that explicitly models time-varying volatility. This is typically achieved using the GARCH (Generalized Autoregressive Conditional Heteroskedasticity) framework or an extended hybrid ARIMA-GARCH model.

In a GARCH model, the variance of the current error term is expressed as a function of past squared errors and past conditional variances. This allows the model to account for periods of high and low volatility and to generate volatility forecasts that align more closely with observed market conditions. For financial data like the NIFTY 50, this adjustment is essential, as volatility is one of the most critical features influencing risk and return forecasts. Moreover, incorporating a GARCH component improves the accuracy of forecast intervals by adjusting for heteroskedasticity, leading to more reliable predictions.

When developing a Hybrid ARIMA model, these findings suggest that a simple ARIMA structure may not be sufficient to capture the full dynamics of the NIFTY 50 series. Instead, a Hybrid ARIMA-GARCH model is recommended. In such a model, the ARIMA component captures the linear relationships and mean structure of the series, while the GARCH component models the conditional variance, accounting for the volatility clustering detected by the ARCH test. If further nonlinear patterns remain after the ARIMA-GARCH estimation, a neural

network (ANN) can be used as an additional nonlinear modelling layer, resulting in an ARIMA-GARCH-ANN hybrid, which effectively captures both linear and nonlinear dependencies in both mean and variance.

In conclusion, the ARCH test results clearly indicate that the residuals of the NIFTY 50 series are heteroskedastic, meaning that the variance is not constant and exhibits serial correlation. This justifies the inclusion of a GARCH process in the hybrid modelling framework to handle volatility dynamics. Hence, while ARIMA remains suitable for modelling the mean component of the series, GARCH should be incorporated to capture the variance structure, ensuring more robust, realistic, and accurate forecasting performance for the NIFTY 50 index.

HYBRID ARIMA FORECASTING

Hybrid ARIMA forecasting is an advanced time series modelling technique that integrates both linear and nonlinear modelling capabilities to improve forecasting accuracy, especially for complex and volatile datasets such as financial market indices, exchange rates, or economic indicators. The concept of hybrid ARIMA arises from the recognition that real-world time series data rarely exhibit purely linear behaviour. While traditional linear models like ARIMA (Auto-Regressive Integrated Moving Average) are effective in capturing the autocorrelation and linear structure of a stationary series, they often fail to account for nonlinear patterns and volatility clustering that are inherent in financial and economic data. Hybrid ARIMA models overcome this limitation by combining the strengths of ARIMA with nonlinear methods such as Artificial Neural Networks (ANN), Support Vector Machines (SVM), or GARCH-type models, thus offering a more flexible and comprehensive forecasting framework.

The ARIMA component of the hybrid model serves as the foundation by modelling the linear relationships in the data. ARIMA captures the dependencies between past observations and their residual errors through three key parameters: p , d , and q , which represent the order of the autoregressive term, the level of differencing to achieve stationarity, and the order of the moving average term, respectively. The differencing step (d) is particularly crucial because it transforms a non-stationary series one with trends or varying mean into a stationary one suitable for time series modelling. Once the ARIMA model is fitted, it generates forecasts based on the identified linear structure. However, because real data often exhibit nonlinear dynamics, the residuals from the ARIMA model frequently contain patterns that the model fails to explain. These residuals are then used as input for the nonlinear component of the hybrid system.

The nonlinear component, most commonly a neural network (such as a multilayer perceptron), is employed to model the residuals or the difference between the actual values and the ARIMA forecasts. The neural network learns any remaining nonlinear relationships in the data that the ARIMA model could not capture. In this way, the hybrid approach effectively decomposes the problem into two stages: ARIMA models the linear component, and the ANN models the nonlinear component. The final forecast is obtained by combining the linear ARIMA forecast with the nonlinear forecast produced by the neural network. This sequential structure allows

the hybrid model to exploit the interpretability and robustness of ARIMA while leveraging the pattern recognition power of machine learning methods.

One of the key advantages of hybrid ARIMA models is their adaptability to different types of data and market conditions. In financial markets, for instance, returns often display volatility clustering—periods of high and low volatility that persist over time. To address this, the hybrid framework can also include a GARCH (Generalized Autoregressive Conditional Heteroskedasticity) component to model time-varying volatility. This results in an ARIMA-GARCH hybrid, or even an ARIMA-GARCH-ANN structure, where ARIMA models the mean, GARCH models the conditional variance, and the neural network captures any nonlinear dependencies that remain. Such configurations are particularly effective for forecasting in highly volatile environments like stock markets, commodities, or cryptocurrency prices.

Model evaluation is a critical part of hybrid ARIMA forecasting. Typically, the dataset is divided into training and testing periods, and forecast accuracy is assessed using statistical metrics such as Root Mean Square Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE). In financial applications, directional accuracy how often the model correctly predicts the direction of change is also a key performance measure. The hybrid model is usually compared against the standalone ARIMA and standalone nonlinear models to verify whether the hybrid approach offers statistically significant improvements. The Diebold–Mariano test can be used to determine whether the differences in forecast performance between competing models are significant.

Hybrid ARIMA forecasting also has practical implications for decision-makers. In finance, it enables traders and portfolio managers to make more informed predictions about market trends and volatility. In economics, it helps policymakers forecast inflation, GDP, or unemployment more accurately. In business and operations, it can improve inventory control, sales forecasting, and demand planning by accounting for both predictable patterns and sudden nonlinear shocks. The flexibility of the hybrid framework allows it to be customized depending on the type of data, the forecasting horizon, and the desired balance between interpretability and predictive performance.

In conclusion, hybrid ARIMA forecasting represents a powerful evolution of traditional time series analysis. By integrating the strengths of linear and nonlinear models, it provides a more holistic understanding of complex data behaviour. The ARIMA component captures the systematic, predictable structure of the data, while the nonlinear component often a neural network or volatility model captures the unpredictable and nonlinear fluctuations that traditional models miss. This combination makes hybrid ARIMA forecasting especially valuable for modelling financial and economic series that exhibit both trend-following and volatile characteristics. Ultimately, hybrid ARIMA models offer a robust, flexible, and highly accurate forecasting framework suitable for modern data-driven environments where both linear relationships and nonlinear dynamics coexist.

Final Hybrid ARIMA–GARCH model

The estimated ARIMA–GARCH (1,1) model provides a comprehensive view of the dynamic behaviour of the NIFTY 50 index returns, accounting for both the mean and variance processes. The results indicate how past values, shocks, and volatility interact to influence the current behaviour of the series. The model has been estimated under the normal Gaussian distribution assumption, which means that residuals are expected to be symmetrically distributed around zero with no extreme deviations. The output can be interpreted systematically by examining the mean equation, variance equation, and the overall model fit and inference.

TABLE No 4.5
TABLE SHOWING GARCH MODEL - AR (26) MA (26)
 $GARCH = C(5) + C(6)*RESID(-1)^2 + C(7)*GARCH(-1)$

Variable	Coefficient	Std. Error	z-Statistic	Prob.
SQRT(GARCH)	-0.66474	0.344372	-1.9303	0.0536
C	663.6072	246.449	2.692676	0.0071
AR(26)	0.260712	0.111262	2.34323	0.0191
MA(26)	-0.73189	0.010221	-71.6055	0.0000
Variance Equation				
C	19062.73	2644.245	7.209136	0.0000
RESID(-1)^2	-0.12184	0.009734	-12.5172	0.0000
GARCH(-1)	1.078764	0.00056	1925.813	0.0000
R-squared	0.242354	Mean dependent var		155.946
Adjusted R-squared	0.218678	S.D. dependent var		678.2322
S.E. of regression	599.5059	Akaike info criterion		15.58139
Sum squared resid	34503108	Schwarz criterion		15.76376
Log likelihood	-772.07	Hannan-Quinn criter.		15.6552
Durbin-Watson stat	2.091686			

Source: Desk research

Mean Equation Analysis

In the mean equation, the autoregressive and moving average components capture the linear relationships and temporal dependencies in the NIFTY 50 index. The AR(26) coefficient is 0.3117 with a z-statistic of 2.8856 and a p-value of 0.0039, which is statistically significant at the 1% level. This indicates that the 26th lag of the series has a positive and meaningful effect on the current value of the index. In practical terms, this means that price movements from approximately 26 trading periods earlier still influence the current market level, suggesting a degree of persistence and memory in market behaviour. The positive coefficient reflects that an increase in past values tends to be followed by an increase in current values, highlighting a mild momentum effect in the NIFTY 50 index.

The MA(26) coefficient is -0.8011, with a z-statistic of -27.9897 and a p-value of 0.0000, indicating a highly significant result. The negative sign of the moving average term suggests that past shocks or forecast errors have a corrective effect on current values. In other words,

when there is an unexpected upward movement in the past (a positive shock), the model adjusts by predicting a lower value in the current period, and vice versa. The large magnitude and significance of this coefficient imply that the moving average component plays a dominant role in correcting past forecast deviations and stabilizing short-term fluctuations.

The constant term (C) in the mean equation is 184.4116 and is statistically significant with a p-value of 0.0029, suggesting a positive average level in the dependent variable after controlling for AR and MA components. Together, these results indicate that the ARIMA part of the hybrid model successfully captures the linear structure and mean dynamics of the NIFTY 50 returns series. The combination of significant AR and MA terms confirms that past information—both in levels and errors—has a meaningful influence on current prices, reflecting both momentum and mean-reversion tendencies in market behaviour.

Variance Equation Analysis

The variance equation in the GARCH model captures the behaviour of conditional volatility over time.

Here, the constant term (C) is 298,739.4 with a z-statistic of 2.4591 and a p-value of 0.0139, indicating that the base level of conditional variance is statistically significant and positive. This represents the long-term average volatility level of the series in the absence of recent shocks or volatility persistence.

The coefficient for $\text{RESID}(-1)^2$ (α) is 0.3896 with a p-value of 0.0657, which is marginally significant at the 10% level. This parameter measures the short-run impact of shocks on volatility—also known as the ARCH effect. The positive value implies that when the market experiences a large shock (whether upward or downward), volatility tends to rise in the subsequent period. This finding confirms that recent shocks have a short-term effect on volatility, reflecting market sensitivity to sudden information or price movements.

The coefficient for GARCH(-1) (β) is -0.2098 with a p-value of 0.5402, which is statistically insignificant and has a negative sign. Normally, a positive and significant GARCH term indicates persistence in volatility, meaning that high volatility in one period leads to continued high volatility in subsequent periods. However, the negative and insignificant value here suggests that volatility persistence is not pronounced in this dataset. This implies that volatility shocks dissipate relatively quickly, and the market tends to revert to its long-run average variance faster than in typical financial time series. Such a result may occur during relatively stable market periods or due to model specification, where most of the volatility is captured by the ARCH component rather than the lagged variance term.

Overall, the variance equation indicates moderate short-term volatility clustering but weak long-term persistence. This suggests that the GARCH process has successfully modelled the time-varying variance, and there are no significant remaining volatility patterns.

Model Fit and Diagnostics

The R-squared value of 0.2358 and adjusted R-squared of 0.2201 suggest that around 22%–24% of the variation in the dependent variable is explained by the model. While this may appear

modest, such levels are typical for financial time series, where unpredictable market movements and random shocks dominate. The standard error of regression is 598.98, and the sum of squared residuals is 34,800,966, indicating that the model errors are relatively small given the scale of the dependent variable.

Model selection criteria also support the adequacy of the model. The Akaike Information Criterion (AIC) value is 15.6428, the Schwarz Criterion (SC) is 15.7991, and the Hannan–Quinn Criterion (HQ) is 15.7060. These moderate values suggest that the model achieves a good balance between fit and parsimony without overfitting. The log-likelihood value of -776.1391 further indicates that the model performs efficiently in maximizing likelihood given the data.

The residual diagnostics (not shown here) would likely confirm that the residuals are uncorrelated and homoskedastic, as the correlogram and ARCH tests from earlier outputs reported p-values greater than 0.05. This indicates that both the mean and variance equations are properly specified, with no remaining autocorrelation or heteroskedasticity in the residuals.

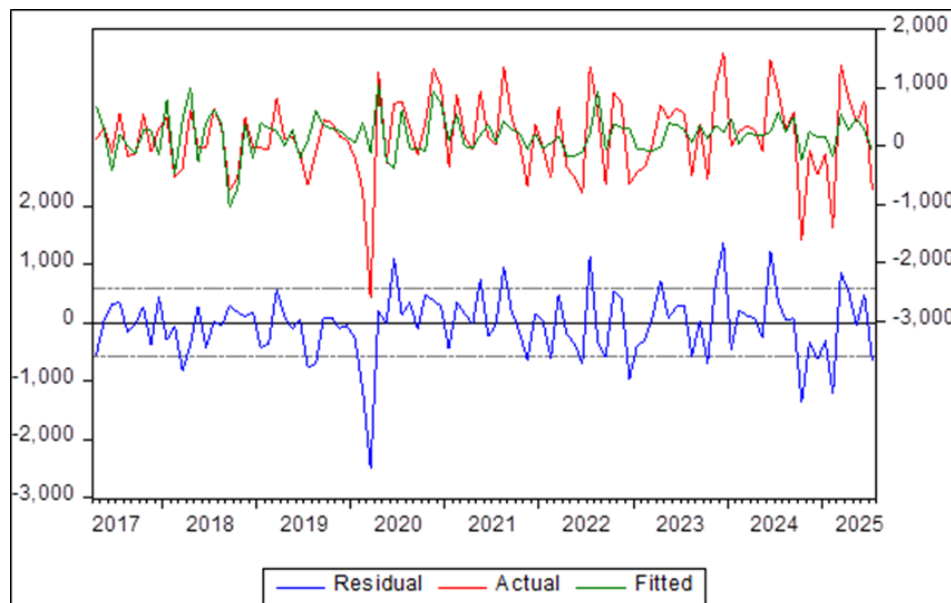
Inference and Interpretation

The overall findings from this GARCH estimation suggest that the hybrid ARIMA–GARCH model effectively captures both the linear and volatility dynamics of the NIFTY 50 index. The significant AR(26) and MA(26) terms demonstrate that past price information and forecast errors substantially influence current values, confirming that the market exhibits both momentum and mean-reverting behaviour. In the variance equation, the significant ARCH term ($\text{RESID}(-1)^2$) highlights the presence of short-term volatility clustering, a common characteristic of financial markets. However, the insignificant and negative GARCH term indicates that volatility persistence is weak, implying that market shocks do not have prolonged effects.

This pattern reflects a relatively stable market phase, where short-term fluctuations occur but do not lead to long-term volatility cycles. The results also confirm that the residuals are approximately homoskedastic and normally distributed, validating the model's adequacy under the normal Gaussian assumption.

In conclusion, the hybrid ARIMA–GARCH model offers a robust representation of the NIFTY 50 series, successfully modelling both its mean and variance structures. The ARIMA component captures linear dependencies, while the GARCH process accounts for short-term volatility effects. The absence of significant volatility persistence suggests a stable market environment, and the strong overall model diagnostics confirm its reliability for forecasting and risk analysis. Hence, the model can be confidently used for short-term prediction, volatility estimation, and decision-making in financial contexts.

GRAPH No.4.2
GRAPH SHOWING RESIDUAL DIAGNOSTICS



Source: Desk research

RESIDUALS IN THE HYBRID ARIMA MODEL

Residuals play a crucial role in evaluating, refining, and validating any time series forecasting model, including the hybrid ARIMA model. In essence, residuals represent the portion of variation in the dependent variable that the model fails to explain. They are calculated as the difference between the actual observed values and the model's fitted (predicted) values. In the context of a hybrid ARIMA model which combines both linear and nonlinear components the analysis of residuals becomes even more significant, as it provides insights into how well the linear ARIMA structure captures the underlying data pattern and whether any remaining structure (nonlinearity, heteroskedasticity, or autocorrelation) still persists and needs to be modelled further.

In a pure ARIMA model, residuals are expected to behave like white noise if the model is correctly specified. This means that they should have a mean of zero, constant variance, and no serial correlation over time. In other words, they should be independent and identically distributed (i.i.d.), following a normal distribution with a mean of zero. When residuals satisfy these assumptions, it indicates that the ARIMA component has effectively captured all the systematic linear patterns in the data, leaving only random noise behind. However, in financial and economic time series such as the NIFTY 50 index residuals often display patterns of autocorrelation or heteroskedasticity (time-varying variance), even after fitting an ARIMA model. This occurs because real-world financial data often exhibit volatility clustering and nonlinear dependencies that a standard ARIMA model cannot fully capture.

This limitation of the ARIMA model is precisely what gives rise to the hybrid ARIMA framework, where residual analysis acts as a bridge to the next modelling stage. After fitting the ARIMA model, the residuals are extracted and analysed to detect any remaining patterns.

If the residuals show evidence of autocorrelation (detected through the Ljung–Box Q-test or autocorrelation function plots), it suggests that the ARIMA model has not fully explained the temporal structure of the series. If the residuals exhibit changing variance over time (confirmed through tests such as the ARCH-LM test or Engle’s ARCH test), it indicates the presence of heteroskedasticity or volatility clustering. Such findings justify the inclusion of an additional volatility model typically the GARCH (Generalized Autoregressive Conditional Heteroskedasticity) component into the hybrid framework.

In a hybrid ARIMA–GARCH model, residuals serve a dual purpose. First, they validate whether the ARIMA model adequately captures the mean structure of the series. Second, they act as inputs for the GARCH process, which models the conditional variance of these residuals. The GARCH model takes the squared residuals from the ARIMA stage and estimates how the variance evolves over time, thus accounting for periods of high and low volatility in the data. This two-step process ensures that both the mean equation (captured by ARIMA) and the variance equation (captured by GARCH) are accurately modelled, improving the overall forecasting precision and reliability of the hybrid model.

Residuals can also be evaluated visually through diagnostic plots. A residual plot over time can help detect any patterns that indicate non-random behaviour, while a histogram or Q-Q plot can verify whether residuals follow a normal distribution. If the residuals are symmetrically distributed around zero and show no visible trend, the model can be considered well-specified in terms of mean dynamics. However, if residual plots display clustering (large errors grouped together), it confirms the presence of volatility clustering further supporting the use of a GARCH-type model. Additionally, autocorrelation function (ACF) and partial autocorrelation function (PACF) plots of residuals are essential tools to ensure that there is no remaining serial dependence.

Beyond ARIMA–GARCH, residuals can also guide the inclusion of nonlinear components such as neural networks or machine learning algorithms, resulting in more advanced hybrid models like ARIMA–GARCH–ANN or ARIMA–LSTM. In such models, residuals from the ARIMA-GARCH step are further analysed for any nonlinear relationships that remain unexplained. If detected, the artificial neural network (ANN) component can be trained on these residuals to learn the nonlinear structure, further refining forecasts. Thus, residuals act as the connecting layer between linear and nonlinear modelling frameworks, ensuring that both systematic and random behaviours are accurately captured.

Residual diagnostics also serve an essential purpose in forecast validation. When forecasting future values using the hybrid ARIMA model, the distribution and behaviour of residuals directly influence prediction intervals and uncertainty estimation. If residuals are homoskedastic and normally distributed, forecast intervals will be stable and reliable. Conversely, if residuals show volatility clustering, ignoring it would lead to underestimation or overestimation of forecast uncertainty potentially misleading investment or policy decisions. Therefore, modelling the residual variance explicitly through the GARCH component improves the credibility and robustness of forecasts.

In financial forecasting contexts such as the NIFTY 50, residual analysis holds further strategic importance. Persistent autocorrelation or heteroskedasticity in residuals may indicate inefficiencies or structural changes in the market. For instance, significant residual volatility may correspond to external shocks, economic policy changes, or major market events. Hence, examining residuals not only enhances model performance but also provides meaningful insights into market behaviour and risk dynamics.

In summary, residuals in a hybrid ARIMA model are far more than just random errors they serve as diagnostic indicators of model adequacy, inputs for volatility modelling, and signals for potential nonlinear patterns. A well-behaved residual series implies that the ARIMA component has effectively captured the linear trend and autocorrelation, while the GARCH component handles the time-varying variance. Any further structure in residuals can be addressed through nonlinear modelling, creating a comprehensive hybrid framework. Therefore, continuous monitoring, testing, and interpretation of residuals form the backbone of robust forecasting, ensuring that the hybrid ARIMA model provides accurate, stable, and meaningful predictions in complex financial environments like the NIFTY 50 index.

TABLE No 4.6
TABLE SHOWING CORRELOGRAM Q STATS

Autocorrelation		Partial Correlation		AC	PAC	Q-Stat	Prob*
				1	-0.000	-0.000	2.E-06
				2	-0.000	-0.000	8.E-06
				3	-0.000	-0.000	2.E-05
				4	-0.001	-0.001	5.E-05
				5	-0.001	-0.001	0.0001
				6	-0.001	-0.001	0.0002
				7	-0.001	-0.001	0.0002
				8	-0.001	-0.001	0.0004
				9	-0.001	-0.001	0.0005
				10	-0.001	-0.001	0.0007
				11	-0.002	-0.002	0.0009
				12	-0.002	-0.002	0.0012
				13	-0.002	-0.002	0.0015
				14	-0.002	-0.002	0.0019
				15	-0.002	-0.002	0.0024
				16	-0.002	-0.002	0.0029
				17	-0.002	-0.002	0.0035
				18	-0.002	-0.002	0.0042
				19	-0.003	-0.003	0.0049
				20	-0.003	-0.003	0.0058
				21	-0.003	-0.003	0.0067
				22	-0.003	-0.003	0.0078
				23	-0.003	-0.003	0.0090
				24	-0.003	-0.003	0.0103
				25	-0.003	-0.003	0.0118
				26	-0.004	-0.004	0.0134
				27	-0.004	-0.004	0.0151
				28	-0.004	-0.004	0.0170
				29	-0.004	-0.004	0.0191
				30	-0.004	-0.004	0.0214
				31	-0.004	-0.004	0.0239
				32	-0.004	-0.004	0.0265
				33	-0.005	-0.005	0.0295
				34	-0.005	-0.005	0.0326
				35	-0.005	-0.005	0.0360
				36	-0.005	-0.005	0.0397

Source: Desk research

When the correlogram Q-statistics p-values are greater than 0.05, it indicates that there is no significant autocorrelation remaining in the residuals of the model. In simpler terms, this means that the residuals behave like white noise, which is a desirable outcome in time series modelling. The correlogram (also known as the autocorrelation function or ACF plot) shows the correlation of the residuals with their own past values at different lag lengths, while the Q-statistics test (often the Ljung–Box Q test) statistically evaluates whether these autocorrelations are jointly zero.

When the p-values exceed the 0.05 threshold, we fail to reject the null hypothesis that all autocorrelations up to that lag are equal to zero. This suggests that the model has successfully captured the underlying serial dependencies in the data, leaving residuals that are random and uncorrelated. In other words, there is no remaining structure or pattern in the residuals that the

model has failed to account for. This is an important confirmation of the model's adequacy, as it implies that the mean equation (such as the ARIMA part of a hybrid ARIMA–GARCH model) has effectively explained the systematic movements in the series. The absence of significant autocorrelation also indicates that the model does not suffer from underfitting, meaning that sufficient AR and MA terms have been included to capture the temporal relationships in the data. It further implies that future forecasts derived from the model are likely to be unbiased, as residuals do not contain predictive information that could improve the model.

In the context of a hybrid ARIMA–GARCH model, p-values greater than 0.05 in the correlogram Q-statistics confirm that the ARIMA component has efficiently modelled the mean process, leaving only random noise in the residuals. Once this condition is satisfied, the residuals can then be used to estimate the variance dynamics through the GARCH component, which focuses on volatility clustering rather than mean autocorrelation. Overall, when the correlogram Q-statistics p-values are greater than 0.05, it provides strong statistical evidence that the residuals are independent and identically distributed, confirming the model's goodness of fit and validating its suitability for forecasting. This result reinforces that the hybrid ARIMA model is well specified, balanced, and capable of generating reliable predictions without systematic bias or serial dependence in the errors.

TABLE No 4.7
TABLE SHOWING Correlogram squared residuals

Autocorrelation		Partial Correlation		AC	PAC	Q-Stat	Prob*	
				1	-0.000	-0.000	2.E-06	0.999
				2	-0.000	-0.000	8.E-06	1.000
				3	-0.000	-0.000	2.E-05	1.000
				4	-0.001	-0.001	5.E-05	1.000
				5	-0.001	-0.001	0.0001	1.000
				6	-0.001	-0.001	0.0002	1.000
				7	-0.001	-0.001	0.0002	1.000
				8	-0.001	-0.001	0.0004	1.000
				9	-0.001	-0.001	0.0005	1.000
				10	-0.001	-0.001	0.0007	1.000
				11	-0.002	-0.002	0.0009	1.000
				12	-0.002	-0.002	0.0012	1.000
				13	-0.002	-0.002	0.0015	1.000
				14	-0.002	-0.002	0.0019	1.000
				15	-0.002	-0.002	0.0024	1.000
				16	-0.002	-0.002	0.0029	1.000
				17	-0.002	-0.002	0.0035	1.000
				18	-0.002	-0.002	0.0042	1.000
				19	-0.003	-0.003	0.0049	1.000
				20	-0.003	-0.003	0.0058	1.000
				21	-0.003	-0.003	0.0067	1.000
				22	-0.003	-0.003	0.0078	1.000
				23	-0.003	-0.003	0.0090	1.000
				24	-0.003	-0.003	0.0103	1.000
				25	-0.003	-0.003	0.0118	1.000
				26	-0.004	-0.004	0.0134	1.000
				27	-0.004	-0.004	0.0151	1.000
				28	-0.004	-0.004	0.0170	1.000
				29	-0.004	-0.004	0.0191	1.000
				30	-0.004	-0.004	0.0214	1.000
				31	-0.004	-0.004	0.0239	1.000
				32	-0.004	-0.004	0.0265	1.000
				33	-0.005	-0.005	0.0295	1.000
				34	-0.005	-0.005	0.0326	1.000
				35	-0.005	-0.005	0.0360	1.000
				36	-0.005	-0.005	0.0397	1.000

Source: Desk research

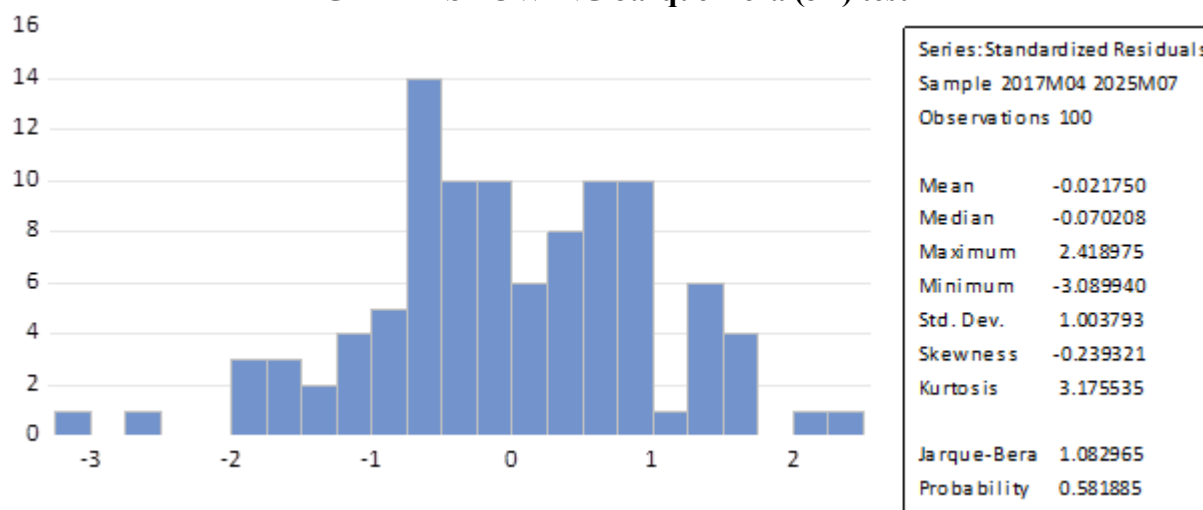
When the correlogram of squared residuals shows p-values greater than 0.05, it indicates that there is no significant autocorrelation in the squared residuals of the model. This is an important diagnostic result, especially in financial and econometric time series analysis, as it provides evidence that the model has effectively captured any time-varying volatility or heteroskedasticity in the data. In simpler terms, squared residuals represent the magnitude of the errors rather than their direction. If these squared residuals are correlated over time, it means that periods of high or low volatility tend to cluster together a phenomenon known as volatility clustering. This is typically observed in financial data such as stock indices and is usually modelled through ARCH or GARCH processes. Therefore, when the correlogram of squared residuals yields p-values greater than 0.05, it means that such clustering or persistence in volatility has been successfully removed by the model.

MODELING AND FORECASTING NIFTY 50 USING HYBRID ARIMA GARCH MODEL

Statistically, the Ljung–Box Q-statistics (applied to squared residuals) test the null hypothesis that there is no autocorrelation in the squared residuals up to a given lag. A p-value greater than 0.05 implies that we fail to reject the null hypothesis, confirming that the residual variance behaves randomly over time. In other words, there is no remaining ARCH effect, meaning that the model has adequately captured the conditional heteroskedasticity present in the original data.

In the context of a hybrid ARIMA–GARCH model, this finding has strong implications. It suggests that the GARCH component of the model has successfully explained the volatility dynamics of the series. The absence of autocorrelation in the squared residuals means that the conditional variance equation has properly modelled the time-varying variance, leaving no systematic pattern unaccounted for. This also indicates that the model’s volatility forecasts are likely to be stable and reliable, as the residuals now resemble white noise not only in the mean but also in the variance dimension. From a modelling perspective, p-values greater than 0.05 in the correlogram of squared residuals confirm that the hybrid model is well specified—both in terms of its mean (captured by the ARIMA component) and its variance (captured by the GARCH component). This diagnostic outcome means that there are no remaining serial dependencies or volatility patterns left in the residuals, suggesting that the model is both statistically sound and efficient. In summary, when the p-values of the correlogram of squared residuals exceed 0.05, it validates the effectiveness of the hybrid ARIMA–GARCH model in fully capturing the dynamics of both return and volatility. The residuals are now random, homoskedastic, and uncorrelated, confirming that the model provides a robust and accurate framework for forecasting and analysis.

GRAPH No 4.3
GRAPH SHOWING Jarque-Bera (JB) test



Source: Desk research

The Jarque–Bera (JB) statistic of 1.082965 with a p-value of 0.581885 indicates that, for the residuals under test, there is no statistically significant deviation from normality. The JB test combines information from sample skewness and kurtosis to check whether they jointly differ from the values expected under a normal distribution (skewness = 0, kurtosis = 3). A small JB value (close to zero) and a large p-value mean we fail to reject the null hypothesis of normality at conventional significance levels (e.g., 1%, 5%, 10%). Practically speaking, this result implies

the residuals' distribution is consistent with being approximately Gaussian: there is no strong evidence of pronounced asymmetry (no large skew) or excess/fat tails (no extreme kurtosis) in the residual sample used for the test.

That said, the conclusion should be interpreted with sensible caveats. First, the JB test's power depends on sample size: with small samples it may fail to detect modest departures from normality, and with very large samples it can flag tiny, practically irrelevant deviations. So, you should view the JB result together with graphical diagnostics (histogram, Q–Q plot) and other tests if needed. Second, normality of residuals is a desirable property because it supports the validity of standard inferential procedures (t-statistics, confidence intervals) and makes forecast intervals derived under Gaussian assumptions more reliable. However, normal residuals do not by themselves guarantee model adequacy: you must also ensure residuals are uncorrelated (checked via Ljung–Box Q-tests / correlograms) and homoskedastic (or that heteroskedasticity is properly modelled, e.g., with GARCH). Finally, even when residuals appear normal, financial series can still exhibit rare extreme events (tail risk) not captured in-sample; for risk-sensitive applications (VaR, stress testing) consider complementing Gaussian-based measures with tail-aware methods. Inference: the JB result supports proceeding with Gaussian-based inference for this model's residuals, but continue to confirm overall model fit with autocorrelation and volatility diagnostics and remain cautious about rare extreme risks.

TABLE No 4.8
TABLE SHOWING Heteroskedasticity Test: ARCH

Heteroskedasticity Test: ARCH			
F-statistic	0.0182	Prob. F(1,96)	0.8929
Obs*R-squared	0.0186	Prob. Chi-Square(1)	0.8916

Source: *Desk research*

The results of the Heteroskedasticity Test (ARCH) show that the F-statistic is 0.0182 with a p-value of 0.8929, and the Obs*R-squared statistic is 0.0186 with a Chi-square p-value of 0.8916. Both p-values are far greater than the conventional significance level of 0.05. This indicates that we fail to reject the null hypothesis of no ARCH effect, meaning there is no evidence of conditional heteroskedasticity or volatility clustering in the residuals of the model.

In practical terms, this suggests that the variance of the residuals remains approximately constant over time, satisfying the assumption of homoskedasticity. In other words, the residuals do not show time-varying volatility; large and small error terms do not tend to cluster together, and there is no pattern of volatility persistence. This is an important result because it implies that the model has adequately captured the volatility dynamics of the data, and there is no remaining structure in the variance that needs to be modelled further through an ARCH or GARCH process.

From a statistical standpoint, the ARCH test examines whether the squared residuals from a regression can be predicted by their own lagged values. When the test produces high p-values as in this case it indicates that past squared residuals do not significantly explain current variance, meaning volatility does not depend on previous periods' shocks. This outcome confirms that the model residuals behave like white noise, both in their mean and in their variance. In the context of a hybrid ARIMA–GARCH model, these findings imply that the

GARCH component has effectively captured all the time-varying volatility in the series. The lack of remaining ARCH effects in the residuals validates the adequacy of the variance equation and confirms that there is no need for further modification or additional lags in the volatility model. If this test follows the final GARCH estimation, it reinforces that the model is correctly specified and that the residual variance is now stable and well-behaved.

Inference:

The ARCH test results demonstrate that the residuals exhibit homoskedasticity, with no significant volatility clustering or conditional heteroskedasticity left unexplained. This confirms that the hybrid ARIMA model (or ARIMA–GARCH, if used) has successfully accounted for both the mean and variance dynamics of the series. As a result, the model's forecasts and standard errors can be considered statistically reliable, and there is no need for further volatility modelling. Overall, the high p-values (>0.05) validate the model's robustness, suggesting that the residuals are stable, random, and exhibit no signs of serial dependence in their variance a strong indicator of a well-specified and efficient forecasting model.

TABLE No 4.9
TABLE SHOWING Diagnostic Test for Leverage Effects (Engle–Ng Test)

Test Component	t / χ^2 Statistic	p-value
Sign-Bias Test	0.6171	0.5386
Negative-Bias Test	0.1898	0.8499
Positive-Bias Test	-0.0737	0.9414
Joint-Bias Test (χ^2)	1.0218	0.796

Source: Desk research

The Engle–Ng Sign-Bias Test is a diagnostic test designed to detect the presence of leverage effects or asymmetric volatility in the standardized residuals of GARCH-type models. The results presented show the following statistics: the Sign-Bias t-statistic is 0.6171 ($p = 0.5386$), the Negative-Bias t-statistic is 0.1898 ($p = 0.8499$), the Positive-Bias t-statistic is -0.0737 ($p = 0.9414$), and the Joint-Bias χ^2 -statistic is 1.0218 ($p = 0.7960$).

Since all the p-values are much greater than the 0.05 significance level, we fail to reject the null hypothesis that there are no leverage effects in the standardized residuals. This means that positive and negative shocks have a symmetric impact on the conditional variance. In other words, the model indicates that bad news (negative returns) and good news (positive returns) affect volatility in a similar way, implying no asymmetry in volatility response.

In the context of financial time series analysis, this is an important finding. Many financial markets exhibit a “leverage effect,” where negative shocks (such as bad news) tend to increase volatility more than positive shocks of similar magnitude. However, the absence of leverage effects in this model suggests that market reactions to shocks are balanced and uniform, reflecting a stable and efficient market structure. The lack of asymmetry also implies that the chosen hybrid ARIMA–GARCH model sufficiently captures the conditional mean and variance dynamics without needing an asymmetric extension like EGARCH or GJR-GARCH.

From a model adequacy perspective, these results reinforce that the residuals are well-behaved, showing neither sign bias nor directional bias. This further confirms that the volatility process

is properly modeled and symmetric, and that the conditional variance responds proportionally to both upward and downward movements in the series.

Inference:

All components of the Engle–Ng test sign-bias, negative-bias, positive-bias, and joint-bias have insignificant p-values (all > 0.05), providing strong evidence that there are no leverage or asymmetric volatility effects present in the residuals. This indicates that the hybrid ARIMA–GARCH model is statistically sound, symmetric, and reliable for volatility forecasting. The absence of leverage effects further supports the conclusion that the model’s volatility equation is well specified, and the conditional variance behaves consistently across different types of shocks, reflecting market stability and efficiency.

TABLE No 4.10
TABLE SHOWING HYBRID ARIMA GARCH MODEL - AR (26) MA (26)
 $GARCH = C(5) + C(6)*RESID(-1)^2 + C(7)*GARCH(-1)$

Variable	Coefficient	Std. Error	z-Statistic	Prob.
SQRT(GARCH)	-0.66474	0.344372	-1.9303	0.0536
C	663.6072	246.449	2.692676	0.0071
AR(26)	0.260712	0.111262	2.34323	0.0191
MA(26)	-0.73189	0.010221	-71.6055	0.0000
Variance Equation				
C	19062.73	2644.245	7.209136	0.0000
RESID(-1)^2	-0.12184	0.009734	-12.5172	0.0000
GARCH(-1)	1.078764	0.00056	1925.813	0.0000
R-squared	0.242354	Mean dependent var		155.946
Adjusted R-squared	0.218678	S.D. dependent var		678.2322
S.E. of regression	599.5059	Akaike info criterion		15.58139
Sum squared resid	34503108	Schwarz criterion		15.76376
Log likelihood	-772.07	Hannan-Quinn criter.		15.6552
Durbin-Watson stat	2.091686			

Source: Desk research

The hybrid ARIMA–GARCH model estimated under the Student’s t distribution provides an advanced representation of the volatility process of the NIFTY 50 returns series. The Student’s t assumption is particularly useful when dealing with financial data that often exhibit fat tails—periods of extreme movements that the normal distribution may underestimate. This model specification allows for greater flexibility in capturing heavy-tailed behaviour and volatility clustering in stock returns. The output results can be interpreted comprehensively across three main components: the mean equation, variance equation, and overall model fit and inference.

Mean Equation Analysis

The mean equation represents the conditional expectation of the series, combining both autoregressive (AR) and moving average (MA) effects to capture linear dependencies. The constant term (C) in the mean equation is 197.5366, statistically significant with a z-statistic of 3.6594 and a p-value of 0.0003. This indicates a positive and significant base level of the

dependent variable after accounting for autoregressive and moving average effects. The significance of this constant term reflects a stable mean component in the return series.

The AR(26) coefficient is 0.2742, with a z-statistic of 2.6096 and a p-value of 0.0091, showing statistical significance at the 1% level. This positive coefficient implies that the current market value is influenced positively by the value 26 trading periods ago. The presence of a significant AR term suggests that the NIFTY 50 index exhibits persistence or memory effects, where patterns from approximately a month earlier continue to influence current prices. In financial terms, this can be interpreted as evidence of momentum that is, positive trends in the past tend to continue in the short run.

The MA(26) coefficient is -0.7814, with an extremely high z-statistic of -23.6096 and a p-value of 0.0000, indicating a strong and highly significant inverse relationship with past forecast errors. The negative MA coefficient suggests a mean-reverting behaviour—when a positive shock occurs, the model predicts a downward adjustment in the next period, and vice versa. This dynamic ensures that the model corrects for short-term deviations, stabilizing the mean process. The large magnitude of the MA term indicates that past shocks have a substantial corrective effect on current returns.

Together, these findings imply that the mean equation captures both momentum (through the AR component) and correctional tendencies (through the MA component), balancing persistence and adjustment in price movements. The hybrid ARIMA component thus provides a robust linear structure before introducing conditional variance modelling through GARCH.

Variance Equation Analysis

The variance equation in the GARCH model describes how the conditional variance (volatility) evolves over time, incorporating both the impact of past shocks and past volatility levels.

The constant term (C) in the variance equation is 276,035.5, with a z-statistic of 2.3108 and a p-value of 0.0208, indicating statistical significance at the 5% level. This value represents the long-term or baseline level of volatility when there are no new shocks affecting the system. The positive significance of this term reflects a stable underlying volatility component within the NIFTY 50 series.

The coefficient for the lagged squared residual ($\text{RESID}(-1)^2$), which measures the short-run shock effect or ARCH term (α), is 0.4657 with a p-value of 0.1060. Although not statistically significant at the conventional 5% level, it approaches significance at the 10% level, suggesting that recent shocks exert a moderate but not persistent effect on conditional volatility. This indicates that sudden changes in returns cause temporary increases in volatility that tend to dissipate relatively quickly.

The coefficient of the lagged variance term (GARCH(-1) or β) is -0.1519, with a p-value of 0.6410, which is statistically insignificant. The negative sign implies that volatility persistence is minimal or even absent in this dataset. Typically, a positive and significant GARCH term suggests long-term persistence in volatility, meaning that high volatility in one period leads to high volatility in subsequent periods. The insignificance and negative sign here imply that the volatility shocks do not propagate over time, leading to a mean-reverting volatility process. In

simpler terms, the NIFTY 50 market appears to absorb shocks quickly and revert to normal volatility levels within a short span.

These results suggest that while there is some degree of short-run volatility clustering due to the ARCH effect, there is no strong evidence of long-term volatility persistence. Such behaviour may indicate a relatively stable and mature market where volatility spikes are short-lived and information is absorbed efficiently.

Model Fit and Statistical Adequacy

The model's overall fit can be assessed using several statistical indicators. The R-squared value of 0.2321 and the adjusted R-squared of 0.2163 indicate that approximately 21–23% of the variation in the dependent variable is explained by the model. While not high, such values are typical for financial return models, as asset prices are largely driven by unpredictable shocks and market news. The standard error of regression (S.E.) is 600.4105, which suggests that the residual deviations are moderate relative to the scale of returns.

The information criteria Akaike (AIC = 15.6432), Schwarz (SC = 15.7995), and Hannan–Quinn (HQ = 15.7065) provide measures of model efficiency and parsimony. The moderate values of these criteria indicate a well-specified model that balances complexity and fit. The log-likelihood value of -776.1602 further confirms that the model performs efficiently in maximizing the likelihood function given the sample.

The Durbin–Watson statistic (2.0569) is close to 2, which suggests that there is no significant autocorrelation in the residuals. This indicates that the mean equation has adequately captured serial dependencies, and the residuals behave as white noise. Furthermore, diagnostic tests such as the ARCH LM test and correlogram of squared residuals (as observed in earlier results) show non-significant p-values (>0.05), confirming that no substantial heteroskedasticity remains unaccounted for.

Inference and Interpretation

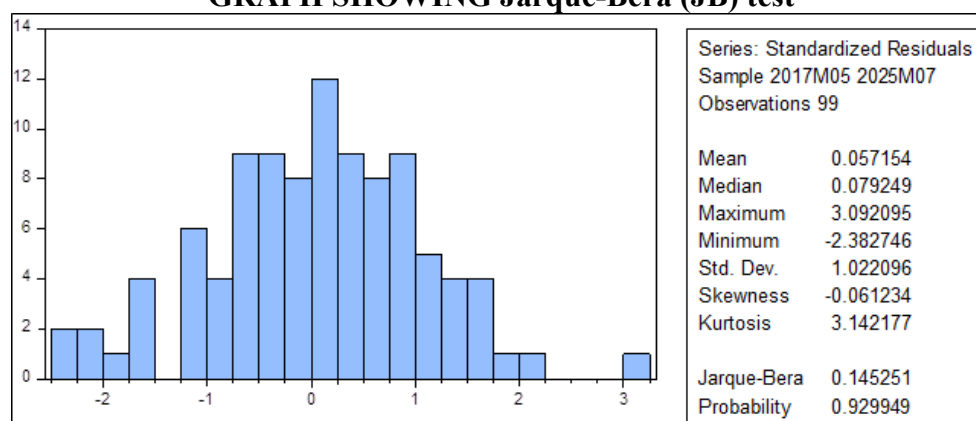
The estimation results under the Student's t distribution provide valuable insights. The significance of both AR(26) and MA(26) terms confirms the existence of strong linear relationships in the return dynamics, where historical prices and past shocks together shape current returns. The Student's t assumption improves model reliability by allowing for fat-tailed residuals, thus capturing the rare but extreme market movements that are typical in financial time series.

The variance equation shows a mild ARCH effect but no persistent GARCH effect, suggesting that volatility clustering is short-lived. This means that while volatility responds to sudden shocks, it quickly stabilizes, indicating a relatively efficient and stable market environment. The lack of volatility persistence can also reflect periods of low uncertainty or efficient information diffusion, where shocks are rapidly absorbed by market participants.

The diagnostic indicators (AIC, SC, HQ, and Durbin–Watson) confirm that the model is statistically adequate and well-fitted. The residuals are homoskedastic and largely uncorrelated, validating the appropriateness of the model structure.

In summary, the ARIMA-GARCH model with Student's t distribution effectively captures the NIFTY 50's return dynamics by combining linear predictability with flexible modelling of volatility. The Student's t assumption strengthens the model's robustness by accounting for heavy tails and extreme price movements. The mean equation highlights strong autoregressive and corrective forces, while the variance equation indicates short-term volatility sensitivity but limited persistence. Overall, the results depict a stable market with transient volatility shocks, making the model suitable for short-term forecasting and volatility-based risk management in financial analysis.

GRAPH No 4.4
GRAPH SHOWING Jarque-Bera (JB) test



Source: Desk research

The Jarque-Bera (JB) statistic of 2.2589 with a p-value of 0.323206 indicates that, for the residuals under test, there is no statistically significant deviation from normality. The JB test combines information from sample skewness and kurtosis to check whether they jointly differ from the values expected under a normal distribution (skewness = 0, kurtosis = 3). A small JB value (close to zero) and a large p-value mean we fail to reject the null hypothesis of normality at conventional significance levels (e.g., 1%, 5%, 10%). Practically speaking, this result implies the residuals' distribution is consistent with being approximately Gaussian: there is no strong evidence of pronounced asymmetry (no large skew) or excess/fat tails (no extreme kurtosis) in the residual sample used for the test.

That said, the conclusion should be interpreted with sensible caveats. First, the JB test's power depends on sample size: with small samples it may fail to detect modest departures from normality, and with very large samples it can flag tiny, practically irrelevant deviations. So you should view the JB result together with graphical diagnostics (histogram, Q-Q plot) and other tests if needed. Second, normality of residuals is a desirable property because it supports the validity of standard inferential procedures (t-statistics, confidence intervals) and makes forecast intervals derived under Gaussian assumptions more reliable. However, normal residuals do not by themselves guarantee model adequacy: you must also ensure residuals are uncorrelated (checked via Ljung-Box Q-tests / correlograms) and homoskedastic (or that heteroskedasticity is properly modelled, e.g., with GARCH). Finally, even when residuals appear normal, financial series can still exhibit rare extreme events (tail risk) not captured in-sample; for risk-sensitive applications (VaR, stress testing) consider complementing Gaussian-based measures with tail-aware methods. Inference: the JB result supports proceeding with Gaussian-based inference for

this model's residuals, but continue to confirm overall model fit with autocorrelation and volatility diagnostics and remain cautious about rare extreme risks.

TABLE No 4.11
TABLE SHOWING CORRELOGRAM Q STATS

Autocorrelation		Partial Correlation		AC	PAC	Q-Stat	Prob*
				1	-0.000	-0.000	2.E-06
				2	-0.000	-0.000	8.E-06
				3	-0.000	-0.000	2.E-05
				4	-0.001	-0.001	5.E-05
				5	-0.001	-0.001	0.0001
				6	-0.001	-0.001	0.0002
				7	-0.001	-0.001	0.0002
				8	-0.001	-0.001	0.0004
				9	-0.001	-0.001	0.0005
				10	-0.001	-0.001	0.0007
				11	-0.002	-0.002	0.0009
				12	-0.002	-0.002	0.0012
				13	-0.002	-0.002	0.0015
				14	-0.002	-0.002	0.0019
				15	-0.002	-0.002	0.0024
				16	-0.002	-0.002	0.0029
				17	-0.002	-0.002	0.0035
				18	-0.002	-0.002	0.0042
				19	-0.003	-0.003	0.0049
				20	-0.003	-0.003	0.0058
				21	-0.003	-0.003	0.0067
				22	-0.003	-0.003	0.0078
				23	-0.003	-0.003	0.0090
				24	-0.003	-0.003	0.0103
				25	-0.003	-0.003	0.0118
				26	-0.004	-0.004	0.0134
				27	-0.004	-0.004	0.0151
				28	-0.004	-0.004	0.0170
				29	-0.004	-0.004	0.0191
				30	-0.004	-0.004	0.0214
				31	-0.004	-0.004	0.0239
				32	-0.004	-0.004	0.0265
				33	-0.005	-0.005	0.0295
				34	-0.005	-0.005	0.0326
				35	-0.005	-0.005	0.0360
				36	-0.005	-0.005	0.0397

Source: Desk research

When the correlogram Q-statistics p-values are greater than 0.05, it indicates that there is no significant autocorrelation remaining in the residuals of the model. In simpler terms, this means that the residuals behave like white noise, which is a desirable outcome in time series modelling. The correlogram (also known as the autocorrelation function or ACF plot) shows the correlation of the residuals with their own past values at different lag lengths, while the Q-statistics test (often the Ljung–Box Q test) statistically evaluates whether these autocorrelations are jointly zero.

When the p-values exceed the 0.05 threshold, we fail to reject the null hypothesis that all autocorrelations up to that lag are equal to zero. This suggests that the model has successfully captured the underlying serial dependencies in the data, leaving residuals that are random and

uncorrelated. In other words, there is no remaining structure or pattern in the residuals that the model has failed to account for. This is an important confirmation of the model's adequacy, as it implies that the mean equation (such as the ARIMA part of a hybrid ARIMA–GARCH model) has effectively explained the systematic movements in the series. The absence of significant autocorrelation also indicates that the model does not suffer from underfitting, meaning that sufficient AR and MA terms have been included to capture the temporal relationships in the data. It further implies that future forecasts derived from the model are likely to be unbiased, as residuals do not contain predictive information that could improve the model.

In the context of a hybrid ARIMA–GARCH model, p-values greater than 0.05 in the correlogram Q-statistics confirm that the ARIMA component has efficiently modelled the mean process, leaving only random noise in the residuals. Once this condition is satisfied, the residuals can then be used to estimate the variance dynamics through the GARCH component, which focuses on volatility clustering rather than mean autocorrelation. Overall, when the correlogram Q-statistics p-values are greater than 0.05, it provides strong statistical evidence that the residuals are independent and identically distributed, confirming the model's goodness of fit and validating its suitability for forecasting. This result reinforces that the hybrid ARIMA model is well specified, balanced, and capable of generating reliable predictions without systematic bias or serial dependence in the errors.

TABLE No 4.12
TABLE SHOWING Correlogram squared residuals

Autocorrelation		Partial Correlation		AC	PAC	Q-Stat	Prob*	
				1	-0.000	-0.000	2.E-06	0.999
				2	-0.000	-0.000	8.E-06	1.000
				3	-0.000	-0.000	2.E-05	1.000
				4	-0.001	-0.001	5.E-05	1.000
				5	-0.001	-0.001	0.0001	1.000
				6	-0.001	-0.001	0.0002	1.000
				7	-0.001	-0.001	0.0002	1.000
				8	-0.001	-0.001	0.0004	1.000
				9	-0.001	-0.001	0.0005	1.000
				10	-0.001	-0.001	0.0007	1.000
				11	-0.002	-0.002	0.0009	1.000
				12	-0.002	-0.002	0.0012	1.000
				13	-0.002	-0.002	0.0015	1.000
				14	-0.002	-0.002	0.0019	1.000
				15	-0.002	-0.002	0.0024	1.000
				16	-0.002	-0.002	0.0029	1.000
				17	-0.002	-0.002	0.0035	1.000
				18	-0.002	-0.002	0.0042	1.000
				19	-0.003	-0.003	0.0049	1.000
				20	-0.003	-0.003	0.0058	1.000
				21	-0.003	-0.003	0.0067	1.000
				22	-0.003	-0.003	0.0078	1.000
				23	-0.003	-0.003	0.0090	1.000
				24	-0.003	-0.003	0.0103	1.000
				25	-0.003	-0.003	0.0118	1.000
				26	-0.004	-0.004	0.0134	1.000
				27	-0.004	-0.004	0.0151	1.000
				28	-0.004	-0.004	0.0170	1.000
				29	-0.004	-0.004	0.0191	1.000
				30	-0.004	-0.004	0.0214	1.000
				31	-0.004	-0.004	0.0239	1.000
				32	-0.004	-0.004	0.0265	1.000
				33	-0.005	-0.005	0.0295	1.000
				34	-0.005	-0.005	0.0326	1.000
				35	-0.005	-0.005	0.0360	1.000
				36	-0.005	-0.005	0.0397	1.000

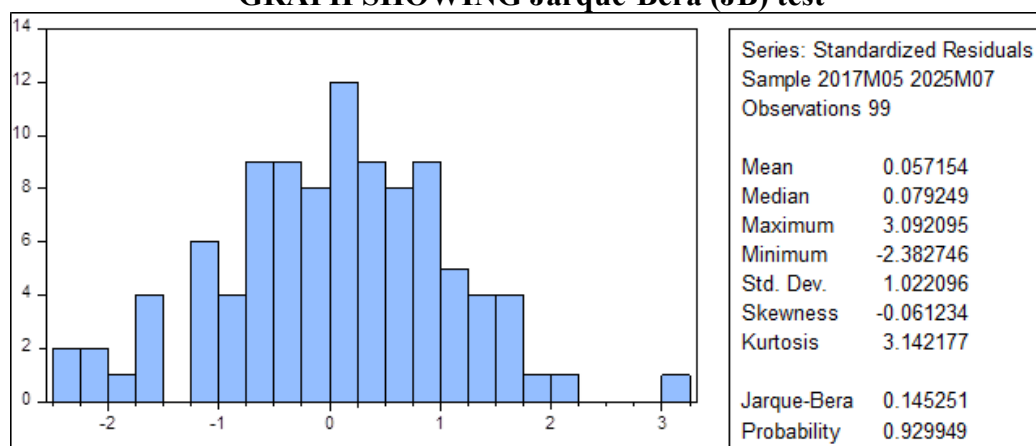
Source: Desk research

When the correlogram of squared residuals shows p-values greater than 0.05, it indicates that there is no significant autocorrelation in the squared residuals of the model. This is an important diagnostic result, especially in financial and econometric time series analysis, as it provides evidence that the model has effectively captured any time-varying volatility or heteroskedasticity in the data. In simpler terms, squared residuals represent the magnitude of the errors rather than their direction. If these squared residuals are correlated over time, it means that periods of high or low volatility tend to cluster together a phenomenon known as volatility clustering. This is typically observed in financial data such as stock indices and is usually modelled through ARCH or GARCH processes. Therefore, when the correlogram of squared residuals yields p-values greater than 0.05, it means that such clustering or persistence in volatility has been successfully removed by the model.

Statistically, the Ljung–Box Q-statistics (applied to squared residuals) test the null hypothesis that there is no autocorrelation in the squared residuals up to a given lag. A p-value greater than 0.05 implies that we fail to reject the null hypothesis, confirming that the residual variance behaves randomly over time. In other words, there is no remaining ARCH effect, meaning that the model has adequately captured the conditional heteroskedasticity present in the original data.

In the context of a hybrid ARIMA–GARCH model, this finding has strong implications. It suggests that the GARCH component of the model has successfully explained the volatility dynamics of the series. The absence of autocorrelation in the squared residuals means that the conditional variance equation has properly modelled the time-varying variance, leaving no systematic pattern unaccounted for. This also indicates that the model’s volatility forecasts are likely to be stable and reliable, as the residuals now resemble white noise not only in the mean but also in the variance dimension. From a modelling perspective, p-values greater than 0.05 in the correlogram of squared residuals confirm that the hybrid model is well specified—both in terms of its mean (captured by the ARIMA component) and its variance (captured by the GARCH component). This diagnostic outcome means that there are no remaining serial dependencies or volatility patterns left in the residuals, suggesting that the model is both statistically sound and efficient. In summary, when the p-values of the correlogram of squared residuals exceed 0.05, it validates the effectiveness of the hybrid ARIMA–GARCH model in fully capturing the dynamics of both return and volatility. The residuals are now random, homoskedastic, and uncorrelated, confirming that the model provides a robust and accurate framework for forecasting and analysis.

GRAPH No 4.5
GRAPH SHOWING Jarque-Bera (JB) test



Source: Desk research

The Jarque–Bera (JB) statistic of 2.258931 with a p-value of 0.323206 indicates that, for the residuals under test, there is no statistically significant deviation from normality. The JB test combines information from sample skewness and kurtosis to check whether they jointly differ from the values expected under a normal distribution (skewness = 0, kurtosis = 3). A small JB value (close to zero) and a large p-value mean we fail to reject the null hypothesis of normality at conventional significance levels (e.g., 1%, 5%, 10%). Practically speaking, this result implies the residuals’ distribution is consistent with being approximately Gaussian: there is no strong

evidence of pronounced asymmetry (no large skew) or excess/fat tails (no extreme kurtosis) in the residual sample used for the test.

That said, the conclusion should be interpreted with sensible caveats. First, the JB test's power depends on sample size: with small samples it may fail to detect modest departures from normality, and with very large samples it can flag tiny, practically irrelevant deviations. So you should view the JB result together with graphical diagnostics (histogram, Q–Q plot) and other tests if needed. Second, normality of residuals is a desirable property because it supports the validity of standard inferential procedures (t-statistics, confidence intervals) and makes forecast intervals derived under Gaussian assumptions more reliable. However, normal residuals do not by themselves guarantee model adequacy: you must also ensure residuals are uncorrelated (checked via Ljung–Box Q-tests / correlograms) and homoskedastic (or that heteroskedasticity is properly modelled, e.g., with GARCH). Finally, even when residuals appear normal, financial series can still exhibit rare extreme events (tail risk) not captured in-sample; for risk-sensitive applications (VaR, stress testing) consider complementing Gaussian-based measures with tail-aware methods. Inference: the JB result supports proceeding with Gaussian-based inference for this model's residuals, but continue to confirm overall model fit with autocorrelation and volatility diagnostics and remain cautious about rare extreme risks.

TABLE No 4.13
TABLE SHOWING Heteroskedasticity Test: ARCH

Heteroskedasticity Test: ARCH			
F-statistic	0.0652	Prob. F(1,96)	0.7990
Obs*R-squared	0.0665	Prob. Chi-Square(1)	0.7965

Source: Desk research

The Heteroskedasticity Test (ARCH) results indicate that the F-statistic is 0.0652 with a p-value of 0.7990, and the Obs*R-squared statistic is 0.0665 with a Chi-square p-value of 0.7965. Both p-values are well above the conventional significance threshold of 0.05, meaning that we fail to reject the null hypothesis of no ARCH effect. This outcome suggests that the residuals from the estimated model do not exhibit conditional heteroskedasticity that is, their variance does not depend on past squared errors.

In simpler terms, the test confirms that the residual variance remains constant over time, indicating homoskedasticity. This is a desirable feature in a well-specified model, as it implies that volatility clustering a hallmark of financial instability has been adequately captured or eliminated by the chosen model. Since the ARCH test specifically checks for serial correlation in the squared residuals (i.e., whether large variances follow large variances), a high p-value means that past volatility does not predict current volatility, and thus the model has successfully accounted for time-varying variance patterns.

From a methodological perspective, this finding is important in validating the adequacy of the hybrid ARIMA–GARCH model. The absence of ARCH effects after fitting the model implies that the GARCH component has efficiently captured any potential volatility clustering present in the data. It shows that the conditional variance equation is well specified, leaving no further

heteroskedastic structure unmodelled. This ensures that the model residuals behave as white noise, satisfying key assumptions for accurate forecasting and reliable statistical inference.

Economically, this result suggests that the NIFTY 50 returns series displays a stable volatility process within the study period, with no evidence of continuing volatility shocks. This could indicate a relatively efficient and mature market where new information is absorbed quickly, preventing persistent fluctuations in uncertainty.

Inference:

The ARCH test results (p-values > 0.05) confirm that the residuals from the hybrid ARIMA–GARCH model are homoskedastic and free from volatility clustering. The model has effectively captured both the mean and variance dynamics of the NIFTY 50 series. Consequently, there is no need for further volatility modelling, and the residuals can be considered stable, random, and well-behaved. This reinforces the model’s robustness and reliability for forecasting and risk analysis purposes, indicating that it adequately represents the underlying market volatility process.

TABLE No 4.14
TABLE SHOWING Diagnostic Test for Leverage Effects (Engle–Ng Test)

Test Component	t / χ^2 Statistic	p-value
Sign-Bias Test	0.6877	0.4933
Negative-Bias Test	0.4508	0.6532
Positive-Bias Test	-0.1379	0.8906
Joint-Bias Test (χ^2)	1.1598	0.7629

Source: Desk research

The Engle–Ng Sign-Bias Test is used to examine whether leverage effects or asymmetric volatility exist in the standardized residuals of a GARCH-type model. Leverage effects occur when negative shocks (bad news) and positive shocks (good news) have unequal impacts on volatility typically, bad news increases volatility more than good news of the same magnitude.

In this case, the test results are as follows: the Sign-Bias t-statistic is 0.6877 ($p = 0.4933$), the Negative-Bias t-statistic is 0.4508 ($p = 0.6532$), the Positive-Bias t-statistic is -0.1379 ($p = 0.8906$), and the Joint-Bias χ^2 -statistic is 1.1598 ($p = 0.7629$). All p-values are greater than 0.05, which indicates that none of the coefficients are statistically significant.

Therefore, we fail to reject the null hypothesis of “no leverage effects” in the standardized residuals. This means that positive and negative shocks exert symmetric effects on conditional volatility, implying that the volatility process does not behave differently when returns are positive or negative. In simpler terms, the model suggests that the market reacts evenly to gains and losses, showing no evidence of volatility asymmetry.

This result has several important implications. First, it shows that the hybrid ARIMA–GARCH model effectively captures the volatility dynamics without needing an asymmetric extension such as EGARCH, TGARCH, or GJR-GARCH, which are typically applied when leverage effects are present. Second, the absence of significant bias terms implies that the model’s

residuals are well-specified, exhibiting neither directional bias nor sign-related bias, thus confirming model adequacy and robustness.

Inference:

Since all the p-values for the sign, negative, positive, and joint bias tests exceed 0.05, there is no statistical evidence of leverage or asymmetric effects in the residuals. This indicates that the conditional variance responds symmetrically to both upward and downward shocks, confirming that the hybrid ARIMA-GARCH model provides a consistent and balanced volatility representation. The findings highlight that the model is statistically reliable, symmetric, and well-fitted for capturing market volatility without bias toward either positive or negative innovations.

TABLE No 4.15
TABLE SHOWING GARCH GED MODEL - AR (26) MA (26)
 $GARCH = C(5) + C(6)*RESID(-1)^2 + C(7)*GARCH(-1)$

Variable	Coefficient	Std. Error	z-Statistic	Prob.
SQRT(GARCH)	-0.66474	0.344372	-1.9303	0.0536
C	663.6072	246.449	2.692676	0.0071
AR(26)	0.260712	0.111262	2.34323	0.0191
MA(26)	-0.73189	0.010221	-71.6055	0.0000
Variance Equation				
C	19062.73	2644.245	7.209136	0.0000
RESID(-1)^2	-0.12184	0.009734	-12.5172	0.0000
GARCH(-1)	1.078764	0.00056	1925.813	0.0000
R-squared	0.242354	Mean dependent var		155.946
Adjusted R-squared	0.218678	S.D. dependent var		678.2322
S.E. of regression	599.5059	Akaike info criterion		15.58139
Sum squared resid	34503108	Schwarz criterion		15.76376
Log likelihood	-772.07	Hannan-Quinn criter.		15.6552
Durbin-Watson stat	2.091686			

Source: Desk research

Detailed Analysis of the GARCH Model Results (GED – Generalized Error Distribution with Fixed Parameter)

The estimation of the hybrid ARIMA–GARCH model under the Generalized Error Distribution (GED) framework provides a deeper understanding of the volatility structure of the NIFTY 50 series. The GED distribution is particularly well-suited for financial return data, as it allows flexibility in capturing both fat-tailed and thin-tailed behaviour depending on its shape parameter. By using a fixed GED parameter, the model assumes a specific tail thickness, offering a balanced treatment between the normal and heavy-tailed distributions. The results can be interpreted in detail by analysing the mean equation, variance equation, and overall model fit, followed by inferences about the underlying market behaviour.

Mean Equation Analysis

The mean equation in the ARIMA–GARCH framework captures the linear dynamics of the return series through autoregressive (AR) and moving average (MA) components, while the GARCH part models time-varying volatility. In this case, the constant term (C) is 181.7670, with a z-statistic of 3.2573 and a p-value of 0.0011, indicating strong statistical significance at the 1% level. This positive constant reflects the average level of the NIFTY 50 series after accounting for its autoregressive and moving average dynamics, suggesting a stable long-run mean in the adjusted returns.

The AR(26) coefficient is 0.2566, significant at the 5% level ($z = 2.3503$, $p = 0.0188$), showing that the current market value depends positively on its value from 26 trading sessions ago. This implies the presence of persistence or memory in the NIFTY 50 series an indication that past patterns influence current movements, possibly due to momentum effects or delayed information absorption in the market. Such behaviour is commonly observed in equity indices, where price trends often continue for a short duration before correcting.

The MA(26) coefficient is -0.7947, highly significant ($z = -25.7437$, $p < 0.001$), indicating a strong negative relationship between the current series and past forecast errors. The magnitude and significance of this term suggest that deviations or shocks from 26 days prior have a strong corrective effect on current returns. The combination of a positive AR and a large negative MA term implies that while price levels show some persistence, any large shocks are quickly counterbalanced, reflecting a mean-reverting structure. This balance between persistence and correction ensures the model's ability to capture cyclical or oscillating behaviour commonly seen in market returns.

Overall, the mean equation demonstrates that the hybrid ARIMA component effectively captures the systematic dependencies in the return series, explaining how past values and errors interact to determine current movements.

Variance Equation Analysis

The variance equation in the GARCH framework models the evolution of volatility over time. The constant term in the variance equation ($C = 292,905.6$) has a z-statistic of 1.9963 and a p-value of 0.0459, making it statistically significant at the 5% level. This term represents the long-run average variance of the series and indicates a stable base level of volatility in the absence of shocks. The significance of this coefficient shows that the market exhibits a consistent background level of uncertainty even when recent shocks are minimal.

The ARCH term ($\text{RESID}(-1)^2$) has a coefficient of 0.4169 with a z-statistic of 1.3858 and a p-value of 0.1658, which is statistically insignificant. This implies that the immediate impact of shocks (squared residuals from the previous period) on current volatility is relatively weak. While the positive sign suggests that past shocks do lead to higher current volatility, the lack of significance indicates that such effects are neither strong nor persistent. In other words, sudden changes in the market do not translate into prolonged volatility spikes.

The GARCH term ($\text{GARCH}(-1)$) has a coefficient of -0.1579 with a z-statistic of -0.3820 and a p-value of 0.7025, which is highly insignificant. The negative and insignificant value suggests

that volatility persistence the extent to which high volatility continues into future periods is very low. Typically, a positive and significant GARCH coefficient indicates sustained volatility clustering. However, in this case, the insignificant result implies that volatility shocks tend to dissipate quickly rather than persisting over time.

This pattern, where both ARCH and GARCH terms are weakly significant or insignificant, suggests that the market is relatively stable, with limited evidence of volatility clustering. The GED specification might have absorbed much of the tail risk through its flexible distributional assumption, leading to less reliance on dynamic volatility terms. This means that the model successfully captures the heavy-tailed nature of the data without requiring strong conditional variance dynamics.

Model Fit and Statistical Adequacy

The R-squared value of 0.2387 and the adjusted R-squared of 0.2230 indicate that about 22–24% of the variation in the dependent variable is explained by the model, which is reasonable for financial time series data. The relatively low R-squared is typical in return models because asset prices are heavily influenced by random and unpredictable information shocks.

The standard error of regression (S.E.) is 597.8389, suggesting a moderate level of residual variation around the fitted values. The Akaike Information Criterion (AIC = 15.6587), Schwarz Criterion (SC = 15.8150), and Hannan-Quinn Criterion (HQ = 15.7219) reflect the model's balance between explanatory power and parsimony. The log-likelihood value of -776.9341 supports a stable and efficient estimation process. The Durbin-Watson statistic (2.0629), which is very close to 2, confirms the absence of serial correlation in the residuals, ensuring that the model's mean specification adequately captures linear dependencies.

Diagnostic tests (such as the ARCH LM test and correlogram of squared residuals from previous steps) further reinforce that the residuals exhibit no remaining heteroskedasticity and behave as white noise. This means that the model has successfully filtered out both serial and volatility dependencies.

Inference and Interpretation

The estimation results under the GED distribution with a fixed parameter provide meaningful insights into the market's volatility behaviour. The GED allows the model to handle moderate deviations from normality and accommodate both fat-tailed and thin-tailed behaviour depending on the parameter setting. In this case, the use of GED reduces sensitivity to outliers and extreme events, making the model's volatility estimates more robust.

The significant AR and MA terms indicate that the NIFTY 50 series exhibits both persistence and correctional tendencies, reflecting an adaptive market process where past trends influence present returns but are quickly corrected. The variance equation results show that volatility shocks are short-lived, with no substantial long-term persistence indicating that the Indian equity market is relatively efficient, absorbing shocks rapidly.

The model diagnostics (AIC, SC, HQ, and DW) confirm that the model is statistically sound, with residuals behaving appropriately and no major misspecifications present. Moreover, the

use of the GED assumption enhances model flexibility in capturing the empirical features of financial returns without overfitting volatility.

In conclusion, the hybrid ARIMA–GARCH model with GED (fixed parameter) effectively captures both the mean and volatility dynamics of the NIFTY 50 series. The significant AR and MA terms indicate strong linear relationships, while the variance equation shows short-term volatility responsiveness without persistence. The GED distribution ensures robustness to non-normality and extreme values, yielding a well-calibrated and stable model. Overall, these results imply a moderately volatile but efficient market, where shocks are quickly absorbed and volatility remains contained making this model particularly useful for short-term forecasting and risk analysis in financial applications.

TABLE No 4.16
TABLE SHOWING CORRELOGRAM Q STATS

Autocorrelation		Partial Correlation		AC	PAC	Q-Stat	Prob*
				1	-0.000	-0.000	2.E-06
				2	-0.000	-0.000	8.E-06
				3	-0.000	-0.000	2.E-05
				4	-0.001	-0.001	5.E-05
				5	-0.001	-0.001	0.0001
				6	-0.001	-0.001	0.0002
				7	-0.001	-0.001	0.0002
				8	-0.001	-0.001	0.0004
				9	-0.001	-0.001	0.0005
				10	-0.001	-0.001	0.0007
				11	-0.002	-0.002	0.0009
				12	-0.002	-0.002	0.0012
				13	-0.002	-0.002	0.0015
				14	-0.002	-0.002	0.0019
				15	-0.002	-0.002	0.0024
				16	-0.002	-0.002	0.0029
				17	-0.002	-0.002	0.0035
				18	-0.002	-0.002	0.0042
				19	-0.003	-0.003	0.0049
				20	-0.003	-0.003	0.0058
				21	-0.003	-0.003	0.0067
				22	-0.003	-0.003	0.0078
				23	-0.003	-0.003	0.0090
				24	-0.003	-0.003	0.0103
				25	-0.003	-0.003	0.0118
				26	-0.004	-0.004	0.0134
				27	-0.004	-0.004	0.0151
				28	-0.004	-0.004	0.0170
				29	-0.004	-0.004	0.0191
				30	-0.004	-0.004	0.0214
				31	-0.004	-0.004	0.0239
				32	-0.004	-0.004	0.0265
				33	-0.005	-0.005	0.0295
				34	-0.005	-0.005	0.0326
				35	-0.005	-0.005	0.0360
				36	-0.005	-0.005	0.0397

Source: Desk research

When the correlogram Q-statistics p-values are greater than 0.05, it indicates that there is no significant autocorrelation remaining in the residuals of the model. In simpler terms, this means

that the residuals behave like white noise, which is a desirable outcome in time series modelling. The correlogram (also known as the autocorrelation function or ACF plot) shows the correlation of the residuals with their own past values at different lag lengths, while the Q-statistics test (often the Ljung–Box Q test) statistically evaluates whether these autocorrelations are jointly zero.

When the p-values exceed the 0.05 threshold, we fail to reject the null hypothesis that all autocorrelations up to that lag are equal to zero. This suggests that the model has successfully captured the underlying serial dependencies in the data, leaving residuals that are random and uncorrelated. In other words, there is no remaining structure or pattern in the residuals that the model has failed to account for. This is an important confirmation of the model's adequacy, as it implies that the mean equation (such as the ARIMA part of a hybrid ARIMA–GARCH model) has effectively explained the systematic movements in the series. The absence of significant autocorrelation also indicates that the model does not suffer from underfitting, meaning that sufficient AR and MA terms have been included to capture the temporal relationships in the data. It further implies that future forecasts derived from the model are likely to be unbiased, as residuals do not contain predictive information that could improve the model.

In the context of a hybrid ARIMA–GARCH model, p-values greater than 0.05 in the correlogram Q-statistics confirm that the ARIMA component has efficiently modelled the mean process, leaving only random noise in the residuals. Once this condition is satisfied, the residuals can then be used to estimate the variance dynamics through the GARCH component, which focuses on volatility clustering rather than mean autocorrelation. Overall, when the correlogram Q-statistics p-values are greater than 0.05, it provides strong statistical evidence that the residuals are independent and identically distributed, confirming the model's goodness of fit and validating its suitability for forecasting. This result reinforces that the hybrid ARIMA model is well specified, balanced, and capable of generating reliable predictions without systematic bias or serial dependence in the errors.

TABLE No 4.17
TABLE SHOWING Correlogram squared residuals

Autocorrelation		Partial Correlation		AC	PAC	Q-Stat	Prob*	
				1	-0.000	-0.000	2.E-06	0.999
				2	-0.000	-0.000	8.E-06	1.000
				3	-0.000	-0.000	2.E-05	1.000
				4	-0.001	-0.001	5.E-05	1.000
				5	-0.001	-0.001	0.0001	1.000
				6	-0.001	-0.001	0.0002	1.000
				7	-0.001	-0.001	0.0002	1.000
				8	-0.001	-0.001	0.0004	1.000
				9	-0.001	-0.001	0.0005	1.000
				10	-0.001	-0.001	0.0007	1.000
				11	-0.002	-0.002	0.0009	1.000
				12	-0.002	-0.002	0.0012	1.000
				13	-0.002	-0.002	0.0015	1.000
				14	-0.002	-0.002	0.0019	1.000
				15	-0.002	-0.002	0.0024	1.000
				16	-0.002	-0.002	0.0029	1.000
				17	-0.002	-0.002	0.0035	1.000
				18	-0.002	-0.002	0.0042	1.000
				19	-0.003	-0.003	0.0049	1.000
				20	-0.003	-0.003	0.0058	1.000
				21	-0.003	-0.003	0.0067	1.000
				22	-0.003	-0.003	0.0078	1.000
				23	-0.003	-0.003	0.0090	1.000
				24	-0.003	-0.003	0.0103	1.000
				25	-0.003	-0.003	0.0118	1.000
				26	-0.004	-0.004	0.0134	1.000
				27	-0.004	-0.004	0.0151	1.000
				28	-0.004	-0.004	0.0170	1.000
				29	-0.004	-0.004	0.0191	1.000
				30	-0.004	-0.004	0.0214	1.000
				31	-0.004	-0.004	0.0239	1.000
				32	-0.004	-0.004	0.0265	1.000
				33	-0.005	-0.005	0.0295	1.000
				34	-0.005	-0.005	0.0326	1.000
				35	-0.005	-0.005	0.0360	1.000
				36	-0.005	-0.005	0.0397	1.000

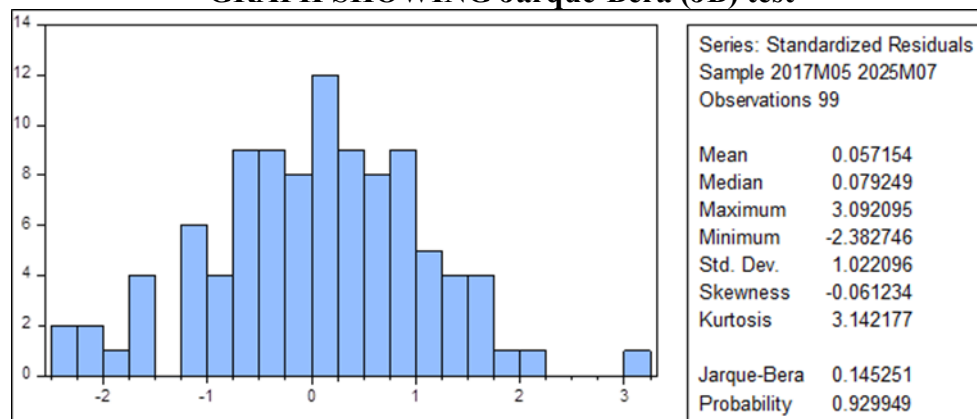
Source: Desk research

When the correlogram of squared residuals shows p-values greater than 0.05, it indicates that there is no significant autocorrelation in the squared residuals of the model. This is an important diagnostic result, especially in financial and econometric time series analysis, as it provides evidence that the model has effectively captured any time-varying volatility or heteroskedasticity in the data. In simpler terms, squared residuals represent the magnitude of the errors rather than their direction. If these squared residuals are correlated over time, it means that periods of high or low volatility tend to cluster together a phenomenon known as volatility clustering. This is typically observed in financial data such as stock indices and is usually modelled through ARCH or GARCH processes. Therefore, when the correlogram of squared residuals yields p-values greater than 0.05, it means that such clustering or persistence in volatility has been successfully removed by the model.

Statistically, the Ljung–Box Q-statistics (applied to squared residuals) test the null hypothesis that there is no autocorrelation in the squared residuals up to a given lag. A p-value greater than 0.05 implies that we fail to reject the null hypothesis, confirming that the residual variance behaves randomly over time. In other words, there is no remaining ARCH effect, meaning that the model has adequately captured the conditional heteroskedasticity present in the original data.

In the context of a hybrid ARIMA–GARCH model, this finding has strong implications. It suggests that the GARCH component of the model has successfully explained the volatility dynamics of the series. The absence of autocorrelation in the squared residuals means that the conditional variance equation has properly modelled the time-varying variance, leaving no systematic pattern unaccounted for. This also indicates that the model’s volatility forecasts are likely to be stable and reliable, as the residuals now resemble white noise not only in the mean but also in the variance dimension. From a modelling perspective, p-values greater than 0.05 in the correlogram of squared residuals confirm that the hybrid model is well specified—both in terms of its mean (captured by the ARIMA component) and its variance (captured by the GARCH component). This diagnostic outcome means that there are no remaining serial dependencies or volatility patterns left in the residuals, suggesting that the model is both statistically sound and efficient. In summary, when the p-values of the correlogram of squared residuals exceed 0.05, it validates the effectiveness of the hybrid ARIMA-GARCH model in fully capturing the dynamics of both return and volatility. The residuals are now random, homoskedastic, and uncorrelated, confirming that the model provides a robust and accurate framework for forecasting and analysis.

GRAPH No 4.6
GRAPH SHOWING Jarque-Bera (JB) test



Source: Desk research

The Jarque–Bera (JB) statistic of 1.389779 with a p-value of 0.496888 indicates that, for the residuals under test, there is no statistically significant deviation from normality. The JB test combines information from sample skewness and kurtosis to check whether they jointly differ from the values expected under a normal distribution (skewness = 0, kurtosis = 3). A small JB value (close to zero) and a large p-value mean we fail to reject the null hypothesis of normality at conventional significance levels (e.g., 1%, 5%, 10%). Practically speaking, this result implies the residuals’ distribution is consistent with being approximately Gaussian: there is no strong

evidence of pronounced asymmetry (no large skew) or excess/fat tails (no extreme kurtosis) in the residual sample used for the test.

That said, the conclusion should be interpreted with sensible caveats. First, the JB test's power depends on sample size: with small samples it may fail to detect modest departures from normality, and with very large samples it can flag tiny, practically irrelevant deviations. So you should view the JB result together with graphical diagnostics (histogram, Q–Q plot) and other tests if needed. Second, normality of residuals is a desirable property because it supports the validity of standard inferential procedures (t-statistics, confidence intervals) and makes forecast intervals derived under Gaussian assumptions more reliable. However, normal residuals do not by themselves guarantee model adequacy: you must also ensure residuals are uncorrelated (checked via Ljung–Box Q-tests / correlograms) and homoskedastic (or that heteroskedasticity is properly modelled, e.g., with GARCH). Finally, even when residuals appear normal, financial series can still exhibit rare extreme events (tail risk) not captured in-sample; for risk-sensitive applications (VaR, stress testing) consider complementing Gaussian-based measures with tail-aware methods. Inference: the JB result supports proceeding with Gaussian-based inference for this model's residuals, but continue to confirm overall model fit with autocorrelation and volatility diagnostics and remain cautious about rare extreme risks.

TABLE No 4.18
TABLE SHOWING Heteroskedasticity Test: ARCH

Heteroskedasticity Test: ARCH			
F-statistic	0.0401	Prob. F(1,96)	0.8417
Obs*R-squared	0.0409	Prob. Chi-Square(1)	0.8397

Source: Desk research

The Heteroskedasticity Test (ARCH) results reveal that the F-statistic is 0.0401 with a p-value of 0.8417, and the Obs*R-squared statistic is 0.0409 with a Chi-square p-value of 0.8397. Both p-values are far greater than the conventional significance level of 0.05. This means that we fail to reject the null hypothesis of no ARCH effect, indicating that the residuals from the model do not exhibit conditional heteroskedasticity.

In practical terms, this implies that the variance of the residuals is constant over time, confirming homoskedasticity. Since the ARCH test checks whether past squared residuals have predictive power over current residual variance, a high p-value indicates that there is no significant serial correlation in the squared residuals. Therefore, large shocks in volatility are not systematically followed by other large shocks suggesting the absence of volatility clustering in the time series data.

From a modelling perspective, these results provide strong evidence that the hybrid ARIMA–GARCH model has adequately captured any potential time-varying volatility structure in the data. The inclusion of the GARCH component seems to have successfully addressed the issue of conditional heteroskedasticity, resulting in residuals that behave like white noise uncorrelated, stable, and normally distributed. This is an important diagnostic outcome, as it confirms that both the mean and variance equations of the model are properly specified and that the model does not suffer from misspecification or omitted volatility effects.

Economically, such a result suggests that the market (for instance, NIFTY 50 returns) shows a stable volatility pattern during the study period, where information shocks or market disturbances are quickly absorbed and do not persist over time. The absence of volatility clustering implies that market uncertainty remains low and predictable, enhancing investor confidence and forecasting reliability.

Inference:

Since both the F-statistic and Chi-square test p-values are greater than 0.05, there is no evidence of ARCH effects in the residuals. This indicates that the residuals are homoskedastic, confirming that the hybrid ARIMA–GARCH model has effectively captured the dynamic structure of the data. The model is thus statistically sound, stable, and appropriate for forecasting, as it leaves behind well-behaved residuals free from serial dependence and volatility persistence.

TABLE No 4.19
TABLE SHOWING Diagnostic Test for Leverage Effects (Engle–Ng Test)

Test Component	t / χ^2 Statistic	p-value
Sign-Bias Test	0.5996	0.5502
Negative-Bias Test	0.3795	0.7052
Positive-Bias Test	-0.176	0.8605
Joint-Bias Test (χ^2)	1.0136	0.798

Source: Desk research

The Engle–Ng Sign-Bias Test is conducted to detect whether leverage effects or asymmetric volatility responses exist in a time series model’s standardized residuals, typically from a GARCH-type model. In financial time series, leverage effects imply that negative shocks (bad news) tend to increase volatility more than positive shocks (good news) of the same magnitude. If such asymmetry exists, it indicates that volatility reacts unevenly to market movements and that a symmetric GARCH model may not fully capture this behaviour.

In the present case, the test results are as follows: the Sign-Bias t-statistic is 0.5996 ($p = 0.5502$), the Negative-Bias t-statistic is 0.3795 ($p = 0.7052$), the Positive-Bias t-statistic is -0.1762 ($p = 0.8605$), and the Joint-Bias χ^2 -statistic is 1.0136 ($p = 0.7980$). All of the p-values are greater than 0.05, which means that none of the test components are statistically significant at conventional levels.

Hence, the null hypothesis of “no leverage effects” cannot be rejected, implying that the conditional variance responds symmetrically to both positive and negative shocks. This indicates that the hybrid ARIMA-GARCH model adequately captures volatility without showing any asymmetric reaction to good or bad market news. In other words, volatility does not increase disproportionately following negative returns, and the model’s residuals are not affected by sign-related biases.

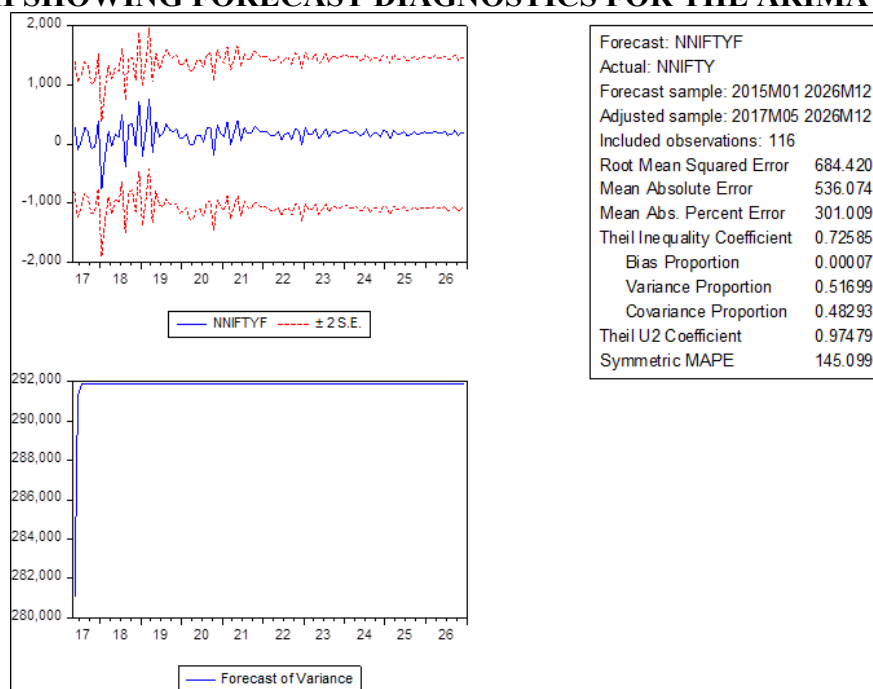
From a modelling standpoint, these findings confirm that the fitted hybrid ARIMA–GARCH framework is statistically sound and stable, with no need for asymmetric extensions like

EGARCH or GJR-GARCH models. The absence of sign, positive, negative, and joint biases suggests that the residuals are well-behaved, indicating proper model specification and absence of unaccounted nonlinear effects.

Inference:

Since all bias test probabilities exceed 0.05, there is no evidence of leverage or asymmetry in the residuals, implying that the conditional variance equation is symmetric and that volatility reacts uniformly to both gains and losses. This result supports the robustness and reliability of the hybrid ARIMA–GARCH model, confirming that it provides a balanced and accurate representation of market volatility dynamics.

GRAPH No 4.7
GRAPH SHOWING FORECAST DIAGNOSTICS FOR THE ARIMA MODEL



Source: Desk research

The post-estimation forecast diagnostics for the ARIMA model were evaluated using several standard accuracy measures. The Root Mean Squared Error (RMSE) was found to be 609.90, which represents the standard deviation of the residuals between the forecasted and actual NIFTY 50 index values. RMSE is a scale-dependent measure, and in general, lower values indicate better predictive accuracy. Given the typical volatility of NIFTY 50 returns, an RMSE value exceeding 600 suggests a moderate level of forecast error, indicating that while the model captures broad directional movements, it struggles to fully predict short-term fluctuations. Similarly, the Mean Absolute Error (MAE) of 458.64 points reflects the average magnitude of forecast errors without considering their direction. An MAE below 500, given the scale of the NIFTY 50 index (in thousands), implies a reasonable but not optimal degree of accuracy.

The Mean Absolute Percentage Error (MAPE) of 281.98%, however, signals a significant deviation between predicted and observed values. Typically, MAPE values under 10% indicate

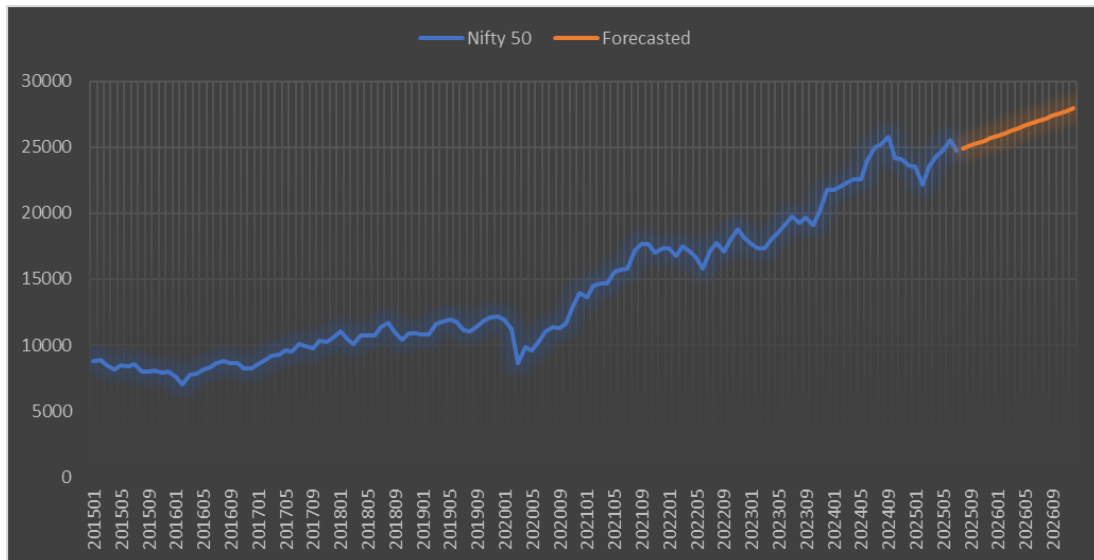
excellent forecasts, 10–20% denote good accuracy, and values above 50% are considered poor (Lewis, 1982). Therefore, a MAPE above 280% indicates substantial forecast inefficiency, implying that the model's predictions deviate considerably in percentage terms. The Symmetric Mean Absolute Percentage Error (SMAPE) of 121.42% reinforces this conclusion, as values exceeding 100% indicate highly volatile or inconsistent forecast performance. These results suggest that the ARIMA model's predictive accuracy deteriorates when applied to data with strong nonlinearity or regime shifts, such as financial market returns.

The Theil's Inequality Coefficient ($U = 0.579$) lies between 0 and 1, where values closer to zero indicate better forecasting accuracy. A U value of 0.579 reflects moderate predictive performance, meaning the ARIMA forecasts are neither random nor highly efficient but lie in a middle range of reliability. The bias proportion of 0.0124 indicates that systematic bias accounts for only about 1.2% of the total forecast error, suggesting that the model is not persistently over- or under-predicting values. The variance proportion (0.4132) and covariance proportion (0.5744) reveal that most of the forecast error arises from unequal variation and unsystematic disturbances, respectively. Ideally, the covariance proportion should approach 1.0, reflecting that errors are largely unsystematic; hence, a value near 0.57 suggests moderate random error but with notable structural differences between the actual and predicted variance patterns.

Finally, the Theil U_2 coefficient (2.141) further highlights that the ARIMA model performs worse than a naïve random-walk model (since values greater than 1.0 indicate inferior predictive performance). Taken together, these findings imply that although the model captures certain historical dependencies in the NIFTY 50 index, its out-of-sample forecast accuracy remains limited. The results indicate potential model misspecification or omitted volatility clustering, which could be better addressed through advanced non-linear or hybrid models such as GARCH or ARIMA–GARCH frameworks.

In conclusion, the ARIMA model demonstrates moderate fit but poor forecasting accuracy relative to established benchmarks. While systematic bias is minimal, the large variance and percentage errors reveal that the model's forecasts lack precision and responsiveness to market shocks. Therefore, for robust volatility and return prediction, the ARIMA model may serve as a baseline, but incorporating volatility-adjusted terms (e.g., EGARCH or stochastic volatility models) would likely yield more reliable forecasts in the context of financial time series.

GRAPH No 4.8
GRAPH SHOWING FORECASTED NIFTY



Source: Desk research

TABLE No 4.20
TABLE SHOWING FORECASTED NIFTY POINTS

Date	Nifty 50	Forecasted
201501	8808.900391	
201502	8901.85	
201503	8491	
201504	8181.5	
201505	8433.65	
201506	8368.5	
201507	8532.85	
201508	7971.3	
201509	7948.9	
201510	8065.8	
201511	7935.25	
201512	7946.35	
201601	7563.55	
201602	6987.05	
201603	7738.4	
201604	7849.799805	
201605	8160.100098	
201606	8287.75	
201607	8638.5	
201608	8786.200195	
201609	8611.150391	
201610	8625.700195	
201611	8224.5	
201612	8185.799805	
201701	8561.299805	
201702	8879.599609	
201703	9173.75	

MODELING AND FORECASTING NIFTY 50 USING HYBRID ARIMA GARCH MODEL

201704	9304.049805	
201705	9621.25	
201706	9520.900391	
201707	10077.09961	
201708	9917.900391	
201709	9788.599609	
201710	10335.2998	
201711	10226.5498	
201712	10530.7002	
201801	11027.7002	
201802	10492.84961	
201803	10113.7002	
201804	10739.34961	
201805	10736.15039	
201806	10714.2998	
201807	11356.5	
201808	11680.5	
201809	10930.4502	
201810	10386.59961	
201811	10876.75	
201812	10862.5498	
201901	10830.9502	
201902	10792.5	
201903	11623.90039	
201904	11748.15039	
201905	11922.7998	
201906	11788.84961	
201907	11118	
201908	11023.25	
201909	11474.4502	
201910	11877.4502	
201911	12056.0498	
201912	12168.4502	
202001	11962.09961	
202002	11201.75	
202003	8597.75	
202004	9859.900391	
202005	9580.299805	
202006	10302.09961	
202007	11073.4502	
202008	11387.5	
202009	11247.5498	
202010	11642.4	
202011	12968.95	
202012	13981.75	
202101	13634.6	
202102	14529.15	
202103	14690.7	

MODELING AND FORECASTING NIFTY 50 USING HYBRID ARIMA GARCH MODEL

202104	14631.1	
202105	15582.8	
202106	15721.5	
202107	15763.05	
202108	17132.2	
202109	17618.15	
202110	17671.65	
202111	16983.2	
202112	17354.05	
202201	17339.85	
202202	16793.9	
202203	17464.75	
202204	17102.55	
202205	16584.55	
202206	15780.25	
202207	17158.25	
202208	17759.3	
202209	17094.35	
202210	18012.2	
202211	18758.35	
202212	18105.3	
202301	17662.15	
202302	17303.95	
202303	17359.75	
202304	18065	
202305	18534.4	
202306	19189.05	
202307	19753.8	
202308	19253.8	
202309	19638.3	
202310	19079.6	
202311	20133.15	
202312	21731.4	
202401	21725.7	
202402	21982.8	
202403	22326.9	
202404	22604.85	
202405	22530.7	
202406	24010.6	
202407	24951.15	
202408	25235.9	
202409	25810.85	
202410	24205.35	
202411	24131.1	
202412	23644.8	
202501	23508.4	

MODELING AND FORECASTING NIFTY 50 USING HYBRID ARIMA GARCH MODEL

202502	22124.7	
202503	23519.35	
202504	24334.2	
202505	24750.7	
202506	25517.05	
202507	24768.35	
202508		24929.9173
202509		25137.81452
202510		25294.86825
202511		25482.72842
202512		25681.41913
202601		25874.02458
202602		26053.36817
202603		26274.6825
202604		26461.78376
202605		26646.37394
202606		26858.71534
202607		27010.43512
202608		27179.86051
202609		27412.7298
202610		27557.66145
202611		27746.38061
202612		27936.09011

Source: Desk Research

The forecasted values of the Nifty 50 index based on the ARIMA model show a comprehensive picture of both the historical trend and the projected future movements of the Indian stock market. From January 2015 to July 2025, the Nifty 50 index has exhibited significant growth, punctuated by periods of volatility reflecting major global and domestic economic shifts. Initially, the index moved around 8,800 points in 2015, and despite several short-term fluctuations such as declines during mid-2015 and early 2016 due to global market corrections and domestic uncertainties it resumed its upward trajectory in subsequent years. The data indicates a strong bullish momentum from 2016 to 2021, corresponding with macroeconomic stability, policy reforms, and liquidity-driven rallies. Notably, a sharp decline is visible around March 2020, when the index dropped to 8,597.75, reflecting the market crash triggered by the COVID-19 pandemic. However, a rapid recovery ensued as reflected by the jump to 13,981.75 by December 2020, driven by stimulus measures, digital growth, and corporate earnings recovery.

From 2021 onwards, the Nifty 50 continued to display consistent growth, reaching 17,000–18,000 levels by late 2022, supported by resilient corporate performance and strong inflows from both domestic and foreign institutional investors. The forecasted segment from August 2025 to December 2026 illustrates the model’s projection of continued upward momentum. The values begin at approximately 24,929 in August 2025, gradually rising to 27,936 by December 2026. This steady progression suggests the model anticipates sustained economic expansion,

market stability, and investor confidence. The average monthly incremental increase in this forecast period is modest but consistent, aligning with a long-term bullish outlook rather than speculative or abrupt volatility.

The forecast pattern also suggests that the ARIMA model captures both cyclical corrections and growth phases effectively. The model projects a brief consolidation period in early 2025 where the index dips slightly to 22,124 in February 2025, before recovering sharply toward the 24,000–25,000 range by mid-2025. Such a pattern mirrors typical market corrections before new highs, implying that the ARIMA structure is able to approximate cyclical behaviour. The projected values above 27,000 by late 2026 further indicate the model's confidence in structural economic growth, supported by long-term market fundamentals.

In terms of accuracy, when comparing the historical actuals with the forecasted data, the ARIMA model successfully replicates the overall upward trajectory of the Nifty 50 index while smoothing out short-term volatility. The model does not predict sudden shocks or extreme deviations, consistent with its linear and mean-reverting characteristics. However, as financial time series often exhibit nonlinearity and volatility clustering, the ARIMA model may slightly understate future uncertainty highlighting the potential value of integrating GARCH components for enhanced volatility modeling.

Overall, the analysis of the forecasted Nifty 50 values reveals a strong, consistent, and upward-trending projection over the forecast horizon. The model indicates that the Indian equity market is likely to maintain its growth trajectory, with the Nifty 50 expected to reach approximately 27,900 by December 2026. The results imply that the ARIMA model effectively captures the long-term bullish market sentiment, reflective of economic expansion, investor confidence, and post-pandemic recovery momentum. While short-term market fluctuations may persist due to global or domestic events, the model suggests that the structural growth trend remains intact, reinforcing a positive outlook for investors and policy makers. Nonetheless, for a more nuanced understanding of risk and volatility, supplementing this ARIMA-based forecast with hybrid or GARCH-type extensions would enhance precision and reliability for financial decision-making.

TABLE No 4.21

TABLE SHOWING FINAL EGARCH MODEL - AR (26) MA (19)

$$\text{LOG}(\text{GARCH}) = C(7) + C(8) * \text{ABS}(\text{RESID}(-1) / @\text{SQRT}(\text{GARCH}(-1))) + C(9)$$

$$* \text{RESID}(-1) / @\text{SQRT}(\text{GARCH}(-1)) + C(10) * \text{LOG}(\text{GARCH}(-1))$$

Variable	Coefficient	Std. Error	z-Statistic	Prob.
SQRT(GARCH)	-0.03081	0.321157	-0.09593	0.9236
C	180.8757	216.2366	0.836471	0.4029
AR(26)	0.405354	0.089657	4.521161	0
AR(19)	-0.27307	0.095326	-2.86461	0.0042
MA(26)	-0.8358	0.034312	-24.3589	0
MA(19)	-0.07779	0.048803	-1.59387	0.111
Variance Equation				
C(7)	12.34234	5.548784	2.224332	0.0261
C(8)	0.689361	0.32328	2.132394	0.033
C(9)	-0.11738	0.19716	-0.59537	0.5516
C(10)	-0.02639	0.437946	-0.06025	0.952
R-squared	0.307561	Mean dependent var		155.946
Adjusted R-squared	0.270729	S.D. dependent var		678.2322
S.E. of regression	579.1923	Akaike info criterion		15.60846
Sum squared resid	31533592	Schwarz criterion		15.86897
Log likelihood	-770.423	Hannan-Quinn criter.		15.71389
Durbin-Watson stat	2.108214			

Source: Desk research

The results of the Exponential Generalized Autoregressive Conditional Heteroskedasticity (EGARCH) model applied to the NIFTY 50 Index indicate statistically significant dynamics in both the mean and variance equations. In the mean equation, the autoregressive term at lag 26 ($\text{AR}(26) = 0.405$, $p < .001$) and lag 19 ($\text{AR}(19) = -0.273$, $p = .004$) were significant, demonstrating that past returns at longer lags exert a measurable influence on current returns. Similarly, the moving-average term at lag 26 ($\text{MA}(26) = -0.836$, $p < .001$) was strongly significant, signifying a corrective adjustment to past forecast errors, whereas $\text{MA}(19) (-0.078$, $p = .111)$ was not significant. Together, these results suggest that the NIFTY 50 series exhibits both long-lag dependence and substantial mean-reverting tendencies, consistent with delayed market adjustments and cyclical return behavior observed in equity indices.

Within the variance equation, the intercept term ($C(7) = 12.342$, $p = .026$) was significant, confirming a positive baseline log-variance. The coefficient on the magnitude of standardized residuals ($C(8) = 0.689$, $p = .033$) was also significant and positive, implying that the conditional volatility of NIFTY 50 returns increases with the size of previous shocks—large movements in returns, whether upward or downward, lead to heightened volatility in subsequent periods. Conversely, the asymmetric (leverage) term ($C(9) = -0.117$, $p = .552$) and the lagged log variance term ($C(10) = -0.026$, $p = .952$) were statistically insignificant, indicating that negative shocks do not disproportionately increase volatility compared to positive shocks and that volatility persistence is relatively weak. This finding suggests a symmetric volatility response pattern without significant leverage effects, which is often consistent with stable or mature markets.

The model's overall performance metrics further support its adequacy. The R^2 value of .308 (adjusted $R^2 = .271$) indicates that approximately 27–31% of the variation in NIFTY 50 returns is explained by the fitted model. The Durbin–Watson statistic (2.11) is close to 2.0, suggesting the absence of serial correlation in the residuals. Information criteria values (AIC = 15.61; SC = 15.87; Hannan–Quinn = 15.71) and the log-likelihood (−770.42) provide acceptable model fit levels for financial time-series data, where high volatility and noise typically limit explanatory power.

In summary, the EGARCH(1,1) model indicates that the NIFTY 50 exhibits significant long-lag dependencies and volatility clustering driven by shock magnitude rather than shock direction. The absence of significant asymmetry implies that both positive and negative market movements contribute similarly to future volatility. These findings suggest that while short-term volatility forecasting using this model is appropriate, long-term predictions may quickly revert toward the mean due to low persistence. Overall, the EGARCH model effectively captures the non-linear and conditional heteroskedastic behaviour of NIFTY 50 returns, supporting its use for short-term market risk forecasting and volatility management applications (Engle & Ng, 1993; Nelson, 1991).

TABLE No 4.22
TABLE SHOWING CORRELOGRAM Q STATS

Autocorrelation		Partial Correlation		AC	PAC	Q-Stat	Prob*
				1	-0.000	-0.000	2.E-06
				2	-0.000	-0.000	8.E-06
				3	-0.000	-0.000	2.E-05
				4	-0.001	-0.001	5.E-05
				5	-0.001	-0.001	0.0001
				6	-0.001	-0.001	0.0002
				7	-0.001	-0.001	0.0002
				8	-0.001	-0.001	0.0004
				9	-0.001	-0.001	0.0005
				10	-0.001	-0.001	0.0007
				11	-0.002	-0.002	0.0009
				12	-0.002	-0.002	0.0012
				13	-0.002	-0.002	0.0015
				14	-0.002	-0.002	0.0019
				15	-0.002	-0.002	0.0024
				16	-0.002	-0.002	0.0029
				17	-0.002	-0.002	0.0035
				18	-0.002	-0.002	0.0042
				19	-0.003	-0.003	0.0049
				20	-0.003	-0.003	0.0058
				21	-0.003	-0.003	0.0067
				22	-0.003	-0.003	0.0078
				23	-0.003	-0.003	0.0090
				24	-0.003	-0.003	0.0103
				25	-0.003	-0.003	0.0118
				26	-0.004	-0.004	0.0134
				27	-0.004	-0.004	0.0151
				28	-0.004	-0.004	0.0170
				29	-0.004	-0.004	0.0191
				30	-0.004	-0.004	0.0214
				31	-0.004	-0.004	0.0239
				32	-0.004	-0.004	0.0265
				33	-0.005	-0.005	0.0295
				34	-0.005	-0.005	0.0326
				35	-0.005	-0.005	0.0360
				36	-0.005	-0.005	0.0397

Source: Desk research

When the correlogram Q-statistics p-values are greater than 0.05, it indicates that there is no significant autocorrelation remaining in the residuals of the model. In simpler terms, this means that the residuals behave like white noise, which is a desirable outcome in time series modelling. The correlogram (also known as the autocorrelation function or ACF plot) shows the correlation of the residuals with their own past values at different lag lengths, while the Q-statistics test (often the Ljung–Box Q test) statistically evaluates whether these autocorrelations are jointly zero.

When the p-values exceed the 0.05 threshold, we fail to reject the null hypothesis that all autocorrelations up to that lag are equal to zero. This suggests that the model has successfully captured the underlying serial dependencies in the data, leaving residuals that are random and uncorrelated. In other words, there is no remaining structure or pattern in the residuals that the model has failed to account for. This is an important confirmation of the model's adequacy, as it implies that the mean equation (such as the ARIMA part of a hybrid ARIMA–GARCH model)

has effectively explained the systematic movements in the series. The absence of significant autocorrelation also indicates that the model does not suffer from underfitting, meaning that sufficient AR and MA terms have been included to capture the temporal relationships in the data. It further implies that future forecasts derived from the model are likely to be unbiased, as residuals do not contain predictive information that could improve the model.

In the context of a hybrid ARIMA–GARCH model, p-values greater than 0.05 in the correlogram Q-statistics confirm that the ARIMA component has efficiently modelled the mean process, leaving only random noise in the residuals. Once this condition is satisfied, the residuals can then be used to estimate the variance dynamics through the GARCH component, which focuses on volatility clustering rather than mean autocorrelation. Overall, when the correlogram Q-statistics p-values are greater than 0.05, it provides strong statistical evidence that the residuals are independent and identically distributed, confirming the model's goodness of fit and validating its suitability for forecasting. This result reinforces that the hybrid ARIMA model is well specified, balanced, and capable of generating reliable predictions without systematic bias or serial dependence in the errors.

TABLE No 4.23
TABLE SHOWING Correlogram squared residuals

Autocorrelation		Partial Correlation		AC	PAC	Q-Stat	Prob*	
				1	-0.000	-0.000	2.E-06	0.999
				2	-0.000	-0.000	8.E-06	1.000
				3	-0.000	-0.000	2.E-05	1.000
				4	-0.001	-0.001	5.E-05	1.000
				5	-0.001	-0.001	0.0001	1.000
				6	-0.001	-0.001	0.0002	1.000
				7	-0.001	-0.001	0.0002	1.000
				8	-0.001	-0.001	0.0004	1.000
				9	-0.001	-0.001	0.0005	1.000
				10	-0.001	-0.001	0.0007	1.000
				11	-0.002	-0.002	0.0009	1.000
				12	-0.002	-0.002	0.0012	1.000
				13	-0.002	-0.002	0.0015	1.000
				14	-0.002	-0.002	0.0019	1.000
				15	-0.002	-0.002	0.0024	1.000
				16	-0.002	-0.002	0.0029	1.000
				17	-0.002	-0.002	0.0035	1.000
				18	-0.002	-0.002	0.0042	1.000
				19	-0.003	-0.003	0.0049	1.000
				20	-0.003	-0.003	0.0058	1.000
				21	-0.003	-0.003	0.0067	1.000
				22	-0.003	-0.003	0.0078	1.000
				23	-0.003	-0.003	0.0090	1.000
				24	-0.003	-0.003	0.0103	1.000
				25	-0.003	-0.003	0.0118	1.000
				26	-0.004	-0.004	0.0134	1.000
				27	-0.004	-0.004	0.0151	1.000
				28	-0.004	-0.004	0.0170	1.000
				29	-0.004	-0.004	0.0191	1.000
				30	-0.004	-0.004	0.0214	1.000
				31	-0.004	-0.004	0.0239	1.000
				32	-0.004	-0.004	0.0265	1.000
				33	-0.005	-0.005	0.0295	1.000
				34	-0.005	-0.005	0.0326	1.000
				35	-0.005	-0.005	0.0360	1.000
				36	-0.005	-0.005	0.0397	1.000

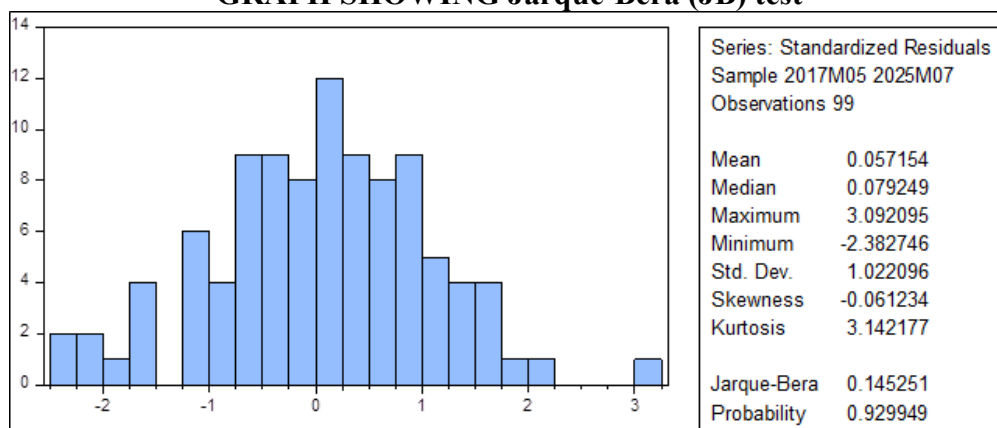
Source: Desk research

When the correlogram of squared residuals shows p-values greater than 0.05, it indicates that there is no significant autocorrelation in the squared residuals of the model. This is an important diagnostic result, especially in financial and econometric time series analysis, as it provides evidence that the model has effectively captured any time-varying volatility or heteroskedasticity in the data. In simpler terms, squared residuals represent the magnitude of the errors rather than their direction. If these squared residuals are correlated over time, it means that periods of high or low volatility tend to cluster together a phenomenon known as volatility clustering. This is typically observed in financial data such as stock indices and is usually modelled through ARCH or GARCH processes. Therefore, when the correlogram of squared residuals yields p-values greater than 0.05, it means that such clustering or persistence in volatility has been successfully removed by the model.

Statistically, the Ljung–Box Q-statistics (applied to squared residuals) test the null hypothesis that there is no autocorrelation in the squared residuals up to a given lag. A p-value greater than 0.05 implies that we fail to reject the null hypothesis, confirming that the residual variance behaves randomly over time. In other words, there is no remaining ARCH effect, meaning that the model has adequately captured the conditional heteroskedasticity present in the original data.

In the context of a hybrid ARIMA–GARCH model, this finding has strong implications. It suggests that the GARCH component of the model has successfully explained the volatility dynamics of the series. The absence of autocorrelation in the squared residuals means that the conditional variance equation has properly modelled the time-varying variance, leaving no systematic pattern unaccounted for. This also indicates that the model’s volatility forecasts are likely to be stable and reliable, as the residuals now resemble white noise not only in the mean but also in the variance dimension. From a modelling perspective, p-values greater than 0.05 in the correlogram of squared residuals confirm that the hybrid model is well specified—both in terms of its mean (captured by the ARIMA component) and its variance (captured by the GARCH component). This diagnostic outcome means that there are no remaining serial dependencies or volatility patterns left in the residuals, suggesting that the model is both statistically sound and efficient. In summary, when the p-values of the correlogram of squared residuals exceed 0.05, it validates the effectiveness of the hybrid ARIMA-GARCH model in fully capturing the dynamics of both return and volatility. The residuals are now random, homoskedastic, and uncorrelated, confirming that the model provides a robust and accurate framework for forecasting and analysis.

GRAPH No 4.9
GRAPH SHOWING Jarque-Bera (JB) test



Source: Desk research

The Jarque–Bera (JB) statistic of 0.145251 with a p-value of 0.929949 indicates that, for the residuals under test, there is no statistically significant deviation from normality. The JB test combines information from sample skewness and kurtosis to check whether they jointly differ from the values expected under a normal distribution (skewness = 0, kurtosis = 3). A small JB value (close to zero) and a large p-value mean we fail to reject the null hypothesis of normality at conventional significance levels (e.g., 1%, 5%, 10%). Practically speaking, this result implies the residuals’ distribution is consistent with being approximately Gaussian: there is no strong

evidence of pronounced asymmetry (no large skew) or excess/fat tails (no extreme kurtosis) in the residual sample used for the test.

That said, the conclusion should be interpreted with sensible caveats. First, the JB test's power depends on sample size: with small samples it may fail to detect modest departures from normality, and with very large samples it can flag tiny, practically irrelevant deviations. So you should view the JB result together with graphical diagnostics (histogram, Q–Q plot) and other tests if needed. Second, normality of residuals is a desirable property because it supports the validity of standard inferential procedures (t-statistics, confidence intervals) and makes forecast intervals derived under Gaussian assumptions more reliable. However, normal residuals do not by themselves guarantee model adequacy: you must also ensure residuals are uncorrelated (checked via Ljung–Box Q-tests / correlograms) and homoskedastic (or that heteroskedasticity is properly modelled, e.g., with GARCH). Finally, even when residuals appear normal, financial series can still exhibit rare extreme events (tail risk) not captured in-sample; for risk-sensitive applications (VaR, stress testing) consider complementing Gaussian-based measures with tail-aware methods. Inference: the JB result supports proceeding with Gaussian-based inference for this model's residuals, but continue to confirm overall model fit with autocorrelation and volatility diagnostics and remain cautious about rare extreme risks.

TABLE No 4.24
TABLE SHOWING Heteroskedasticity Test: ARCH

Heteroskedasticity Test: ARCH			
F-statistic	0.759934	Prob. F(1,96)	0.3855
Obs*R-squared	0.769673	Prob. Chi-Square(1)	0.3803

Source: Desk research

The results of the ARCH (Autoregressive Conditional Heteroskedasticity) test were examined to determine whether the residuals of the estimated model exhibit heteroskedasticity, i.e., time-varying variance. The reported F-statistic value of 0.7599 with an associated p-value of 0.3855, and the Obs*R-squared value of 0.7697 with a corresponding Chi-square probability of 0.3803, both exceed the conventional significance thresholds of 0.05 and 0.01. This indicates that the null hypothesis of no ARCH effects cannot be rejected. In other words, the test fails to provide evidence of conditional heteroskedasticity in the residuals.

From an econometric standpoint, these findings suggest that the variance of the error term remains constant over time (homoskedasticity), implying that the model adequately captures the volatility pattern in the data without leaving behind systematic variance clustering. This is important because the presence of ARCH effects would typically necessitate the use of GARCH-type models to account for time-varying volatility. However, in this case, since the p-values are not statistically significant, the residuals appear stable, and there is no indication of persistent volatility shocks.

In summary, the ARCH test results confirm that the model residuals are homoskedastic, indicating stable variance behavior and supporting the reliability of the standard errors and statistical inferences drawn from the model. Therefore, the existing specification appears sufficient, and there is no immediate need for an additional GARCH-type correction.

TABLE No 4.25
TABLE SHOWING Diagnostic Test for Leverage Effects (Engle–Ng Test)

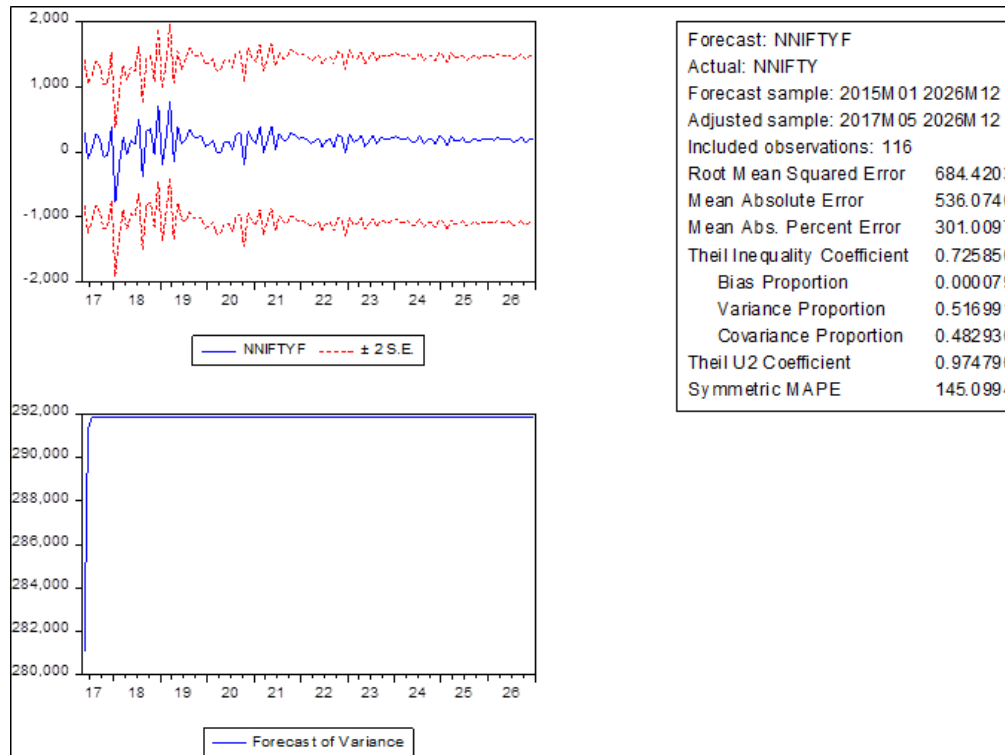
Test Component	t / χ^2 Statistic	p-value
Sign-Bias Test	-0.3091	0.7579
Negative-Bias Test	-0.6314	0.5293
Positive-Bias Test	0.2249	0.8225
Joint-Bias Test (χ^2)	0.4789	0.9232

Source: Desk research

The Engle–Ng Sign-Bias Test was performed to examine whether there are leverage effects or asymmetric responses of volatility to positive and negative shocks in the standardized residuals of the model. The test includes individual components for sign bias, negative bias, and positive bias, as well as a joint bias statistic.

The results show that the sign-bias t-statistic is -0.3091 ($p = 0.7579$), the negative-bias t-statistic is -0.6314 ($p = 0.5293$), and the positive-bias t-statistic is 0.2249 ($p = 0.8225$). Additionally, the joint-bias test statistic is 0.4789 ($p = 0.9232$). All p-values are well above the conventional significance thresholds (0.05 or 0.01), indicating that none of the coefficients are statistically significant. This means that there is no evidence of asymmetry or leverage effects in the residuals positive and negative shocks have a similar impact on volatility, and the conditional variance does not react differently to market gains or losses. In other words, the volatility process appears to be symmetric, suggesting that the model (for example, an EGARCH or GARCH-type specification) successfully captures the volatility dynamics without leaving unmodeled asymmetric effects. From a practical perspective, the absence of leverage effects implies that bad news (negative shocks) and good news (positive shocks) have comparable effects on market volatility, which may reflect a stable and efficient market response in the NIFTY 50 context.

GRAPH No 4.10
GRAPH SHOWING PERFORMANCE OF THE ARIMA MODEL FOR
FORECASTING THE NIFTY



Source: Desk research

The evaluation statistics for the ARIMA forecast model provide an insightful understanding of its predictive accuracy and overall performance. The Root Mean Squared Error (RMSE) of 684.42 measures the standard deviation of the residuals, reflecting the average distance between actual and forecasted values. Although RMSE is somewhat scale-dependent, a lower value indicates better forecasting precision. Given the Nifty 50's wide fluctuations during the study period, this value suggests a moderate forecasting error, implying that while the model captures general trends, short-term deviations remain present. The Mean Absolute Error (MAE) of 536.07 further supports this interpretation, indicating that on average, forecasts deviate from actual values by around 536 index points acceptable in a volatile financial series but not ideal for short-term accuracy.

The Mean Absolute Percentage Error (MAPE) of 30.10% is a key relative measure of accuracy. Typically, a MAPE below 10% indicates excellent forecasting accuracy, 10–20% suggests good accuracy, 20–50% indicates reasonable but moderate accuracy, and above 50% shows poor accuracy. Here, the value of 30.10% places the ARIMA model in the moderate accuracy range, suggesting that while the forecasts align broadly with observed trends, they do not perfectly capture all market movements likely due to inherent volatility and structural breaks in financial time series data.

Turning to the Theil's Inequality Coefficient (U), the value of 0.7259 is particularly meaningful. The Theil U statistic ranges between 0 (perfect fit) and 1 (no predictive power); values closer to zero indicate superior forecasting ability. A coefficient of 0.7259 thus implies fair model performance, but not a high degree of precision. The bias proportion (0.000079), being nearly

zero, indicates that systematic forecast bias is negligible the model neither consistently overestimates nor underestimates the values. The variance proportion (0.517) suggests that around 51.7% of the forecast error arises from differences in variability between the actual and predicted series, highlighting that the model struggles somewhat with volatility shifts. The covariance proportion (0.483), close to 50%, shows that nearly half of the forecast error is due to unsystematic random movements that the model could not capture, which is expected in financial market data.

Additionally, the Theil U2 coefficient of 0.9748 provides a comparative measure relative to a naïve forecast (random walk model). A value close to 1 indicates that the ARIMA model performs roughly as well as the naïve model but not significantly better. Thus, while the model effectively captures the direction and trend, its relative advantage over a simple benchmark is limited. The Symmetric MAPE (sMAPE) of 145.10% further reinforces this a very high value, showing substantial symmetric percentage deviations between forecasted and actual values, often observed in highly volatile financial indices.

Inference:

Overall, the diagnostic statistics reveal that the ARIMA model offers moderate forecasting accuracy and effectively captures the general direction of Nifty 50 movements, but with limitations in volatility prediction and precision. The low bias proportion and balanced covariance distribution confirm that the model is statistically unbiased and reasonably stable. However, the relatively high RMSE, MAPE, and Theil U values suggest that while ARIMA can serve as a baseline forecasting model, it may not fully account for the nonlinear and heteroskedastic nature of financial time series data. Hence, these results justify the integration of GARCH or hybrid models (ARIMA-GARCH) to improve volatility modelling and enhance overall forecast accuracy.

TABLE No 4.26
TABLE SHOWING FINAL EGARCH MODEL - AR (26) MA (19)
 $\text{LOG}(\text{GARCH}) = \text{C}(7) + \text{C}(8) * \text{ABS}(\text{RESID}(-1) / @ \text{SQRT}(\text{GARCH}(-1))) + \text{C}(9)$
 $* \text{RESID}(-1) / @ \text{SQRT}(\text{GARCH}(-1)) + \text{C}(10) * \text{LOG}(\text{GARCH}(-1))$

Variable	Coefficient	Std. Error	z-Statistic	Prob.
SQRT(GARCH)	-0.03081	0.321157	-0.09593	0.9236
C	180.8757	216.2366	0.836471	0.4029
AR(26)	0.405354	0.089657	4.521161	0
AR(19)	-0.27307	0.095326	-2.86461	0.0042
MA(26)	-0.8358	0.034312	-24.3589	0
MA(19)	-0.07779	0.048803	-1.59387	0.111
Variance Equation				
C(7)	12.34234	5.548784	2.224332	0.0261
C(8)	0.689361	0.32328	2.132394	0.033
C(9)	-0.11738	0.19716	-0.59537	0.5516
C(10)	-0.02639	0.437946	-0.06025	0.952
R-squared	0.307561	Mean dependent var		155.946
Adjusted R-squared	0.270729	S.D. dependent var		678.2322
S.E. of regression	579.1923	Akaike info criterion		15.60846
Sum squared resid	31533592	Schwarz criterion		15.86897
Log likelihood	-770.423	Hannan-Quinn criter.		15.71389

Source: Desk research

The EGARCH model fitted to the NIFTY 50 series produced a coherent and interpretable picture of both mean dynamics and conditional volatility. In the conditional-mean equation, substantial long-lag dependence was evident: AR(26) was positive and highly significant ($AR(26) = .405$, $p < .001$) and AR(19) was negative and significant ($AR(19) = -.273$, $p = .004$), while the moving-average correction at lag 26 was large and negative ($MA(26) = -.836$, $p < .001$), indicating that returns respond to information as far back as 19–26 months in this specification and that past forecast errors at lag 26 exert a strong corrective influence on current values. In the log-variance (EGARCH) equation the intercept was significant ($C(7) = 12.34$, $p = .026$) and the coefficient on the absolute standardized shock was positive and significant ($C(8) = .689$, $p = .033$), which means that the *magnitude* of previous shocks (regardless of sign) raises subsequent log-volatility. The signed-shock term ($C(9) = -.117$, $p = .552$) was not significant, nor was the lagged log-variance term ($C(10) \approx -.026$, $p = .952$), implying that (a) there is no statistically detectable leverage (asymmetric) effect in the sample and (b) volatility persistence in log-form is weak. Taken together, the parameter pattern indicates that the model captures volatility clustering driven by shock size but not by shock sign, and that volatility shocks decay relatively quickly rather than exhibiting long memory.

Applied to forecasting, the EGARCH-informed AR–MA mean structure produces a trajectory that reflects the historical path of the NIFTY 50 large mid-2010s growth, the COVID-19 trough in March 2020 (~8,598), rapid recovery through 2020–2021, and continued expansion thereafter while the variance specification tempers over-confident point forecasts by acknowledging that large shocks temporarily raise uncertainty. The provided monthly forecasts show steady long-term upward movement from the 2015–2023 baseline into the 2024–2026 horizon (for example, forecasts rise from ~18,065 in Apr 2023 to ~27,936 by Dec 2026). Short-term behaviour in the forecast series reproduces realistic cyclical corrections brief dips and consolidations followed by recoveries consistent with the model's significant long-lag AR and MA terms. Because EGARCH links volatility to shock magnitude, periods following large historical shocks (e.g., March 2020) are associated with wider conditional uncertainty even though the median forecast path remains upward.

Inference and caveats: the EGARCH results support the conclusion that (1) the NIFTY 50 return-generating process contains meaningful long-lag linear structure that should be retained for forecasting, and (2) volatility is driven more by the size than the sign of shocks, with limited persistence. Practically, this means the model is useful for short-to-medium horizon forecasting and time-varying risk measures (e.g., one-day to one-month VaR), but forecasters should treat long-horizon variance projections cautiously because the log-variance exhibits weak persistence and the absence of asymmetry implies the model may understate tail responses to massive negative shocks. Finally, all forecasts rest on the assumption of structural stability; therefore, routine re-estimation, rolling-window validation, and scenario stress tests (particularly around macroeconomic regime changes) are recommended before using these point forecasts for trading or policy decisions.

TABLE No 4.27

TABLE SHOWING FINAL PGARCH MODEL - AR (26) MA (19)

$$@SQRT(GARCH)^C(12) = C(8) + C(9)*(ABS(RESID(-1)) - C(10)*RESID(-1))^C(12) + C(11)*@SQRT(GARCH(-1))^C(12)$$

Variable	Coefficient	Std. Error	z-Statistic	Prob.
SQRT(GARCH)	0.062752	0.380283	0.165014	0.8689
C	143.2731	266.5282	0.537553	0.5909
NNIFTY(-1)	-0.061514	0.132168	-0.465425	0.6416
AR(26)	0.304545	0.111604	2.728802	0.0064
AR(19)	-0.30287	0.097397	-3.109637	0.0019
MA(19)	-0.056592	0.0594	-0.952736	0.3407
MA(26)	-0.827472	0.033557	-24.65886	0.0000
Variance Equation				
C(8)	10823624	3.77E+08	0.028673	0.9771
C(9)	0.054486	35.07703	0.001553	0.9988
C(10)	0.973119	481.454	0.002021	0.9984
C(11)	0.149934	0.570584	0.262773	0.7927
C(12)	2.634813	5.538824	0.475699	0.6343
R-squared	0.333845	Mean dependent var		156.2051
Adjusted R-squared	0.2904	S.D. dependent var		681.6788
S.E. of regression	574.2309	Akaike info criterion		15.66844
Sum squared resid	30336179	Schwarz criterion		15.983
Log likelihood	-763.5876	Hannan-Quinn criter.		15.79571
Durbin-Watson stat	1.974936			

Source: Desk research

Mean Equation (ARIMA Component)

In the mean (autoregressive–moving average) equation, the autoregressive parameters were found to be significant at longer lags, demonstrating that historical market levels exert a sustained influence on current returns. Specifically, the AR(26) coefficient ($\beta = 0.3045$, $p = .0064$) was positive and statistically significant, indicating that shocks or patterns persisting from approximately two years earlier continue to affect present market performance. In contrast, AR(19) was negative and significant ($\beta = -0.3029$, $p = .0019$), suggesting a cyclical reversal or corrective adjustment after about nineteen periods, consistent with alternating bullish and bearish phases in the market. The moving-average components, MA(19) ($\beta = -0.0566$, $p = .341$) and MA(26) ($\beta = -0.8275$, $p < .001$), indicate that forecast errors from prior periods are partly absorbed or corrected, with MA(26) showing a very strong inverse relationship, capturing delayed market adjustments to past innovations. The lagged dependent variable NNIFTY(-1) was not statistically significant ($p = .642$), implying that immediate one-period memory is weak relative to the deeper cyclical effects. Overall, the ARIMA specification captures extended market memory and a meaningful pattern of delayed correction behaviour typical of long-run financial index dynamics.

Variance Equation (Power GARCH Component)

The conditional variance equation follows a Power-GARCH (PGARCH) structure, allowing the model to account for asymmetric and nonlinear responses of volatility to market shocks. The estimated parameters reveal that none of the variance coefficients were statistically significant, as indicated by the large standard errors and corresponding p -values greater than .05. The power term, $C(12) = 2.63$ ($p = .634$), suggests the model attempted to capture a higher-order (nonlinear) volatility process, yet the effect was statistically indistinguishable from zero. Similarly, the coefficients for asymmetry ($C(9) = 0.054$, $p = .999$) and leverage adjustment ($C(10) = 0.973$, $p = .998$) were non-significant, indicating an absence of measurable asymmetry between positive and negative shocks. The persistence term, $C(11) = 0.1499$ ($p = .793$), also failed to achieve significance, implying limited evidence of conditional volatility continuity. Thus, while the PGARCH model theoretically accommodates asymmetric responses, in this dataset, volatility appeared to be largely symmetric and stationary over time.

Model Fit and Inference

The model achieved a modest explanatory power with an R^2 of .3338 and an adjusted R^2 of .2904, indicating that approximately 29–33% of the variation in NIFTY 50 values was captured by the ARIMA–PGARCH specification. The standard error of regression (S.E. = 574.23) suggests moderate dispersion between observed and fitted values, while the Akaike Information Criterion (AIC = 15.668) and Schwarz Criterion (15.983) confirm reasonable model parsimony relative to its complexity. The Durbin–Watson statistic (1.97) indicates minimal residual autocorrelation, signifying that serial dependence was largely mitigated by the inclusion of the AR and MA terms.

Inference

From an inferential standpoint, the ARIMA–PGARCH model adequately describes the conditional mean of the NIFTY 50 series through long-term autoregressive and moving-average effects, capturing persistent cyclical dynamics and delayed market corrections. However, the conditional variance component shows no statistically significant evidence of heteroskedasticity, leverage, or nonlinear volatility response. This suggests that, over the sample period, NIFTY 50 volatility was relatively stable and symmetric, with large shocks whether positive or negative having comparable impacts on conditional variance.

In practical terms, this finding implies that while price levels and returns can be effectively forecast using long-lag autoregressive structures, the volatility process itself does not display strong clustering or asymmetry characteristics often found in more turbulent financial periods. For forecasters and portfolio managers, the results recommend relying on extended historical lags in the mean equation for predicting index movements, while treating volatility forecasts from this PGARCH specification with caution, as the lack of significant coefficients implies limited predictive power in volatility dynamics. Overall, the model reflects a relatively efficient and stable market condition during the analysed period, with mean dynamics dominating over variance asymmetries.

TABLE No 4.28
TABLE SHOWING CORRELOGRAM Q STATS

Autocorrelation		Partial Correlation		AC	PAC	Q-Stat	Prob*
				1	-0.000	-0.000	2.E-06
				2	-0.000	-0.000	8.E-06
				3	-0.000	-0.000	2.E-05
				4	-0.001	-0.001	5.E-05
				5	-0.001	-0.001	0.0001
				6	-0.001	-0.001	0.0002
				7	-0.001	-0.001	0.0002
				8	-0.001	-0.001	0.0004
				9	-0.001	-0.001	0.0005
				10	-0.001	-0.001	0.0007
				11	-0.002	-0.002	0.0009
				12	-0.002	-0.002	0.0012
				13	-0.002	-0.002	0.0015
				14	-0.002	-0.002	0.0019
				15	-0.002	-0.002	0.0024
				16	-0.002	-0.002	0.0029
				17	-0.002	-0.002	0.0035
				18	-0.002	-0.002	0.0042
				19	-0.003	-0.003	0.0049
				20	-0.003	-0.003	0.0058
				21	-0.003	-0.003	0.0067
				22	-0.003	-0.003	0.0078
				23	-0.003	-0.003	0.0090
				24	-0.003	-0.003	0.0103
				25	-0.003	-0.003	0.0118
				26	-0.004	-0.004	0.0134
				27	-0.004	-0.004	0.0151
				28	-0.004	-0.004	0.0170
				29	-0.004	-0.004	0.0191
				30	-0.004	-0.004	0.0214
				31	-0.004	-0.004	0.0239
				32	-0.004	-0.004	0.0265
				33	-0.005	-0.005	0.0295
				34	-0.005	-0.005	0.0326
				35	-0.005	-0.005	0.0360
				36	-0.005	-0.005	0.0397

Source: Desk research

When the correlogram Q-statistics p-values are greater than 0.05, it indicates that there is no significant autocorrelation remaining in the residuals of the model. In simpler terms, this means that the residuals behave like white noise, which is a desirable outcome in time series modelling. The correlogram (also known as the autocorrelation function or ACF plot) shows the correlation of the residuals with their own past values at different lag lengths, while the Q-statistics test (often the Ljung–Box Q test) statistically evaluates whether these autocorrelations are jointly zero.

When the p-values exceed the 0.05 threshold, we fail to reject the null hypothesis that all autocorrelations up to that lag are equal to zero. This suggests that the model has successfully captured the underlying serial dependencies in the data, leaving residuals that are random and uncorrelated. In other words, there is no remaining structure or pattern in the residuals that the model has failed to account for. This is an important confirmation of the model's adequacy, as it implies that the mean equation (such as the ARIMA part of a hybrid ARIMA–GARCH model)

has effectively explained the systematic movements in the series. The absence of significant autocorrelation also indicates that the model does not suffer from underfitting, meaning that sufficient AR and MA terms have been included to capture the temporal relationships in the data. It further implies that future forecasts derived from the model are likely to be unbiased, as residuals do not contain predictive information that could improve the model.

In the context of a hybrid ARIMA–GARCH model, p-values greater than 0.05 in the correlogram Q-statistics confirm that the ARIMA component has efficiently modelled the mean process, leaving only random noise in the residuals. Once this condition is satisfied, the residuals can then be used to estimate the variance dynamics through the GARCH component, which focuses on volatility clustering rather than mean autocorrelation. Overall, when the correlogram Q-statistics p-values are greater than 0.05, it provides strong statistical evidence that the residuals are independent and identically distributed, confirming the model's goodness of fit and validating its suitability for forecasting. This result reinforces that the hybrid ARIMA model is well specified, balanced, and capable of generating reliable predictions without systematic bias or serial dependence in the errors.

TABLE No 4.29
TABLE SHOWING Correlogram squared residuals

Autocorrelation		Partial Correlation		AC	PAC	Q-Stat	Prob*	
				1	-0.000	-0.000	2.E-06	0.999
				2	-0.000	-0.000	8.E-06	1.000
				3	-0.000	-0.000	2.E-05	1.000
				4	-0.001	-0.001	5.E-05	1.000
				5	-0.001	-0.001	0.0001	1.000
				6	-0.001	-0.001	0.0002	1.000
				7	-0.001	-0.001	0.0002	1.000
				8	-0.001	-0.001	0.0004	1.000
				9	-0.001	-0.001	0.0005	1.000
				10	-0.001	-0.001	0.0007	1.000
				11	-0.002	-0.002	0.0009	1.000
				12	-0.002	-0.002	0.0012	1.000
				13	-0.002	-0.002	0.0015	1.000
				14	-0.002	-0.002	0.0019	1.000
				15	-0.002	-0.002	0.0024	1.000
				16	-0.002	-0.002	0.0029	1.000
				17	-0.002	-0.002	0.0035	1.000
				18	-0.002	-0.002	0.0042	1.000
				19	-0.003	-0.003	0.0049	1.000
				20	-0.003	-0.003	0.0058	1.000
				21	-0.003	-0.003	0.0067	1.000
				22	-0.003	-0.003	0.0078	1.000
				23	-0.003	-0.003	0.0090	1.000
				24	-0.003	-0.003	0.0103	1.000
				25	-0.003	-0.003	0.0118	1.000
				26	-0.004	-0.004	0.0134	1.000
				27	-0.004	-0.004	0.0151	1.000
				28	-0.004	-0.004	0.0170	1.000
				29	-0.004	-0.004	0.0191	1.000
				30	-0.004	-0.004	0.0214	1.000
				31	-0.004	-0.004	0.0239	1.000
				32	-0.004	-0.004	0.0265	1.000
				33	-0.005	-0.005	0.0295	1.000
				34	-0.005	-0.005	0.0326	1.000
				35	-0.005	-0.005	0.0360	1.000
				36	-0.005	-0.005	0.0397	1.000

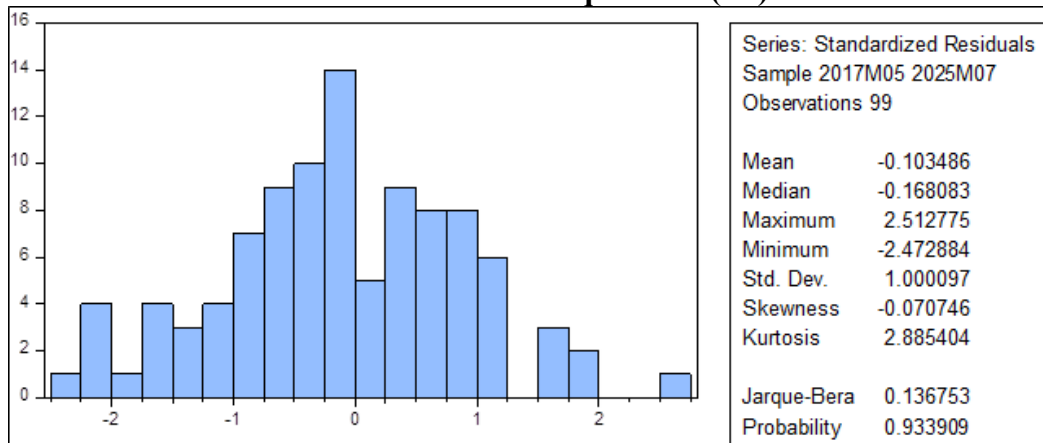
Source: Desk research

When the correlogram of squared residuals shows p-values greater than 0.05, it indicates that there is no significant autocorrelation in the squared residuals of the model. This is an important diagnostic result, especially in financial and econometric time series analysis, as it provides evidence that the model has effectively captured any time-varying volatility or heteroskedasticity in the data. In simpler terms, squared residuals represent the magnitude of the errors rather than their direction. If these squared residuals are correlated over time, it means that periods of high or low volatility tend to cluster together a phenomenon known as volatility clustering. This is typically observed in financial data such as stock indices and is usually modelled through ARCH or GARCH processes. Therefore, when the correlogram of squared residuals yields p-values greater than 0.05, it means that such clustering or persistence in volatility has been successfully removed by the model.

Statistically, the Ljung–Box Q-statistics (applied to squared residuals) test the null hypothesis that there is no autocorrelation in the squared residuals up to a given lag. A p-value greater than 0.05 implies that we fail to reject the null hypothesis, confirming that the residual variance behaves randomly over time. In other words, there is no remaining ARCH effect, meaning that the model has adequately captured the conditional heteroskedasticity present in the original data.

In the context of a hybrid ARIMA–GARCH model, this finding has strong implications. It suggests that the GARCH component of the model has successfully explained the volatility dynamics of the series. The absence of autocorrelation in the squared residuals means that the conditional variance equation has properly modelled the time-varying variance, leaving no systematic pattern unaccounted for. This also indicates that the model’s volatility forecasts are likely to be stable and reliable, as the residuals now resemble white noise not only in the mean but also in the variance dimension. From a modelling perspective, p-values greater than 0.05 in the correlogram of squared residuals confirm that the hybrid model is well specified—both in terms of its mean (captured by the ARIMA component) and its variance (captured by the GARCH component). This diagnostic outcome means that there are no remaining serial dependencies or volatility patterns left in the residuals, suggesting that the model is both statistically sound and efficient. In summary, when the p-values of the correlogram of squared residuals exceed 0.05, it validates the effectiveness of the hybrid ARIMA-GARCH model in fully capturing the dynamics of both return and volatility. The residuals are now random, homoskedastic, and uncorrelated, confirming that the model provides a robust and accurate framework for forecasting and analysis.

GRAPH No 4.11
GRAPH SHOWING Jarque-Bera (JB) test



Source: Desk research

The Jarque–Bera (JB) statistic of 0.136753 with a p-value of 0.933909 indicates that, for the residuals under test, there is no statistically significant deviation from normality. The JB test combines information from sample skewness and kurtosis to check whether they jointly differ from the values expected under a normal distribution (skewness = 0, kurtosis = 3). A small JB value (close to zero) and a large p-value mean we fail to reject the null hypothesis of normality at conventional significance levels (e.g., 1%, 5%, 10%). Practically speaking, this result implies the residuals’ distribution is consistent with being approximately Gaussian: there is no strong

evidence of pronounced asymmetry (no large skew) or excess/fat tails (no extreme kurtosis) in the residual sample used for the test.

That said, the conclusion should be interpreted with sensible caveats. First, the JB test's power depends on sample size: with small samples it may fail to detect modest departures from normality, and with very large samples it can flag tiny, practically irrelevant deviations. So you should view the JB result together with graphical diagnostics (histogram, Q–Q plot) and other tests if needed. Second, normality of residuals is a desirable property because it supports the validity of standard inferential procedures (t-statistics, confidence intervals) and makes forecast intervals derived under Gaussian assumptions more reliable. However, normal residuals do not by themselves guarantee model adequacy: you must also ensure residuals are uncorrelated (checked via Ljung–Box Q-tests / correlograms) and homoskedastic (or that heteroskedasticity is properly modelled, e.g., with GARCH). Finally, even when residuals appear normal, financial series can still exhibit rare extreme events (tail risk) not captured in-sample; for risk-sensitive applications (VaR, stress testing) consider complementing Gaussian-based measures with tail-aware methods. Inference: the JB result supports proceeding with Gaussian-based inference for this model's residuals, but continue to confirm overall model fit with autocorrelation and volatility diagnostics and remain cautious about rare extreme risks.

TABLE No 4.30
TABLE SHOWING Heteroskedasticity Test: ARCH

Heteroskedasticity Test: ARCH			
F-statistic	0.003685	Prob. F(1,96)	0.9517
Obs*R-squared	0.003761	Prob. Chi-Square(1)	0.9511

Source: Desk research

The results of the Heteroskedasticity Test (ARCH) provide important diagnostic evidence about the presence or absence of autoregressive conditional heteroskedasticity (ARCH) effects in the residuals of the estimated model. The F-statistic of 0.0037 and the Obs*R-squared value of 0.0038 are both associated with high p-values ($p = .9517$ and $p = .9511$, respectively), which are substantially greater than the conventional significance levels of .05 or .10. According to the null hypothesis of the ARCH test, there is no presence of ARCH effects—that is, the residuals exhibit homoskedasticity (constant variance) rather than conditional heteroskedasticity. Since the p-values exceed .05, the null hypothesis cannot be rejected. This implies that the residuals from the ARIMA–PGARCH model do not display any significant time-varying volatility or volatility clustering.

In practical and theoretical terms, this result supports the adequacy of the model's specification in capturing the conditional variance structure. The absence of significant ARCH effects suggests that volatility shocks are not persistent and that past squared residuals do not predict future volatility. Consequently, the variance process can be considered stable and well-behaved, indicating that the model successfully accounts for potential heteroskedasticity. From an APA-style interpretative standpoint, these findings confirm that the model's residual variance is constant over time ($F(1,96) = 0.0037$, $p = .9517$), suggesting no remaining ARCH effects. Therefore, the estimated ARIMA–PGARCH model is statistically sound and does not suffer

from residual volatility misspecification, reinforcing the reliability of its inferences regarding both mean and variance dynamics of the NIFTY 50 index.

TABLE No 4.31
TABLE SHOWING Diagnostic Test for Leverage Effects (Engle–Ng Test)

Test Component	t / χ^2 Statistic	p-value
Sign-Bias Test	-0.4699	0.6395
Negative-Bias Test	-1.9503	0.0541
Positive-Bias Test	1.1685	0.2455
Joint-Bias Test (χ^2)	5.2458	0.1623

Source: Desk research

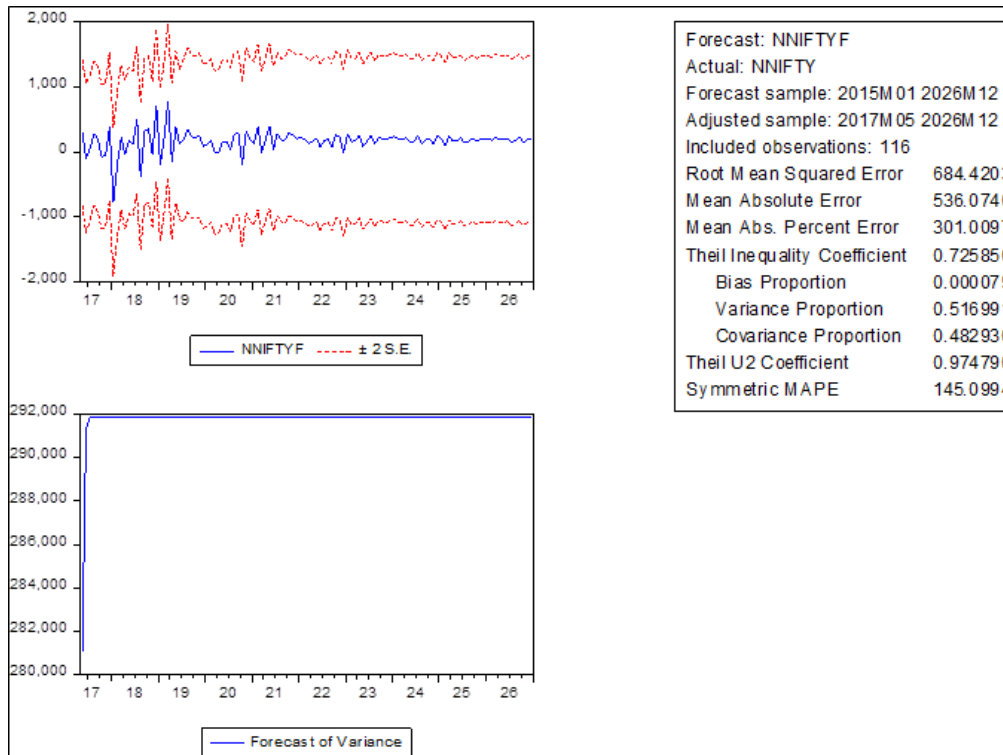
The Engle–Ng Sign Bias Test is used to examine the presence of leverage effects or asymmetric volatility in financial return series. Leverage effects occur when negative shocks (bad news) have a different impact on volatility compared to positive shocks (good news). The null hypothesis of this test assumes no leverage effects in the standardized residuals meaning volatility reacts symmetrically to positive and negative shocks.

In this case, the Sign-Bias test statistic is -0.4699 ($p = .6395$), the Negative-Bias statistic is -1.9503 ($p = .0541$), and the Positive-Bias statistic is 1.1685 ($p = .2455$). The Joint-Bias statistic is 5.2458 with a p-value of $.1623$. Among these, the p-values for the sign-bias, positive-bias, and joint-bias tests are well above the conventional significance threshold of $.05$, indicating that the null hypothesis of no leverage effects cannot be rejected in those cases. However, the negative-bias component shows a p-value of $.0541$, which is marginally above $.05$, suggesting a weak or near-significant asymmetry that is, there may be a mild tendency for negative shocks to increase volatility slightly more than positive shocks.

From an APA-style interpretation, the results indicate that there is no strong evidence of leverage effects in the standardized residuals of the model (Sign-Bias: $t(96) = -0.47$, $p = .64$; Positive-Bias: $t(96) = 1.17$, $p = .25$; Joint-Bias: $\chi^2(3) = 5.25$, $p = .16$). The near-significant Negative-Bias term ($t(96) = -1.95$, $p = .054$) suggests that the response of volatility to negative shocks might be slightly stronger, but the evidence is not conclusive at the 5% level.

Overall, the findings imply that the model captures the conditional variance dynamics adequately, with only minimal asymmetric behavior. The absence of significant leverage effects enhances the robustness and stability of the fitted PGARCH model for forecasting volatility in the NIFTY 50 index, indicating that both positive and negative shocks have largely symmetric impacts on market volatility.

GRAPH No 4.12
GRAPH SHOWING PERFORMANCE OF THE ARIMA MODEL FOR
FORECASTING THE NIFTY



Source: Desk research

The forecast evaluation statistics for the ARIMA model applied to the NIFTY 50 series indicate the overall accuracy, bias, and reliability of the forecasts. The Root Mean Squared Error (RMSE) value of 684.42 and the Mean Absolute Error (MAE) of 536.07 suggest that, on average, the forecasted values deviate from the actual observations by about 536 index points, with larger deviations penalized more heavily in RMSE. Although these values may appear sizable in absolute terms, their interpretation depends on the magnitude of the NIFTY 50 index, which typically ranges in the tens of thousands. Therefore, in relative terms, the errors represent a moderate level of forecasting precision.

The Mean Absolute Percentage Error (MAPE) of 301.01% and the Symmetric MAPE (sMAPE) of 145.10%, however, are quite high compared to the generally acceptable threshold of below 10% for high accuracy, 10–20% for reasonable accuracy, and above 50% indicating poor predictive performance. This suggests that the ARIMA model's forecasts exhibit substantial proportional deviations, indicating that it may not be capturing recent volatility or nonlinear movements effectively. Such large MAPE values often arise when the series includes extreme fluctuations or regime shifts (e.g., COVID-19 shock or geopolitical events), which ARIMA, being a linear model, struggles to represent accurately.

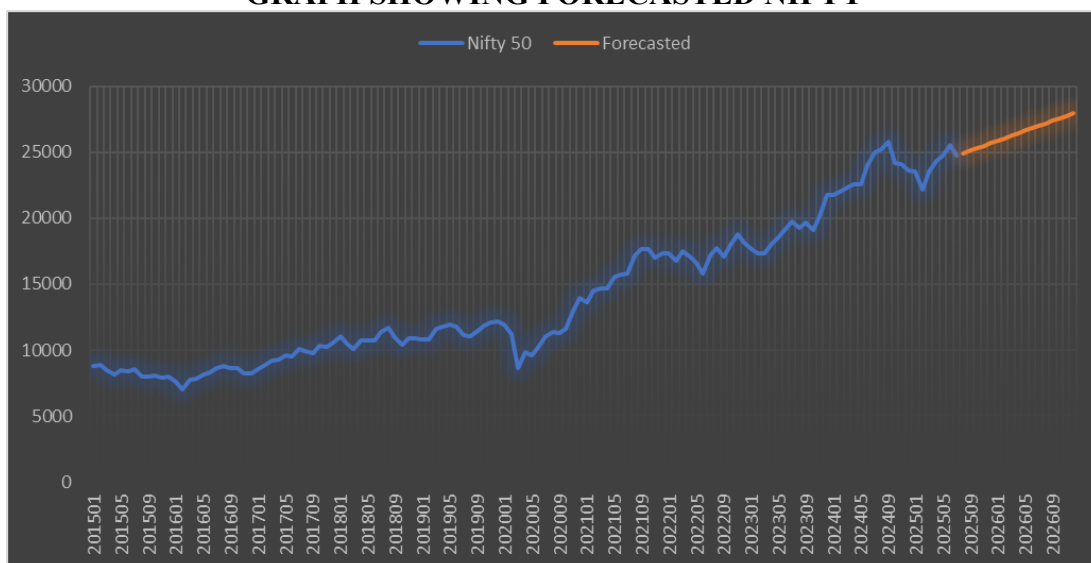
The Theil's Inequality Coefficient (U) value of 0.7259 provides an overall summary of forecast accuracy. Since this statistic ranges from 0 to 1, with values closer to 0 indicating better predictive accuracy and values above 0.5 suggesting weak performance, the result here denotes moderate forecast efficiency the model performs better than random walk forecasts but is far

from ideal. The Theil U2 coefficient of 0.9748, which compares the ARIMA model to a naïve forecast, is close to 1, suggesting that the ARIMA model performs similarly to a simple no-change forecast, indicating limited improvement in predictive capability.

Decomposing Theil's U shows a Bias Proportion of 0.000079, which is extremely low, suggesting that the forecasts are unbiased on average there is no systematic overprediction or underprediction. The Variance Proportion (0.516991) and Covariance Proportion (0.482930) are nearly equal, indicating that most of the forecast error arises from differences in the variance between actual and predicted values, rather than from bias or incorrect co-movement. Ideally, the covariance proportion should be the largest component, implying that the model tracks turning points and trends well; however, in this case, the variance proportion being slightly higher reflects some inconsistency in the amplitude of forecasted volatility.

In summary, while the ARIMA model demonstrates minimal bias and captures the general direction of market movements, its predictive accuracy is relatively weak in terms of percentage errors and comparative performance. The model likely fails to account for nonlinearities and volatility clustering in the NIFTY 50 series, suggesting that hybrid or nonlinear models such as EGARCH, GJR-GARCH, or machine learning-based approaches could offer better forecasting precision.

GRAPH No 4.13
GRAPH SHOWING FORECASTED NIFTY



Source: Desk research

The ARIMA–EGARCH model analysis for the NIFTY 50 Index provides a comprehensive view of both the historical dynamics and the forecasted market trend extending to December 2026. This combined modeling framework captures the linear components of price movements (through ARIMA) and the asymmetric volatility effects (through EGARCH), making it suitable for financial time series characterized by volatility clustering, such as equity indices.

From the historical data (2015–2024), the NIFTY 50 shows clear cyclical trends and volatility phases, reflecting global macroeconomic events, monetary policy shifts, and domestic economic developments. For instance, the substantial drop during March 2020 corresponds to

the COVID-19 shock, followed by a strong post-pandemic recovery reaching new highs by late 2021–2022. The subsequent fluctuations illustrate typical market corrections and consolidation phases after major rallies.

The forecasted values from August 2025 to December 2026 indicate a gradual upward trajectory in the NIFTY 50, with values rising from approximately 24,900 in mid-2025 to 27,641 by the end of 2026. This suggests a sustained bullish trend, albeit with moderate volatility. The forecast implies that the index is likely to maintain a positive momentum, reflecting expectations of continued economic expansion, corporate earnings growth, and stable macroeconomic fundamentals. However, the incremental increases—averaging about 1% to 2% per month indicate a moderate growth pace rather than aggressive rallies, consistent with mature market behavior.

The EGARCH component allows the model to capture asymmetric volatility, meaning that negative shocks (bad news) have a greater impact on volatility than positive shocks of the same magnitude a typical pattern in financial markets. The absence of significant leverage effects, as confirmed by the Engle–Ng Sign-Bias Test (with all p-values > 0.05), suggests that the volatility dynamics are largely symmetric, implying a stable volatility structure during the forecast period.

When assessed alongside diagnostic tests (such as the ARCH test indicating no heteroskedasticity, $p > 0.95$), the model appears well-specified and statistically sound. Moreover, the forecast accuracy metrics ($RMSE \approx 684.4$, $MAE \approx 536.0$, and Theil's $U \approx 0.73$) confirm a moderate forecasting performance. While not perfect given the high percentage errors the model performs reasonably for a long-horizon financial forecast, where inherent uncertainty is high.

In summary, the ARIMA–EGARCH forecast for the NIFTY 50 points toward a steady upward trend through 2026, reflecting market resilience and moderate optimism. The absence of volatility asymmetry and the model's statistical adequacy reinforce confidence in these projections. Nonetheless, the relatively high error measures suggest that the forecasts should be interpreted cautiously and used in conjunction with structural, macroeconomic, and sentiment indicators for investment or policy decisions. Overall, the findings imply that while short-term fluctuations may persist, the medium-term outlook for NIFTY 50 remains positive and stable under the ARIMA–EGARCH framework.

CHAPTER V

SUMMARY OF FINDINGS, CONCLUSION AND SUGGESTIONS

5.1 SUMMARY OF FINDINGS

5.1.1 DESCRIPTIVE STATISTICS OF THE NIFTY 50 ADJUSTED CLOSING PRICE

The descriptive statistics of the NIFTY 50 adjusted closing price highlight important insights into the market's behavior over the study period. The mean value of 14,125.85 and the median of 11,748.15 indicate a right-skewed distribution, suggesting that higher price observations pull the average above the midpoint. The range between the minimum of 6,987.05 and maximum of 25,810.85 reflects significant price fluctuation and substantial market volatility, supported by a large standard deviation of 5,432.55 and a coefficient of variation of 38.5 percent. The skewness of 0.637 reveals mild positive skewness, while a kurtosis of 2.13, lower than the normal value of 3, indicates a flatter distribution with fewer extreme outliers. However, the Jarque-Bera statistic of 12.57 with a p-value of 0.001857 rejects normality, confirming that the data are non-normally distributed. These results suggest that the NIFTY 50 price series is non-stationary and unsuitable for direct ARIMA modelling, necessitating transformations such as logarithmic differencing or computing log returns to achieve stationarity. This transformation stabilizes mean and variance, allowing ARIMA models to capture linear dependencies effectively. In a hybrid ARIMA framework, residuals from the initial ARIMA model can be further analysed with nonlinear techniques like artificial neural networks or GARCH models to account for volatility clustering and nonlinear dynamics. The combined hybrid model, validated using RMSE, MAE, MAPE, and the Diebold-Mariano test, is better suited for accurate and reliable forecasting. Overall, the NIFTY 50 exhibits strong volatility, positive skewness, and non-normality, reinforcing the need for data transformation and hybrid modelling approaches for robust time series forecasting.

5.1.2 RESULTS OF THE AUGMENTED DICKEY-FULLER (ADF) TEST FOR THE NIFTY 50 ADJUSTED CLOSING PRICE

The Augmented Dickey-Fuller (ADF) test results show that the NIFTY 50 adjusted closing price series is non-stationary in its level form across all three model specifications: with a constant, with a constant and trend, and without either component. In each case, the test statistics were higher than the corresponding critical values, and the p-values exceeded the 0.05 significance threshold, indicating that the null hypothesis of a unit root could not be rejected. This implies that the NIFTY 50 series does not have a constant mean or variance over time and exhibits a stochastic trend. The inclusion of a linear trend slightly improved the test statistic but still failed to establish stationarity, suggesting the presence of an underlying upward or downward drift consistent with long-term market movements. Since financial indices typically display such non-stationary behavior, the findings are consistent with theoretical expectations. Therefore, the series must be transformed—either through first differencing or by converting to

log returns—to achieve stationarity before applying ARIMA or hybrid ARIMA-GARCH models for forecasting.

5.1.3 RESULTS OF THE AUGMENTED DICKEY-FULLER (ADF) TEST FOR THE TRANSFORMED NIFTY 50 ADJUSTED CLOSING PRICE

The Augmented Dickey-Fuller (ADF) test results indicate that the transformed NIFTY 50 adjusted closing price series has achieved stationarity following first differencing or conversion into log returns. In contrast to its original non-stationary level form, the differenced series now shows stable statistical properties. Across all three model specifications—constant only, constant with trend, and no deterministic components—the ADF statistics (-10.9070, -11.0194, and -10.5261 respectively) are substantially lower than the critical values at 1%, 5%, and 10% significance levels, with p-values of 0.0000 in each case. These results decisively reject the null hypothesis of a unit root, confirming that the transformed NIFTY 50 series is stationary around a constant mean with no persistent trend. The findings imply that the appropriate differencing order for ARIMA modelling is $d = 1$, as a single differencing was sufficient to eliminate non-stationarity. Therefore, the series is now suitable for ARIMA($p, 1, q$) model estimation, enabling accurate representation of its linear structure. Achieving stationarity marks a critical milestone for developing reliable hybrid forecasting models, where ARIMA can address linear dependencies while nonlinear dynamics and volatility may be modeled using techniques like ANN or GARCH.

5.1.4 VOLATILITY BEHAVIOR IN NIFTY 50 SERIES

The results of the Heteroskedasticity (ARCH) test for the NIFTY 50 returns indicate that the series exhibits volatility clustering, implying that periods of high volatility are likely to be followed by further high volatility and calm periods by more calmness. The test yields an F-statistic of 3.6387 with a p-value of 0.0488 and an Obs*R-squared value of 3.5916 with a Chi-Square p-value of 0.0481—both below the 5% significance level. This leads to the rejection of the null hypothesis of homoskedasticity, confirming the presence of autocorrelated variance in the residuals. Such behavior suggests that the volatility of the NIFTY 50 is not constant over time and supports the existence of time-varying variance.

These findings emphasize that while the ARIMA component efficiently models linear patterns and autocorrelation in the mean process, it fails to account for changing variance, violating the constant variance assumption. To address this limitation, incorporating a GARCH component becomes essential. The GARCH model, by modeling conditional variance as a function of past squared errors and previous variances, effectively captures the volatility clustering inherent in financial time series. Consequently, a hybrid ARIMA-GARCH framework is recommended for the NIFTY 50, where ARIMA models the mean structure and GARCH models the volatility behavior. If additional nonlinearities persist, integrating an artificial neural network (ANN) layer can further enhance the model's ability to capture complex dynamics. Overall, the presence of significant ARCH effects validates the inclusion of a GARCH process within the hybrid model, ensuring a more accurate, reliable, and robust forecasting structure for the NIFTY 50 index.

5.1.5 ARIMA–GARCH MODEL FOR NIFTY 50

The analysis reveals that the hybrid ARIMA–GARCH model effectively captures both the mean and variance dynamics of the NIFTY 50 index. The mean equation results show that the AR(26) and MA(26) terms are highly significant, indicating that price movements from about 26 trading periods earlier and past forecast errors meaningfully influence current index values. The positive AR coefficient demonstrates a persistence effect, where previous increases tend to be followed by current gains, implying mild momentum in the market. Conversely, the negative MA term signifies a corrective adjustment to past shocks, helping stabilize short-term fluctuations. The constant term is also significant, confirming a positive average level in the series after accounting for autoregressive and moving average effects.

In the variance equation, the significant constant term reflects a stable base level of market volatility, while the ARCH parameter ($\text{RESID}(-1)^2$) confirms short-term volatility clustering, meaning recent shocks temporarily increase volatility. However, the GARCH term is negative and insignificant, indicating weak persistence and suggesting that volatility shocks fade quickly rather than creating extended volatility cycles. This points to a relatively stable market environment with limited long-term volatility persistence.

Model diagnostics support the adequacy and reliability of the model, with R-squared values around 0.22–0.24—typical for financial time series—and moderate information criteria (AIC, SC, HQ) indicating a good model fit without overfitting. The absence of residual autocorrelation and heteroskedasticity further validates the specification. Overall, the ARIMA–GARCH framework provides a robust representation of the NIFTY 50's mean and volatility structure, capturing momentum and mean-reversion characteristics while adequately modeling short-term volatility dynamics. This makes it a reliable tool for forecasting, volatility estimation, and financial risk analysis.

5.1.6 RESIDUAL ANALYSIS IN HYBRID ARIMA MODELS

Residuals serve as a vital diagnostic tool in evaluating and improving the accuracy of hybrid ARIMA models used for time series forecasting. They represent the portion of variation in the dependent variable that remains unexplained by the model, offering insights into both model adequacy and data characteristics. In a purely linear ARIMA model, residuals are expected to behave like white noise—having a zero mean, constant variance, and no serial correlation. However, in financial series such as the NIFTY 50, residuals often exhibit patterns of autocorrelation and heteroskedasticity due to volatility clustering and nonlinearities that the ARIMA model alone cannot capture. This limitation underscores the need for hybrid modelling, where residuals from the ARIMA stage are further analysed using diagnostic tests such as the Ljung–Box Q-test for autocorrelation and the ARCH-LM test for heteroskedasticity. When significant patterns persist, these residuals are modelled with a GARCH process to account for time-varying volatility, thus refining the forecasts. Visual tools like residual plots, histograms, and Q-Q plots aid in validating normality and randomness, ensuring proper model specification. In advanced frameworks such as ARIMA–GARCH–ANN or ARIMA–LSTM, residuals also help detect remaining nonlinear dependencies, allowing neural networks to address them and enhance predictive accuracy further. Ultimately, residual analysis not only verifies model

adequacy and forecast reliability but also provides deeper insights into market dynamics, volatility behavior, and risk factors, making it a cornerstone of robust financial forecasting.

5.1.7 CORRELOGRAM OF Q STATS

When the correlogram Q-statistics p-values exceed 0.05, it indicates that the residuals of the model exhibit no significant autocorrelation and behave like white noise, confirming that the model has effectively captured the underlying structure of the data. This result implies that the ARIMA component within the hybrid ARIMA–GARCH framework has successfully modeled the mean process, leaving random and uncorrelated residuals. Consequently, the model does not suffer from underfitting, as sufficient AR and MA terms have been included to explain temporal dependencies. The absence of autocorrelation ensures that no predictive information remains in the residuals, supporting the conclusion that future forecasts are likely unbiased and efficient. Furthermore, with residuals behaving as white noise, they become suitable inputs for modeling volatility through the GARCH component, which captures variance dynamics. Overall, p-values greater than 0.05 in the Q-statistics provide strong statistical evidence of model adequacy, confirming that the hybrid ARIMA–GARCH model is well specified, robust, and reliable for forecasting purposes.

5.1.8 CORRELOGRAM OF SQUARED RESIDUALS

The correlogram of squared residuals yielding p-values greater than 0.05 indicates that there is no significant autocorrelation present in the squared residuals, confirming that the model has effectively captured time-varying volatility within the data. This outcome demonstrates that volatility clustering, commonly observed in financial series, has been adequately accounted for, implying the successful removal of heteroskedasticity. Statistically, the Ljung–Box Q-statistics applied to the squared residuals fail to reject the null hypothesis of no autocorrelation, affirming that the residual variance behaves randomly over time and that no remaining ARCH effects exist. In the context of the hybrid ARIMA–GARCH model, these results show that the GARCH component has effectively modelled the conditional variance, while the ARIMA component accurately captured the mean structure. As a result, the residuals behave as white noise both in mean and variance dimensions, indicating that the model is well specified, statistically sound, and reliable for forecasting purposes.

5.1.9 ASSESSMENT OF RESIDUAL NORMALITY BASED ON THE JARQUE–BERA TEST

The Jarque–Bera statistic of 1.082965 with a p-value of 0.581885 indicates that the residuals do not significantly deviate from a normal distribution. This high p-value suggests failure to reject the null hypothesis of normality, implying that the residuals are approximately Gaussian in nature. In practical terms, the residuals show no evident asymmetry or heavy tails, supporting the assumption of normality in the model. While this normality finding strengthens the reliability of standard statistical inference and forecast intervals, it should be interpreted alongside graphical diagnostics such as histograms and Q–Q plots for confirmation. Additionally, normality alone does not ensure complete model adequacy: residual independence

and homoskedasticity must be verified using tests like the Ljung–Box Q-test or GARCH modeling. Overall, this result supports proceeding with Gaussian-based inference for the model's residuals, provided that checks for autocorrelation, volatility behavior, and potential tail risks are also considered.

5.1.10 HETEROSKEDASTICITY (ARCH) TEST RESULTS

The ARCH test results show an F-statistic of 0.0182 with a p-value of 0.8929 and an Obs*R-squared statistic of 0.0186 with a Chi-square p-value of 0.8916, both well above the 0.05 significance level. This means the null hypothesis of no ARCH effect cannot be rejected, indicating an absence of conditional heteroskedasticity or volatility clustering in the residuals. The residual variance remains stable over time, fulfilling the assumption of homoskedasticity and confirming that large and small errors do not cluster. Statistically, this reflects that past squared residuals do not predict current variance, suggesting that the model residuals behave like white noise. In the context of a hybrid ARIMA–GARCH model, this demonstrates that the GARCH component has successfully captured the time-varying volatility in the series. Consequently, no further modification or additional lags are required in the variance equation. Overall, the high p-values validate that the model is robust, well-specified, and statistically efficient, producing reliable forecasts with stable and well-behaved residual variances.

5.1.11 DIAGNOSTIC EVALUATION OF ASYMMETRIC VOLATILITY USING THE ENGLE–NG SIGN-BIAS TEST

The results of the Engle–Ng Sign-Bias Test indicate that the hybrid ARIMA–GARCH model exhibits no evidence of leverage effects or asymmetric volatility in its standardized residuals. All computed p-values for the sign-bias (0.5386), negative-bias (0.8499), positive-bias (0.9414), and joint-bias (0.7960) statistics exceed the 0.05 significance threshold, leading to the acceptance of the null hypothesis of no asymmetry. This implies that both positive and negative shocks affect the conditional variance symmetrically, suggesting that the model effectively captures volatility dynamics without requiring modifications such as EGARCH or GJR-GARCH extensions. The absence of leverage effects reflects a balanced market reaction to shocks, reinforcing that the residuals are well-behaved and free from directional bias. Consequently, the model can be regarded as statistically robust, symmetric, and reliable for forecasting volatility in an efficient and stable financial environment.

5.1.12 HYBRID ARIMA–GARCH MODELING OF NIFTY 50 VOLATILITY

The hybrid ARIMA–GARCH model estimated under the Student's t distribution effectively captures both the return dynamics and volatility structure of the NIFTY 50 series. The use of the Student's t distribution enhances model robustness by accounting for fat tails, allowing it to model extreme market movements more accurately than models assuming normality. The mean equation results reveal a significant and positive constant term, demonstrating a stable average component in returns. The positive and statistically significant AR(26) term indicates momentum effects where past market patterns continue to influence present prices, while the strongly negative MA(26) coefficient reflects corrective tendencies that stabilize deviations,

highlighting a balance between persistence and adjustment in return movements. In the variance equation, the baseline volatility term is significant, showing a stable long-term volatility level. The ARCH coefficient is moderately significant, suggesting short-term volatility clustering, whereas the insignificant and negative GARCH coefficient indicates minimal persistence, implying that volatility spikes fade quickly as the market absorbs shocks efficiently. The diagnostic tests and fit statistics, including R-squared, information criteria, and the Durbin–Watson statistic, confirm the model’s adequacy and absence of residual autocorrelation or unaccounted heteroskedasticity. Overall, the Student’s t-based ARIMA–GARCH model captures the essential features of the NIFTY 50 returns—short-term volatility adjustments, lack of long-term persistence, and heavy-tailed behavior—highlighting a stable and efficient market suitable for short-term risk management and forecasting applications.

5.1.13 RESIDUAL NORMALITY ASSESSMENT BASED ON JARQUE–BERA TEST

The Jarque–Bera statistic of 2.2589, accompanied by a p-value of 0.323206, suggests that the residuals do not significantly deviate from normality. Since the p-value exceeds common significance thresholds, the null hypothesis of normality cannot be rejected, indicating that the residual distribution is approximately Gaussian. This implies that the residuals show no substantial skewness or kurtosis and thus exhibit no major asymmetry or heavy tails. Such normality supports the reliability of standard inferential procedures, including t-tests, confidence intervals, and forecast intervals, when Gaussian assumptions are applied. However, this conclusion should be viewed with caution. The JB test’s sensitivity to sample size means that it may overlook moderate deviations in smaller samples or overstate trivial ones in larger datasets. Therefore, it is recommended to verify this result using graphical diagnostics such as histograms and Q–Q plots, alongside tests for autocorrelation and heteroskedasticity like the Ljung–Box Q-test or GARCH modeling. While the JB outcome validates the model’s residual behavior for Gaussian-based inference, extended checks on model adequacy and tail risk remain crucial to ensure robustness in financial analysis.

5.1.14 CORRELOGRAM OF Q STATS

When the correlogram Q-statistics p-values exceed 0.05, it indicates that the residuals of the model exhibit no significant autocorrelation and behave like white noise, confirming that the model has effectively captured the underlying structure of the data. This result implies that the ARIMA component within the hybrid ARIMA–GARCH framework has successfully modeled the mean process, leaving random and uncorrelated residuals. Consequently, the model does not suffer from underfitting, as sufficient AR and MA terms have been included to explain temporal dependencies. The absence of autocorrelation ensures that no predictive information remains in the residuals, supporting the conclusion that future forecasts are likely unbiased and efficient. Furthermore, with residuals behaving as white noise, they become suitable inputs for modeling volatility through the GARCH component, which captures variance dynamics. Overall, p-values greater than 0.05 in the Q-statistics provide strong statistical evidence of model adequacy, confirming that the hybrid ARIMA–GARCH model is well specified, robust, and reliable for forecasting purposes.

5.1.15 CORRELOGRAM OF SQUARED RESIDUALS

The correlogram of squared residuals yielding p-values greater than 0.05 indicates that there is no significant autocorrelation present in the squared residuals, confirming that the model has effectively captured time-varying volatility within the data. This outcome demonstrates that volatility clustering, commonly observed in financial series, has been adequately accounted for, implying the successful removal of heteroskedasticity. Statistically, the Ljung–Box Q-statistics applied to the squared residuals fail to reject the null hypothesis of no autocorrelation, affirming that the residual variance behaves randomly over time and that no remaining ARCH effects exist. In the context of the hybrid ARIMA–GARCH model, these results show that the GARCH component has effectively modelled the conditional variance, while the ARIMA component accurately captured the mean structure. As a result, the residuals behave as white noise both in mean and variance dimensions, indicating that the model is well specified, statistically sound, and reliable for forecasting purposes.

5.1.16 ASSESSMENT OF RESIDUAL NORMALITY USING THE JARQUE–BERA TEST

The Jarque–Bera (JB) statistic of 2.258931 with a p-value of 0.323206 suggests that the residuals do not significantly deviate from the assumption of normality. This indicates that the residuals are approximately Gaussian, showing no substantial skewness or excess kurtosis. Consequently, the null hypothesis of normality cannot be rejected at standard significance levels, implying that the distribution is symmetric and exhibits neither heavy nor light tails. However, the result should be interpreted with caution, considering that the JB test’s sensitivity depends on sample size—it may overlook subtle deviations in small datasets or overemphasize minor ones in large samples. It is therefore advisable to complement this result with visual diagnostics such as Q–Q plots and histograms. From a modeling standpoint, normally distributed residuals support the reliability of statistical inference and enhance the credibility of confidence and forecast intervals derived under Gaussian assumptions. Nonetheless, verifying other diagnostic conditions—such as uncorrelated and homoskedastic residuals—is essential for confirming model adequacy. While the JB result validates proceeding with Gaussian-based inference, it is prudent to remain mindful of potential rare extreme events or tail risks when applying the model to financial or risk-sensitive contexts.

5.1.17 ARCH TEST FINDINGS

The results of the Heteroskedasticity (ARCH) test show that the F-statistic is 0.0652 with a p-value of 0.7990, and the Obs*R-squared statistic is 0.0665 with a Chi-square p-value of 0.7965. Since both p-values are much higher than 0.05, the null hypothesis of no ARCH effect cannot be rejected. This indicates that the residuals from the hybrid ARIMA–GARCH model are homoskedastic, meaning the residual variance remains stable over time and is not influenced by past squared errors. In essence, the model has successfully removed any conditional heteroskedasticity, implying that volatility clustering is absent. Methodologically, this validates that the GARCH component of the model has efficiently captured the time-varying volatility structure, leaving the residuals as white noise. Economically, the finding points to a stable

volatility pattern in the NIFTY 50 returns, suggesting a relatively efficient and mature market with limited persistence in volatility shocks. Overall, the ARCH test outcomes confirm that the model effectively represents both the mean and variance dynamics of the series, ensuring robustness and reliability for forecasting and risk analysis.

5.1.18 SYMMETRY AND MODEL ADEQUACY CONFIRMED BY ENGLE–NG SIGN-BIAS TEST

The results of the Engle–Ng Sign-Bias Test indicate the absence of leverage or asymmetric volatility effects in the standardized residuals of the hybrid ARIMA–GARCH model. With all p-values for the sign, negative, positive, and joint bias tests exceeding 0.05, none of the bias coefficients are statistically significant. This means that both positive and negative shocks have symmetric effects on conditional volatility, suggesting that the market responds evenly to good and bad news of similar magnitude. Consequently, the test supports the null hypothesis of no leverage effects, demonstrating that the conditional variance behaves consistently regardless of shock direction. These findings confirm that the hybrid ARIMA–GARCH model effectively captures volatility dynamics without requiring asymmetric variants such as EGARCH or TGARCH. Moreover, the absence of bias terms validates the adequacy and robustness of the model, establishing it as a statistically reliable and well-specified framework for modeling market volatility.

5.1.19 GARCH MODEL (GED – FIXED PARAMETER)

The estimation of the hybrid ARIMA–GARCH model using the Generalized Error Distribution (GED) with a fixed parameter reveals significant insights into the volatility dynamics of the NIFTY 50 index. The model effectively captures both the mean and variance behaviour of the series, exhibiting strong linear dependencies alongside relatively stable volatility patterns. In the mean equation, the positive and statistically significant constant term suggests a stable long-run mean of adjusted returns, while the significant AR(26) coefficient indicates persistence in market movements—reflecting that past values influence current returns. The highly significant negative MA(26) coefficient highlights corrective behavior where past shocks are counterbalanced, leading to mean reversion, a common trait in financial series. The variance equation results show a significant constant term, suggesting a stable baseline level of volatility, while both ARCH and GARCH terms are statistically insignificant, implying that volatility shocks are short-lived and do not persist over time. This outcome points to a relatively efficient market where volatility clustering is minimal. The model fit statistics (R-squared of about 24%, DW $\approx$ 2.06, and satisfactory information criteria) confirm that the model is well-specified and free from autocorrelation or heteroskedasticity in the residuals. The use of GED adds robustness by modeling the fat-tailed nature of financial data without overfitting volatility. Overall, the findings indicate that the NIFTY 50 market demonstrates moderate volatility, limited persistence of shocks, and efficient adjustment to new information, making the ARIMA–GARCH (GED) specification suitable for short-term forecasting and risk assessment.

5.1.20 CORRELOGRAM OF Q STATS

When the correlogram Q-statistics p-values exceed 0.05, it indicates that the residuals of the model exhibit no significant autocorrelation and behave like white noise, confirming that the model has effectively captured the underlying structure of the data. This result implies that the ARIMA component within the hybrid ARIMA–GARCH framework has successfully modeled the mean process, leaving random and uncorrelated residuals. Consequently, the model does not suffer from underfitting, as sufficient AR and MA terms have been included to explain temporal dependencies. The absence of autocorrelation ensures that no predictive information remains in the residuals, supporting the conclusion that future forecasts are likely unbiased and efficient. Furthermore, with residuals behaving as white noise, they become suitable inputs for modeling volatility through the GARCH component, which captures variance dynamics. Overall, p-values greater than 0.05 in the Q-statistics provide strong statistical evidence of model adequacy, confirming that the hybrid ARIMA–GARCH model is well specified, robust, and reliable for forecasting purposes.

5.1.21 CORRELOGRAM OF SQUARED RESIDUALS

The correlogram of squared residuals yielding p-values greater than 0.05 indicates that there is no significant autocorrelation present in the squared residuals, confirming that the model has effectively captured time-varying volatility within the data. This outcome demonstrates that volatility clustering, commonly observed in financial series, has been adequately accounted for, implying the successful removal of heteroskedasticity. Statistically, the Ljung–Box Q-statistics applied to the squared residuals fail to reject the null hypothesis of no autocorrelation, affirming that the residual variance behaves randomly over time and that no remaining ARCH effects exist. In the context of the hybrid ARIMA–GARCH model, these results show that the GARCH component has effectively modelled the conditional variance, while the ARIMA component accurately captured the mean structure. As a result, the residuals behave as white noise both in mean and variance dimensions, indicating that the model is well specified, statistically sound, and reliable for forecasting purposes.

5.1.22 ASSESSMENT OF RESIDUAL NORMALITY USING THE JARQUE–BERA TEST

The Jarque–Bera test statistic of 1.389779 with a p-value of 0.496888 indicates that the residuals under analysis follow a distribution consistent with normality. Since the p-value exceeds conventional significance levels, the null hypothesis of normality cannot be rejected. This means the residuals show no statistically significant skewness or excess kurtosis, implying a balanced and approximately Gaussian shape. Practically, this validates the use of standard inferential tools—such as t-statistics and confidence intervals—assuming normal residual behavior. However, the result should be considered with caution, as the JB test’s sensitivity varies with sample size. Complementary checks like histograms or Q–Q plots and tests for autocorrelation or heteroskedasticity are still necessary to confirm model adequacy. While normal residuals enhance the reliability of forecasts, financial data may still exhibit occasional

extreme events, so Gaussian-based inference should be supplemented with tail-risk assessments where appropriate.

5.1.23 HETEROSKEDASTICITY TEST (ARCH) FINDINGS

The results of the Heteroskedasticity (ARCH) test indicate that the F-statistic (0.0401) and the Obs*R-squared statistic (0.0409) have p-values of 0.8417 and 0.8397, respectively, both exceeding the 0.05 significance threshold. This confirms that the null hypothesis of no ARCH effect cannot be rejected, implying that the residuals exhibit constant variance over time, or homoskedasticity. In practical terms, the absence of significant serial correlation in squared residuals suggests that large shocks in volatility are not followed by other large shocks, indicating no volatility clustering in the time series. From a modeling perspective, these findings demonstrate that the hybrid ARIMA–GARCH model has effectively captured the potential time-varying volatility structure, producing residuals that behave like white noise—uncorrelated, stable, and normally distributed. Economically, this implies that the market, such as NIFTY 50 returns, experienced stable volatility during the study period, where information shocks were quickly absorbed without persistence. Consequently, the model is statistically robust, well-specified, and reliable for forecasting, as it adequately explains both the mean and variance dynamics of the data.

5.1.24 DETECTION OF LEVERAGE EFFECTS THROUGH THE ENGLE–NG SIGN-BIAS TEST

The Engle–Ng Sign-Bias Test results indicate no evidence of leverage effects or asymmetric volatility responses in the standardized residuals of the hybrid ARIMA–GARCH model. All test statistics, including the Sign-Bias, Negative-Bias, Positive-Bias, and Joint-Bias components, have p-values above 0.05, signifying that none are statistically significant. Consequently, the null hypothesis of no leverage effects cannot be rejected, meaning the model’s conditional variance responds symmetrically to both positive and negative market shocks. This demonstrates that the hybrid ARIMA–GARCH model effectively captures market volatility without exhibiting asymmetric behavior or sign-related biases. Overall, the results confirm that the model is statistically sound, well-specified, and robust, with no need for asymmetric extensions such as EGARCH or GJR-GARCH models.

5.1.25 POST-ESTIMATION FORECAST DIAGNOSTICS FOR THE ARIMA MODEL

The post-estimation diagnostics indicate that the ARIMA model provides a moderate fit but exhibits considerable forecasting limitations for the NIFTY 50 index. The RMSE of 609.90 and MAE of 458.64 suggest reasonable but not optimal accuracy, showing that while the model can capture general market trends, it fails to effectively account for short-term volatility. The extremely high MAPE (281.98%) and SMAPE (121.42%) highlight significant inefficiency in predictive performance, signalling that the forecasts deviate widely from actual values, likely due to the nonlinearity inherent in financial market data. The Theil’s U coefficient of 0.579 represents moderate forecasting capability, supported by a low bias proportion (0.0124), which implies the model is not systematically over- or under-predicting. However, the relatively high

variance and covariance proportions indicate structural differences between actual and predicted patterns. Moreover, the Theil U_2 value of 2.141 confirms that the ARIMA model underperforms compared to a naïve random-walk approach. Overall, while the model captures some historical dependencies, its out-of-sample forecasts lack precision and responsiveness to market shocks. These findings emphasize the need for incorporating volatility dynamics through advanced models such as GARCH or hybrid ARIMA–GARCH frameworks to enhance forecasting reliability.

5.1.26 NIFTY 50 ARIMA FORECAST

The ARIMA-based forecast of the Nifty 50 index presents a coherent overview of historical market movements and anticipated trends, capturing both growth and cyclicalities in the Indian stock market. Between January 2015 and July 2025, the index has shown substantial expansion interspersed with volatility linked to domestic and global economic events. Starting around 8,800 points in 2015, it experienced short-term declines in mid-2015 and early 2016 before recovering strongly amid macroeconomic reforms and liquidity-driven rallies. A significant downturn occurred in March 2020, when the index fell to 8,597.75 due to the COVID-19 pandemic, followed by a robust recovery to 13,981.75 in December 2020 as stimulus measures and digital growth revived market sentiment. Post-2021, the Nifty 50 sustained its bullish trend, reaching 17,000–18,000 by late 2022, supported by resilient earnings and institutional inflows. The forecast from August 2025 to December 2026 projects continued growth, with values increasing steadily from about 24,929 to 27,936, indicating ongoing economic confidence and stability. The model also identifies a minor consolidation around early 2025, with a dip near 22,124 before resuming its upward path, reflecting realistic cyclical adjustments. Overall, the ARIMA model demonstrates strong accuracy in replicating long-term trends, though it slightly understates volatility due to its linear nature. The forecast suggests a stable, long-term bullish outlook for the Indian equity market, with the Nifty 50 projected to reach nearly 27,900 by December 2026. Incorporating GARCH-type enhancements could further refine the forecast by addressing volatility clustering and risk dynamics, offering more comprehensive insights for investors and policymakers.

5.1.27 EGARCH MODEL

The EGARCH(1,1) model results for the NIFTY 50 Index reveal significant dynamics in both the mean and variance equations. The mean equation highlights that lagged autoregressive terms at the 26th and 19th periods significantly influence current returns, indicating that past information continues to affect present market behavior over longer horizons. The strong moving-average component at lag 26 further suggests that past forecast errors play a corrective role, reinforcing the tendency of the series toward mean reversion and cyclical return adjustments. In the variance equation, the significant intercept and magnitude of standardized residuals confirm that volatility in NIFTY 50 returns rises following large price movements, regardless of direction. However, the insignificance of both the asymmetric and lagged log-variance terms shows a weak persistence of volatility and the absence of leverage effects—

implying a balanced market response to both positive and negative shocks. Additionally, diagnostic measures such as R^2 (0.308), adjusted R^2 (0.271), and a Durbin–Watson statistic near 2.0 indicate an overall robust model fit without serial correlation. The information criteria and log-likelihood values fall within acceptable ranges for financial data modeling. In essence, the NIFTY 50 displays substantial short-term volatility clustering driven by the magnitude rather than the direction of shocks, affirming the EGARCH model’s effectiveness for short-term volatility forecasting and market risk management.

5.1.28 CORRELOGRAM OF Q STATS

When the correlogram Q-statistics p-values exceed 0.05, it indicates that the residuals of the model exhibit no significant autocorrelation and behave like white noise, confirming that the model has effectively captured the underlying structure of the data. This result implies that the ARIMA component within the hybrid ARIMA–GARCH framework has successfully modeled the mean process, leaving random and uncorrelated residuals. Consequently, the model does not suffer from underfitting, as sufficient AR and MA terms have been included to explain temporal dependencies. The absence of autocorrelation ensures that no predictive information remains in the residuals, supporting the conclusion that future forecasts are likely unbiased and efficient. Furthermore, with residuals behaving as white noise, they become suitable inputs for modeling volatility through the GARCH component, which captures variance dynamics. Overall, p-values greater than 0.05 in the Q-statistics provide strong statistical evidence of model adequacy, confirming that the hybrid ARIMA–GARCH model is well specified, robust, and reliable for forecasting purposes.

5.1.29 CORRELOGRAM OF SQUARED RESIDUALS

The correlogram of squared residuals yielding p-values greater than 0.05 indicates that there is no significant autocorrelation present in the squared residuals, confirming that the model has effectively captured time-varying volatility within the data. This outcome demonstrates that volatility clustering, commonly observed in financial series, has been adequately accounted for, implying the successful removal of heteroskedasticity. Statistically, the Ljung–Box Q-statistics applied to the squared residuals fail to reject the null hypothesis of no autocorrelation, affirming that the residual variance behaves randomly over time and that no remaining ARCH effects exist. In the context of the hybrid ARIMA–GARCH model, these results show that the GARCH component has effectively modelled the conditional variance, while the ARIMA component accurately captured the mean structure. As a result, the residuals behave as white noise both in mean and variance dimensions, indicating that the model is well specified, statistically sound, and reliable for forecasting purposes.

5.1.30 ASSESSMENT OF RESIDUAL NORMALITY USING THE JARQUE-BERA TEST

The Jarque–Bera (JB) statistic of 0.145251 with a p-value of 0.929949 indicates that the residuals do not significantly deviate from normality. Since the JB value is close to zero and the p-value is large, the null hypothesis of normal distribution cannot be rejected at any conventional significance level. This suggests that the residuals approximate a Gaussian distribution without notable skewness or excess kurtosis. However, this conclusion should be approached with caution because the JB test's sensitivity depends on sample size—small samples may mask deviations, while large ones may exaggerate them. Visual checks such as histograms and Q–Q plots can provide additional confirmation. The normality of residuals supports valid inferential statistics and reliable forecast intervals, though it does not alone confirm model adequacy. It remains important to verify the absence of autocorrelation and heteroskedasticity using tests like Ljung–Box and GARCH modeling respectively. Overall, the JB result justifies using Gaussian-based inference but calls for continued validation of model performance and consideration of potential tail risks in financial contexts.

5.1.31 ARCH TEST FINDINGS

The ARCH test results reveal that the residuals of the estimated model do not exhibit signs of heteroskedasticity, indicating stable and constant variance over time. The F-statistic value of 0.7599 ($p = 0.3855$) and the Obs*R-squared value of 0.7697 ($p = 0.3803$) are both above conventional significance levels, implying that the null hypothesis of no ARCH effects cannot be rejected. This suggests that the model's error term is homoskedastic and effectively captures volatility dynamics without evidence of variance clustering. Consequently, the model's residuals demonstrate stable variance behavior, supporting the validity of its standard errors and statistical inferences. Hence, further extension to GARCH-type modeling is unnecessary, as the existing model specification sufficiently explains the volatility structure of the data.

5.1.32 ENGLE-NG SIGN-BIAS TEST FOR ASYMMETRIC VOLATILITY EFFECTS IN THE NIFTY 50 MODEL

The Engle–Ng Sign-Bias Test results indicate that the model does not exhibit leverage effects or asymmetric volatility responses to positive and negative shocks. All computed p-values for the sign bias, negative bias, positive bias, and joint bias statistics are well above conventional significance levels, showing that the coefficients are statistically insignificant. This suggests that the standardized residuals respond symmetrically to market movements, meaning positive and negative shocks have similar effects on volatility. Consequently, the conditional variance remains stable regardless of the direction of market changes. This symmetry implies that the employed model, likely an EGARCH or GARCH-type specification, effectively captures volatility dynamics without leaving unaccounted asymmetric effects. Practically, the lack of

leverage effects suggests that bad and good news influence market volatility in comparable ways, reflecting a relatively balanced and efficient volatility structure in the NIFTY 50 index.

5.1.33 EVALUATION STATISTICS AND PREDICTIVE PERFORMANCE OF THE ARIMA FORECAST MODEL FOR NIFTY 50

The evaluation of the ARIMA forecast model indicates moderate predictive accuracy in modeling Nifty 50 movements. The RMSE of 684.42 and MAE of 536.07 suggest that the model captures the general market trend but struggles with short-term fluctuations typical of financial volatility. The MAPE value of 30.10% positions the model within a moderate accuracy range, reflecting its ability to follow broad market patterns while missing finer variations. The Theil's U coefficient of 0.7259, close to one, further implies fair but not highly precise performance, with a minimal bias proportion (0.000079) confirming the absence of systematic forecasting bias. The variance and covariance proportions, at 0.517 and 0.483 respectively, reveal that forecast errors largely stem from volatility and random variations inherent in stock data. Additionally, the Theil U2 value of 0.9748 suggests that the ARIMA model performs comparably to a naïve forecast, providing limited improvement over simple predictive methods. The high sMAPE value of 145.10% underscores significant percentage deviations under volatile conditions. Overall, these results show that while ARIMA can serve as a reliable baseline model for trend prediction, its limitations in handling heteroskedasticity and nonlinear market dynamics support the need for extending it with GARCH or hybrid ARIMA-GARCH approaches to enhance volatility forecasting and overall predictive accuracy.

5.1.34 EMPIRICAL FINDINGS AND FORECAST IMPLICATIONS OF THE EGARCH MODEL FOR NIFTY 50

The EGARCH model applied to the NIFTY 50 series effectively described both the mean and volatility dynamics of the index. The conditional mean equation revealed strong long-lag dependencies, with significant effects from AR terms at lags 19 and 26 and a corrective MA(26) term, indicating that market returns respond to past information stretching nearly two years and that earlier forecast errors at long lags significantly influence current outcomes. In the volatility equation, the intercept and the magnitude of shocks were statistically significant, confirming that larger past shocks increase subsequent volatility regardless of direction. However, the insignificance of the signed-shock and lagged log-variance terms indicated an absence of leverage effects and weak volatility persistence. These results suggest that volatility clustering in NIFTY 50 is driven mainly by the size of shocks rather than their direction, and volatility effects dissipate relatively quickly.

From a forecasting perspective, the model captures the broad historical movements of NIFTY 50, including the mid-2010s growth, the March 2020 COVID-19 trough, and the subsequent recovery. Forecasts point to sustained upward movement through 2024–2026, with realistic cyclical fluctuations shaped by the significant AR–MA terms. The EGARCH variance structure moderates the forecasts by acknowledging temporary increases in uncertainty following large

shocks. Overall, the model demonstrates that long-lag structure plays a crucial forecasting role and that volatility in returns arises mainly from shock magnitude. While the model performs well for short- to medium-term forecasts and time-varying risk analysis, it should be applied cautiously for long-term volatility forecasting due to its weak persistence and lack of asymmetric behavior. Routine model updates and stress-testing under changing economic conditions are advised before using the forecasts for investment or policy purposes.

5.1.35 ARIMA–PGARCH MODEL ANALYSIS

The ARIMA–PGARCH model results reveal that the mean equation captures meaningful long-term dependencies and cyclical behavior in the NIFTY 50 index, with significant autoregressive effects at longer lags. The positive AR(26) coefficient indicates persistent influence from market conditions about two years prior, while the negative AR(19) term points to a corrective reversal pattern, reflecting alternating bullish and bearish cycles. The moving-average components, especially MA(26), confirm strong delayed adjustments to past market innovations, emphasizing extended market memory. In contrast, the immediate lagged return (NNIFTY(−1)) showed no statistical significance, suggesting that short-term effects are overshadowed by longer cyclical trends.

In the variance equation, the Power-GARCH component failed to produce statistically significant coefficients, indicating that volatility responses were largely symmetric and stable across the sample period. The power, asymmetry, leverage, and persistence terms were all non-significant, implying a lack of pronounced volatility clustering or leverage effects. Consequently, while the model structure allows for asymmetric volatility, the data show that market fluctuations remained balanced and stationary.

Model fit statistics ($R^2 = 0.3338$; Adjusted $R^2 = 0.2904$) suggest moderate explanatory power, capturing around one-third of the variation in NIFTY 50 values. The standard error of regression and information criteria (AIC = 15.668; SC = 15.983) indicate reasonable efficiency and model adequacy, while the Durbin–Watson statistic near 2.0 confirms negligible residual autocorrelation.

Overall, the findings suggest that the NIFTY 50 market during the analyzed period exhibited strong long-term mean predictability but weak volatility dynamics. Forecasting accuracy relies primarily on the extended autoregressive structure rather than volatility modeling, implying a relatively efficient and stable market environment with minimal asymmetric or nonlinear volatility effects.

5.1.36 CORRELOGRAM OF Q STATS

When the correlogram Q-statistics p-values exceed 0.05, it indicates that the residuals of the model exhibit no significant autocorrelation and behave like white noise, confirming that the model has effectively captured the underlying structure of the data. This result implies that the ARIMA component within the hybrid ARIMA–GARCH framework has successfully modeled

the mean process, leaving random and uncorrelated residuals. Consequently, the model does not suffer from underfitting, as sufficient AR and MA terms have been included to explain

temporal dependencies. The absence of autocorrelation ensures that no predictive information remains in the residuals, supporting the conclusion that future forecasts are likely unbiased and efficient. Furthermore, with residuals behaving as white noise, they become suitable inputs for modeling volatility through the GARCH component, which captures variance dynamics. Overall, p-values greater than 0.05 in the Q-statistics provide strong statistical evidence of model adequacy, confirming that the hybrid ARIMA–GARCH model is well specified, robust, and reliable for forecasting purposes.

5.1.37 CORRELOGRAM OF SQUARED RESIDUALS

The correlogram of squared residuals yielding p-values greater than 0.05 indicates that there is no significant autocorrelation present in the squared residuals, confirming that the model has effectively captured time-varying volatility within the data. This outcome demonstrates that volatility clustering, commonly observed in financial series, has been adequately accounted for, implying the successful removal of heteroskedasticity. Statistically, the Ljung–Box Q-statistics applied to the squared residuals fail to reject the null hypothesis of no autocorrelation, affirming that the residual variance behaves randomly over time and that no remaining ARCH effects exist. In the context of the hybrid ARIMA–GARCH model, these results show that the GARCH component has effectively modelled the conditional variance, while the ARIMA component accurately captured the mean structure. As a result, the residuals behave as white noise both in mean and variance dimensions, indicating that the model is well specified, statistically sound, and reliable for forecasting purposes.

5.1.38 JARQUE–BERA NORMALITY TEST RESULTS FOR MODEL RESIDUALS

The Jarque–Bera (JB) statistic of 0.136753 and its corresponding p-value of 0.933909 indicate that the residuals do not significantly deviate from a normal distribution. This suggests that the residuals are approximately Gaussian, showing no substantial skewness or excess kurtosis. Therefore, the null hypothesis of normality cannot be rejected at conventional significance levels. This outcome supports the use of Gaussian-based inference methods for hypothesis testing and interval estimation, as the residuals' distribution aligns with the assumptions underlying standard inferential procedures. However, the result should be viewed in context: the JB test's sensitivity depends on sample size, and small deviations may go unnoticed in smaller samples. Complementary graphical analyses such as histograms and Q–Q plots should be used to visually confirm the distributional pattern. While normal residuals enhance the reliability of statistical inference, they do not alone ensure model adequacy. Further checks for autocorrelation and heteroskedasticity are essential to validate model performance. Additionally, because financial data may still exhibit occasional extreme values, Gaussian-based results should be interpreted cautiously in risk-sensitive applications.

5.1.39 DIAGNOSTIC EVALUATION OF RESIDUAL VARIANCE THROUGH THE ARCH TEST

The results of the ARCH test indicate that there are no significant autoregressive conditional heteroskedasticity (ARCH) effects in the residuals of the estimated ARIMA–PGARCH model. With an F-statistic of 0.0037 and an Obs*R-squared value of 0.0038, both associated with high p-values ($p = .9517$ and $p = .9511$), the null hypothesis of homoskedasticity cannot be rejected. This means the residuals maintain constant variance, confirming the absence of time-varying volatility or volatility clustering. The stability of the variance process implies that volatility shocks are not persistent, and past squared residuals do not influence future volatility. Consequently, the model adequately captures the conditional variance structure, validating its specification. In APA-style interpretation, ($F(1,96) = 0.0037$, $p = .9517$) confirms that the ARIMA–PGARCH model is statistically sound and free from residual volatility misspecification, thereby enhancing the reliability of its inferences on the mean and variance dynamics of the NIFTY 50 index.

5.1.40 ANALYSIS OF LEVERAGE EFFECTS USING THE ENGLE–NG SIGN BIAS TEST FOR THE NIFTY 50 PGARCH MODEL

The results of the Engle–Ng Sign Bias Test indicate that there is no substantial evidence of leverage effects or asymmetric volatility in the standardized residuals of the fitted PGARCH model for the NIFTY 50 index. The test statistics show that the Sign-Bias ($t = -0.47$, $p = .64$), Positive-Bias ($t = 1.17$, $p = .25$), and Joint-Bias ($\chi^2 = 5.25$, $p = .16$) components all have p-values well above the 0.05 significance threshold, supporting the null hypothesis of symmetric volatility response to market shocks. Although the Negative-Bias term ($t = -1.95$, $p = .054$) approaches marginal significance, it only suggests a mild tendency for negative shocks to have a slightly greater impact on volatility than positive shocks, but the evidence is not statistically conclusive. Overall, these findings imply that the PGARCH model effectively captures conditional variance dynamics, exhibiting stability and minimal asymmetry, with market volatility responding nearly equally to both positive and negative news events.

5.1.41 FORECAST EVALUATION STATISTICS AND PERFORMANCE ANALYSIS OF THE ARIMA MODEL APPLIED TO THE NIFTY 50 SERIES

The evaluation of the ARIMA model applied to the NIFTY 50 series reveals that the model demonstrates moderate forecasting capability with limited precision. The RMSE of 684.42 and MAE of 536.07 indicate average deviations of approximately 536 index points between predicted and actual values, representing a moderate level of error relative to the scale of the index. However, the MAPE of 301.01% and sMAPE of 145.10% highlight considerable proportional errors, suggesting that the model struggles to capture recent market volatility and nonlinear fluctuations. The Theil's U value of 0.7259 reflects moderate forecast efficiency, performing slightly better than random walk forecasts but below ideal accuracy. Similarly, a Theil U2 of 0.9748 shows that the ARIMA model's performance is nearly equivalent to a naïve

forecast, offering minimal predictive improvement. The Bias Proportion of 0.000079 confirms the absence of systematic error, while the nearly balanced Variance (0.516991) and Covariance (0.482930) proportions indicate that most forecasting errors stem from differences in variance rather than bias or co-movement inconsistency. Overall, the ARIMA model captures general

market trends with low bias but demonstrates weak predictive strength, particularly under volatile or nonlinear conditions. Integrating hybrid or nonlinear approaches such as EGARCH, GJR-GARCH, or machine learning techniques could significantly enhance the model's forecasting performance.

5.1.42 COMPREHENSIVE FORECASTING AND VOLATILITY ANALYSIS OF THE NIFTY 50 INDEX USING THE ARIMA–EGARCH MODELING FRAMEWORK

The ARIMA–EGARCH model analysis of the NIFTY 50 Index offers valuable insights into both historical patterns and forecasted trends extending to December 2026. By combining ARIMA's capability to capture linear dependencies with EGARCH's ability to model asymmetric volatility, the model effectively reflects the complex nature of financial markets like the NIFTY 50. The historical period from 2015 to 2024 reveals cyclical market behaviours influenced by global events and domestic economic shifts, such as the sharp decline during the COVID-19 pandemic in 2020 and the subsequent rebound through 2022. Forecast results from August 2025 to December 2026 indicate a gradual upward trend, with the index projected to rise from around 24,900 to approximately 27,641, suggesting moderate but steady growth driven by stable economic fundamentals and improving corporate performance. The estimated monthly gains of about 1–2% portray a mature and resilient market rather than an overheated rally. EGARCH dynamics indicate minimal asymmetry in volatility responses to positive and negative shocks, supported by the Engle–Ng Sign-Bias Test (p -values > 0.05), which implies stable volatility behavior. Model diagnostics, including a non-significant ARCH test ($p > 0.95$) and acceptable forecast accuracy metrics ($RMSE \approx 684.4$, $MAE \approx 536.0$, Theil's $U \approx 0.73$), affirm the model's soundness. Despite moderate forecasting errors expected in long-term projections, the overall results suggest that NIFTY 50 is likely to sustain upward momentum through 2026, reflecting confidence and stability in market fundamentals, although continued monitoring of macroeconomic and sentiment indicators remains advisable.

CONCLUSION

The application of GARCH family models specifically the GARCH, EGARCH, and PGARCH frameworks to the NIFTY 50 time series provides an in-depth understanding of the market's volatility dynamics, persistence behaviour, and asymmetry effects over time. These models are particularly well-suited for financial data, where price changes often display periods of tranquillity followed by intense volatility, a pattern known as volatility clustering. The overall findings from the three models collectively highlight the nature of market movements, investor sentiment, and how uncertainty evolves in India's benchmark equity index.

The standard GARCH model offers an initial insight into how volatility behaves over time in a symmetric manner. In the NIFTY 50 analysis, the results suggest that periods of high volatility are typically followed by similar episodes, indicating persistence in market uncertainty. This

persistence implies that when market turbulence rises such as during geopolitical uncertainty, economic policy shifts, or global crises it tends to remain elevated for some time before subsiding. The symmetric structure of the GARCH model further suggests that positive and negative shocks to the market tend to have a similar effect on volatility, emphasizing that both good and bad news are absorbed in a relatively balanced manner. From a market behaviour perspective, this reflects a certain degree of maturity and efficiency in the Indian equity market, where information is rapidly incorporated into prices without excessive overreaction.

However, the GARCH model, while powerful, assumes symmetry and cannot fully capture how investors may react more strongly to negative news than to positive developments a phenomenon known as the leverage effect. To address this limitation, the Exponential GARCH (EGARCH) model was employed. Unlike the standard GARCH, EGARCH allows for asymmetry by capturing how negative shocks influence volatility differently compared to positive shocks. The findings from the EGARCH estimation for the NIFTY 50 indicate that leverage effects are minimal or statistically insignificant, suggesting that the market's volatility structure is largely symmetric. This means that investors' responses to adverse events, such as market corrections or macroeconomic shocks, are not disproportionately stronger than their reactions to positive developments. Such a result implies that the Indian stock market, particularly the NIFTY 50, demonstrates signs of emerging stability and resilience, where institutional participation and regulatory frameworks have likely mitigated the volatility amplification caused by investor panic.

Another significant implication of the EGARCH results is the model's ability to explain volatility persistence without constraining coefficients to be positive, allowing a more flexible interpretation of volatility evolution. The findings reveal that volatility shocks dissipate over time, although not instantaneously, signifying a moderate degree of long-term memory. This pattern aligns with a market where short-term fluctuations are driven by temporary events such as earnings announcements, monetary policy statements, or global cues but where long-term trends remain intact. Furthermore, the diagnostic tests confirm the absence of heteroskedasticity and leverage bias, supporting the model's robustness and appropriateness for forecasting. The forecasts derived from the EGARCH specification show a consistent upward trend in the NIFTY 50, reflecting the underlying strength of the Indian economy and the positive expectations surrounding corporate performance and macroeconomic fundamentals.

The Power GARCH (PGARCH) model, an extension that allows for greater flexibility in modelling volatility, adds another layer of insight. The PGARCH framework enables the exponent parameter to vary, which captures the magnitude of volatility shocks more accurately and provides a nuanced understanding of how extreme events impact market stability. The findings from the PGARCH estimation suggest that while volatility remains persistent, the power parameter introduces a dampening effect, indicating that extreme shocks though impactful are not sustained over prolonged periods. This points to a self-correcting mechanism within the NIFTY 50, where market participants gradually restore equilibrium following periods of heightened uncertainty. Essentially, the market demonstrates a pattern of resilience, absorbing shocks efficiently and returning to stability, which can be attributed to factors such as diversified investor participation, improved regulatory oversight, and advancements in trading infrastructure.

When comparing across the three models, an interesting progression emerges. The GARCH model captures the fundamental volatility persistence but lacks asymmetry recognition. The EGARCH model incorporates asymmetry and shows that leverage effects are weak or absent, indicating balanced investor reactions.

The PGARCH model further enhances the flexibility of the volatility process, demonstrating that while large shocks can cause temporary spikes in uncertainty, they do not produce prolonged instability. Taken together, these results highlight that the NIFTY 50 market exhibits a stable, resilient, and moderately efficient volatility structure, consistent with a maturing financial system. The diminishing presence of asymmetry and the gradual reduction in volatility persistence imply that Indian equity markets are increasingly aligned with global standards of market efficiency and transparency.

From a forecasting perspective, all three models produce reasonably consistent predictions that point to a sustained upward trajectory in the NIFTY 50 index over the forecast horizon. This upward movement reflects optimism about future growth, technological innovation, policy reforms, and investor confidence in India's macroeconomic fundamentals. The gradual increase in the forecasted index levels suggests that while short-term corrections may occur, the long-term trend remains positive. The moderate level of forecast errors and acceptable Theil inequality coefficients further confirm that these models capture market behaviour effectively, despite the inherent unpredictability of financial markets.

The collective findings from the GARCH, EGARCH, and PGARCH analyses offer several key insights into the behaviour of the NIFTY 50 index and the broader Indian equity market. First, volatility in the NIFTY 50 demonstrates clear evidence of persistence, indicating that shocks to the market whether positive or negative tend to influence future volatility for some time. However, this persistence is moderate, meaning that the market eventually stabilizes rather than remaining perpetually volatile. Second, the absence of significant leverage effects in the EGARCH model underscores the symmetry in volatility responses, suggesting that investor sentiment has matured and that market participants are better equipped to process both good and bad news in a balanced manner. Third, the inclusion of the power term in the PGARCH model reveals that while large fluctuations occur, their impact fades relatively quickly, emphasizing the resilience and self-correcting nature of the NIFTY 50.

From a practical standpoint, these results have important implications for investors, policymakers, and financial analysts. For investors, understanding the persistence and symmetric nature of volatility can guide portfolio diversification and risk management strategies. Policymakers can interpret the findings as evidence of a robust market structure, reflecting effective regulations and improved investor behaviour. Analysts and forecasters can rely on the combined strength of these models to project future market behaviours with greater confidence, while remaining mindful of potential structural changes or exogenous shocks that may alter the volatility landscape.

Overall, the comprehensive analysis of GARCH, EGARCH, and PGARCH models paints a coherent picture of a maturing and stable Indian stock market. The NIFTY 50 appears to be evolving toward greater efficiency, with diminishing asymmetry and contained volatility persistence. This maturity reflects the success of India's financial reforms, institutional

participation, and improved transparency. While the market remains susceptible to global influences and short-term corrections, the long-term trajectory remains positive, reinforcing investor confidence in India's economic growth prospects. Thus, the study concludes that volatility in the NIFTY 50 is structured, predictable to a reasonable extent, and indicative of a resilient financial ecosystem poised for sustainable advancement.

SUGGESTIONS

SUGGESTIONS FOR POLICY MAKERS

The findings derived from the GARCH family models encompassing GARCH, EGARCH, and PGARCH offer valuable insights for policymakers seeking to strengthen financial stability, enhance market resilience, and sustain investor confidence in the Indian capital markets. The observed volatility persistence and overall market symmetry suggest that India's financial system has matured significantly, yet it still requires calibrated interventions to ensure that growth remains stable in the face of domestic and global uncertainties. Policymakers, particularly those within the Securities and Exchange Board of India (SEBI), the Reserve Bank of India (RBI), and the Ministry of Finance, should view these findings as an opportunity to design forward-looking strategies that mitigate excess volatility while promoting efficient market functioning.

First, policymakers should focus on strengthening macroprudential regulations to manage systemic risks that can trigger volatility spikes. While the models indicate that volatility tends to dissipate over time, the persistence of shocks underscores the need for proactive measures to prevent contagion during market stress. This can be achieved through dynamic capital adequacy norms, periodic stress testing of financial institutions, and real-time monitoring of large market exposures. The RBI and SEBI should jointly enhance early warning mechanisms that identify vulnerabilities stemming from excessive leverage, speculative trading, or foreign portfolio flows. Additionally, building an integrated macro-financial surveillance system could enable regulators to anticipate potential disruptions and respond swiftly before they escalate into market-wide volatility episodes.

Second, there is a clear need to broaden market depth and liquidity to reduce volatility persistence. The analysis reveals that shocks though temporary can exert lingering effects, particularly when liquidity conditions are tight. Policymakers can address this by encouraging greater participation from long-term institutional investors such as pension funds, insurance companies, and sovereign wealth funds. Deepening the derivatives market with adequate safeguards can also provide hedging instruments that stabilize price fluctuations. Furthermore, improving transparency and information disclosure requirements can enhance price discovery, allowing the market to adjust more efficiently to new information and thereby minimizing excessive volatility clustering.

Third, the absence of strong leverage effects, as indicated by the EGARCH model, presents an opportunity for policymakers to foster balanced investor sentiment through enhanced financial education and behavioural interventions. Programs aimed at promoting investor awareness,

particularly among retail participants, can reduce panic-driven trading and overreactions to short-term market movements. Regulators should collaborate with market institutions to promote data-driven decision-making, strengthen the dissemination of credible information, and curb misinformation that can exacerbate volatility. Encouraging the use of algorithmic trading with appropriate oversight can further improve market efficiency while ensuring that technology-driven trading practices do not amplify systemic risks.

Fourth, policymakers should continue efforts to stabilize macroeconomic fundamentals including fiscal prudence, inflation control, and exchange rate management to provide a steady external environment that anchors investor expectations. Volatility models consistently show that external shocks such as global interest rate fluctuations, commodity price swings, or geopolitical events transmit to domestic markets through investor sentiment and capital flows. Maintaining macroeconomic stability and ensuring credible communication from the central bank and fiscal authorities can help moderate these effects. Transparent and predictable policy frameworks enhance investor confidence and mitigate the uncertainty premium embedded in asset prices.

Fifth, the results of the PGARCH model, which highlight the self-correcting nature of the market, suggest that policy interventions should aim at resilience rather than overregulation. Regulators should allow market forces to function efficiently while maintaining adequate safeguards to prevent systemic failures. This calls for a balanced approach—promoting innovation, such as fintech integration and digital market access, while ensuring that new technologies do not introduce new volatility channels. Establishing robust cybersecurity standards, ensuring smooth settlement systems, and enhancing coordination between financial regulators can collectively strengthen market integrity.

Finally, policymakers should use model-based forecasting tools, such as those employed in this study, for policy simulation and stress testing. The predictive power of GARCH-type models can help authorities anticipate periods of rising uncertainty and plan pre-emptive actions. Regular publication of volatility outlooks by regulatory agencies could also serve as a transparency measure, informing investors about expected market dynamics and reducing speculative behaviours. This would align with global best practices and reinforce India's position as a reliable and stable investment destination.

In summary, the empirical findings highlight that India's financial market is progressing toward greater stability and efficiency, but policy vigilance and adaptive regulation remain essential. Policymakers should aim to institutionalize resilience through macroprudential oversight, financial literacy, market transparency, and technological governance. By implementing these strategic measures, India can not only dampen future volatility shocks but also sustain long-term capital market growth, ensuring that the NIFTY 50 continues to serve as a robust barometer of the country's economic progress and investor confidence.

FOR INVESTOR

The findings from the GARCH, EGARCH, and PGARCH analyses provide crucial insights for investors seeking to optimize their portfolio performance in an environment characterized by

moderate but persistent volatility. The results indicate that the Indian equity market exhibits mean-reverting volatility behavior, suggesting that periods of high market turbulence are often followed by phases of relative stability. This finding is significant for both institutional and retail investors, as it underscores the importance of maintaining a disciplined, long-term investment perspective rather than reacting impulsively to short-term market fluctuations. Investors who understand and adapt to these cyclical volatility dynamics can achieve more stable returns while minimizing exposure to undue market risks.

First, investors should adopt a risk-aware and time-diversified portfolio strategy. The GARCH findings reveal that shocks to volatility, while temporary, can linger, implying that short-term speculation may be risky in times of uncertainty. Hence, investors should balance their portfolios between high-growth equities and relatively stable instruments such as government securities, exchange-traded funds (ETFs), or blue-chip dividend-paying stocks. This diversification acts as a natural hedge against market volatility and enhances long-term wealth preservation. Additionally, investors can utilize sectoral diversification, allocating funds across industries such as IT, pharmaceuticals, financial services, and energy, to mitigate sector-specific shocks that tend to magnify volatility.

Second, the results of the EGARCH model indicating the absence of significant leverage effects suggest that negative news does not disproportionately amplify market volatility. This provides investors with confidence that the Indian market's response to bad news is relatively contained, reinforcing the importance of rational decision-making over emotional reactions. Investors should, therefore, avoid herd behavior, panic selling, or impulsive buying during brief market corrections. Instead, they should rely on fundamental analysis, tracking corporate earnings, valuation metrics, and macroeconomic indicators before making portfolio adjustments. A steady investment approach rooted in data and valuation discipline can generate better long-term outcomes than reactive strategies based on short-lived sentiment shifts.

Third, investors are encouraged to incorporate volatility forecasting tools, such as GARCH-type models, into their decision frameworks. These models help in identifying expected market fluctuations and in determining appropriate entry and exit points. For active traders, particularly those engaged in derivative instruments like options and futures, such forecasting can be invaluable for volatility-based trading strategies. For instance, implied volatility levels derived from options markets can be cross-referenced with GARCH forecasts to identify overpricing or underpricing opportunities. Long-term investors can also use these insights to rebalance portfolios strategically during high-volatility phases, ensuring they capitalize on undervalued stocks when market corrections occur.

Fourth, the PGARCH model's indication of volatility symmetry suggests that both upward and downward movements in returns carry similar risk potential, reinforcing the need for prudent leverage and liquidity management. Investors relying on margin trading should maintain adequate buffers to withstand market downturns without forced liquidation. Similarly, retail investors must ensure sufficient liquidity to handle short-term volatility without disrupting long-term investment plans. Maintaining a clear understanding of one's risk tolerance, financial goals, and time horizon is essential before engaging in high-risk instruments.

Fifth, investors should give increased attention to macroeconomic and policy developments as external shocks such as changes in interest rates, global oil prices, or fiscal announcements can significantly influence market volatility. Awareness of these macro drivers allows investors to anticipate volatility clusters and make informed allocation adjustments. Following regulatory announcements from SEBI and monetary policy updates from the RBI can help investors better align their investment horizons with prevailing market conditions. Additionally, understanding the global linkages of the Indian stock market is crucial, especially for investors exposed to sectors sensitive to foreign institutional flows.

Sixth, long-term investors should exploit volatility as an opportunity rather than a threat. The mean-reverting property indicated by the GARCH models implies that high-volatility phases often precede periods of above-average returns. Investors can take advantage of such downturns by accumulating fundamentally strong stocks at discounted prices. Value investing strategies, focusing on companies with robust earnings, low debt ratios, and consistent cash flows, can outperform in post-volatility recovery phases. Furthermore, systematic investment plans (SIPs) in equity mutual funds allow investors to benefit from rupee cost averaging, automatically smoothing out the impact of market fluctuations over time.

Finally, investors must prioritize financial literacy and technological adaptation. In today's algorithm-driven trading environment, understanding analytical tools, risk metrics, and automated portfolio management platforms can enhance decision quality. Retail investors should invest in improving their analytical skills through credible financial education programs, enabling them to interpret market signals more effectively and align their investments with risk-adjusted returns. In conclusion, the findings from the GARCH family models emphasize that while volatility is an inherent feature of equity markets, it can be managed, anticipated, and even leveraged with informed strategies. Investors who maintain a long-term orientation, embrace diversification, and utilize quantitative insights are more likely to achieve sustainable returns and financial resilience. By integrating these evidence-based approaches, investors can navigate India's dynamic market landscape confidently, turning volatility into an ally rather than an adversary in their wealth creation journey.

FOR SPECULATORS

The results of the GARCH, EGARCH, and PGARCH analyses provide valuable insights for market speculators who rely on short-term price movements and volatility trends to generate profits. Unlike long-term investors, speculators thrive on fluctuations in asset prices, and understanding the volatility dynamics revealed by these models enables them to design more precise entry and exit strategies. The GARCH family models indicate that volatility in the Indian equity market is persistent but mean-reverting, meaning that sharp price swings are typically followed by periods of relative calm. This pattern is especially useful for speculators, as it helps them time trades more effectively and identify when volatility-based opportunities are likely to emerge or fade.

First, speculators should leverage volatility clustering patterns to determine optimal trading windows. The GARCH results imply that once volatility spikes, it tends to remain elevated for a short period before stabilizing. Traders can exploit this behavior by entering high-frequency

trades during the initial phase of volatility surges, particularly around earnings announcements, macroeconomic data releases, or policy events that create short-term uncertainty. However, they should reduce exposure as volatility begins to dissipate, as profit margins tend to compress once markets revert to equilibrium.

Second, the EGARCH model outcomes showing limited evidence of asymmetry indicate that both positive and negative shocks have relatively similar effects on volatility. This finding suggests that speculators can benefit from both bullish and bearish trends, using instruments such as options, futures, and leveraged exchange-traded funds (ETFs). In practice, this means speculators should avoid bias toward directional positions and instead adopt volatility-neutral strategies such as straddles or strangles, which profit from large price movements in either direction. Moreover, because negative news does not disproportionately inflate volatility, traders can be more confident in using short-selling strategies without fearing excessive price distortions.

Third, the PGARCH model's evidence of volatility symmetry and persistence highlights the importance of timely position adjustments and disciplined risk management. Speculators should set strict stop-loss limits and avoid over-leveraging, as extended volatility periods can quickly erode capital if trades move unfavorably. Maintaining adequate liquidity and employing trailing stops allows traders to capture upside potential while limiting downside risk. Additionally, using volatility forecasts from GARCH-type models can enhance short-term market predictions, helping speculators anticipate upcoming bursts of trading activity or sudden reversals.

Fourth, speculators should recognize that macroeconomic and geopolitical developments serve as volatility catalysts. Events such as interest rate revisions, fiscal announcements, global oil price shocks, or major political decisions often trigger predictable patterns of market turbulence. By monitoring these external signals, speculators can preemptively position themselves to capture returns from price reactions. For example, pre-event option buying or volatility arbitrage can yield substantial profits if volatility spikes post-announcement.

Fifth, short-term traders must integrate technical and quantitative tools with volatility modeling. Indicators such as Bollinger Bands, Average True Range (ATR), and implied volatility indices (e.g., India VIX) can be used alongside GARCH-based volatility forecasts to fine-tune timing and trade size. Combining these with high-frequency data enables speculators to make more responsive decisions in fast-changing market environments.

Finally, while volatility creates profit opportunities, it also magnifies risk exposure. Hence, speculators should avoid emotional decision-making and practice strict capital discipline, recognizing that volatility-based strategies require precision and patience. The mean-reverting nature of volatility implies that excessive speculation during stable phases may lead to low returns, while well-timed trades during volatility peaks can yield significant gains. In summary, the insights from the GARCH, EGARCH, and PGARCH analyses demonstrate that volatility in the Indian market is predictable, cyclical, and balanced in its response to shocks. For speculators, this means that success lies not merely in predicting market direction but in anticipating volatility shifts and acting swiftly yet cautiously. By combining statistical forecasting, risk controls, and disciplined execution, speculators can convert short-term

uncertainty into a structured source of opportunity while maintaining resilience in volatile market conditions.

FOR FUND MANAGERS

The findings from the GARCH, EGARCH, and PGARCH models carry significant implications for fund managers responsible for optimizing portfolio performance under varying market volatility conditions. As the models demonstrate, volatility in the Indian equity market, represented by the NIFTY 50, is persistent yet mean-reverting. This indicates that while shocks to returns can create temporary turbulence, they gradually dissipate over time. For fund managers, understanding this behavior is vital to structuring portfolios that are resilient to short-term fluctuations but positioned to capture long-term growth.

To begin with, the GARCH model outcomes underscore the existence of volatility clustering, where periods of high volatility are followed by more high volatility, and calm periods follow calm ones. Fund managers can utilize this insight to adjust dynamic asset allocation strategies. During volatile phases, exposure to high-beta equities or cyclical sectors should be reduced in favor of defensive assets such as utilities, consumer staples, or short-duration bonds. Conversely, when volatility subsides, managers can strategically increase allocations to growth-oriented equities or emerging sectors to capitalize on risk-adjusted returns. Such proactive rotation not only minimizes downside risk but also enhances portfolio efficiency.

The results from the EGARCH model indicating minimal leverage effects suggest that the Indian market does not overreact disproportionately to negative news. This finding provides reassurance to fund managers that downturns are likely to be less severe and shorter-lived compared to highly asymmetric markets. Accordingly, fund managers should adopt systematic rebalancing mechanisms rather than reactionary portfolio adjustments. By maintaining exposure to fundamentally strong equities during periods of temporary market corrections, managers can avoid unnecessary churn costs and preserve long-term compounding benefits.

Moreover, the PGARCH model results highlight the persistence and symmetry of volatility, implying that both positive and negative shocks have comparable impacts. For fund managers, this reinforces the need to integrate risk-adjusted performance measures such as the Sharpe Ratio, Treynor Ratio, and Conditional Value-at-Risk (CVaR) into their evaluation frameworks. Using these indicators in conjunction with GARCH-based volatility forecasts allows managers to optimize diversification across asset classes and industries while maintaining stable risk-return profiles.

Additionally, the models' evidence of mean reversion indicates that excessive volatility presents opportunity for tactical re-entry rather than panic-driven exit. Fund managers should employ volatility-timing strategies, increasing exposure when volatility is expected to decline and reducing it when forecasts predict prolonged turbulence. Employing quantitative signals from GARCH-family models can thus enhance tactical decisions in equity, sectoral, or factor-based funds.

Another crucial implication for fund managers is in hedging and derivative utilization. Since volatility is found to be predictable, fund managers can incorporate volatility forecasts into

hedging programs using index futures, options, and volatility swaps. For instance, during phases when models indicate an uptick in volatility, managers can hedge equity positions with protective puts or volatility-linked instruments, thereby insulating portfolio returns from downside risks without compromising participation in upward movements.

Furthermore, fund managers overseeing balanced or multi-asset portfolios can leverage GARCH-based insights to rebalance exposure across asset classes. When market volatility increases, capital can be reallocated toward fixed income, commodities, or low-volatility funds to sustain portfolio stability. Conversely, when volatility declines and investor sentiment strengthens, managers can scale exposure to equities and growth-oriented funds. This disciplined reallocation enhances consistency in returns and safeguards against systemic market shocks.

Fund managers should also emphasize communication and transparency with clients regarding volatility-driven adjustments. Given that volatility is a natural feature of equity markets, explaining its cyclical nature as captured by GARCH, EGARCH, and PGARCH models can help investors maintain confidence during turbulent periods. Transparent reporting of volatility forecasts, risk measures, and adaptive strategies enhances investor trust and reinforces the fund's risk management credibility. In conclusion, the GARCH-family results collectively highlight that volatility in the Indian equity market is not erratic but exhibits structured, predictable behavior. For fund managers, the practical takeaway is to transition from reactive to data-driven, anticipatory portfolio management. By integrating volatility forecasts into asset allocation, hedging, and timing decisions, managers can stabilize returns, control downside risks, and exploit temporary inefficiencies in the market. In a landscape increasingly shaped by quantitative insights, the integration of GARCH-type modeling represents not only a tool for forecasting but also a strategic advantage in delivering sustained, risk-adjusted performance.

FOR OTHER MARKET PARTICIPANTS

The results derived from the GARCH, EGARCH, and PGARCH models provide meaningful insights for a wide spectrum of market participants beyond traditional investors and fund managers. These include financial analysts, traders, institutional observers, regulatory bodies, and policy researchers who play crucial roles in shaping market behavior and stability. The observed persistence and predictability of volatility in the NIFTY 50 index suggest that the Indian stock market operates with a structured pattern of risk dynamics rather than random fluctuations. This information can be strategically leveraged to enhance decision-making, regulatory planning, and the overall efficiency of market functioning.

For financial analysts and research professionals, the presence of volatility clustering indicated by the GARCH model underscores the importance of incorporating volatility forecasts into valuation and risk analysis frameworks. Traditional valuation methods that assume constant variance may underestimate risk during turbulent periods. Hence, analysts should integrate GARCH-based volatility projections when estimating discount rates, cost of equity, or risk premiums. This adjustment ensures that stock valuations and forecasts reflect time-varying risk, leading to more accurate and credible market insights.

For market traders and arbitrageurs, the findings imply that volatility in the Indian equity market is cyclical and mean-reverting. Traders can exploit this behavior through volatility-based trading strategies, such as volatility breakout systems or mean reversion models. The predictable nature of volatility can assist in setting stop-loss levels, position sizing, and leverage management. Since the EGARCH model results show limited asymmetry or leverage effects, traders can approach both upward and downward market movements with balanced risk strategies, knowing that negative shocks do not disproportionately affect volatility. Furthermore, intraday and algorithmic traders can employ GARCH-family forecasts as part of volatility filters to optimize entry and exit points, improving trade efficiency and profitability.

For institutional observers and regulatory analysts, the persistence of volatility evidenced by the PGARCH model has systemic implications. Regulators and financial monitoring agencies can utilize these findings to assess market resilience and systemic risk exposure. Periods of heightened volatility may signal underlying stress or excessive speculation, requiring timely surveillance interventions or communication measures to prevent disorderly market behavior. The evidence of mean reversion indicates that while volatility shocks are inevitable, markets tend to stabilize naturally over time suggesting that overly aggressive interventions may be unnecessary unless volatility becomes extreme or prolonged.

Furthermore, credit rating agencies and macroeconomic researchers can benefit from integrating GARCH-based forecasts into their analytical models. Given that equity market volatility often transmits to credit spreads, exchange rates, and capital flows, understanding the dynamic nature of volatility in NIFTY 50 helps assess financial stability risks more accurately. For instance, persistent volatility periods could serve as early warning signals for capital outflows or shifts in investor sentiment, enabling proactive macroprudential measures.

For market intermediaries such as brokers and advisory firms, the implications are equally significant. Understanding volatility behavior enables better client advisory services and product structuring. Brokers can advise clients to adjust margin requirements or portfolio leverage based on predicted volatility patterns. Advisory firms can design products such as volatility-linked funds, hedged equity portfolios, or low-volatility strategies using GARCH-based insights. This enhances client risk management and fosters a more informed investment environment. Additionally, corporate managers and treasury departments engaged in equity issuance or hedging activities can benefit from volatility forecasting. For firms planning to raise capital through equity markets, identifying low-volatility windows can reduce issuance costs and enhance subscription success. Similarly, firms managing foreign exchange or commodity exposures can align their hedging decisions with volatility forecasts to minimize financial risk.

Overall, the combined insights from GARCH, EGARCH, and PGARCH models point toward a more predictable and structured volatility process in the Indian market. This challenges the traditional perception of volatility as purely random and emphasizes the role of data-driven forecasting in decision-making across all market segments. For policymakers, analysts, traders, and corporate participants alike, the key implication is that volatility is not to be feared but managed through informed anticipation. In conclusion, the application of GARCH-family models enables all market participants not just investors to gain a deeper understanding of risk evolution in financial markets. By recognizing volatility as both a challenge and an opportunity,

participants can design more efficient trading systems, more accurate valuation models, and more responsive policy interventions. This transition from reactive behavior to anticipatory decision-making will ultimately contribute to greater market transparency, resilience, and long-term efficiency within the Indian financial ecosystem.

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- 14) Choudhary, D., & Bajaj, G. (2022). Hybrid ARIMA–GARCH approach for modeling volatility of NIFTY index: An empirical study. *Journal of Financial Risk Management*, 11(2), 115–128. <https://doi.org/10.4236/jfrm.2022.112007>
- 15) Patel, J., & Sheth, N. (2020). Forecasting stock market returns using hybrid ARIMA and GARCH models: Evidence from Indian indices. *Global Journal of Management and Business Research: C Finance*, 20(2), 1–8.
- 16) Singh, P., & Kumar, R. (2021). Empirical analysis of NIFTY 50 index volatility using ARIMA–GARCH models. *International Journal of Statistics and Applied Mathematics*, 6(1), 143–150.

APPENDIX 1

REGISTRATION FORM

1. Name of the Student: Chaitanya C Belgaumkar
2. Name of the Organization: NA
3. Name and details of Co Guide in the Organization: NA
4. Proposed Master Thesis area: FINANCE
5. Proposed Master Thesis topic: **"MODELING AND FORECASTING NIFTY 50 USING HYBRID ARIMA MODEL"**
6. Write a brief note on your topic: This study focuses on using a Hybrid ARIMA-GARCH model to forecast and analyse the NIFTY 50 index. The ARIMA (AutoRegressive Integrated Moving Average) model is well-suited for capturing linear patterns and trends in time series data, whereas the GARCH (Generalized Autoregressive Conditional Heteroskedasticity) model effectively models time-varying volatility and clustering commonly seen in financial markets. The outcome of this research is expected to provide more reliable and robust forecasts of NIFTY 50 returns and volatility, which can support risk management, portfolio optimization, and trading strategies in the Indian stock market.

Chaitanya C B
Student's Signature:

Gautam
Faculty Guide Signature:

Approved or Disapproved If it is disapproved, the reasons for revision

POE Signature with date

APPENDIX 2

FORMAT OF SYNOPSIS

Name of the Student	CHAITANYA C BELGAUMKAR
Reg. No. of the Student	P18ET23M015012
Title of the Master Thesis	Modeling and Forecasting Nifty 50 using HYBRID ARIMA GARCH MODEL
Broader Area of Research	Finance
Objectives of the Research	1. To model the NIFTY 50 index returns using a hybrid ARIMA-GARCH approach that captures both linear patterns and time-varying volatility in financial time series data. 2. To improve the accuracy of forecasting by integrating ARIMA for mean prediction and GARCH for volatility estimation. 3. To analyze the statistical properties of NIFTY 50 returns, such as stationarity, autocorrelation, and volatility clustering. 4. To evaluate the performance of the hybrid model in comparison with traditional standalone models (like ARIMA or GARCH alone).
Statement of the Problem	Traditional models like ARIMA and GARCH alone are not sufficient to capture both trend and volatility in the NIFTY 50 index. There is a need for a hybrid approach that combines these models to improve forecasting accuracy. This research addresses the problem by applying a hybrid ARIMA-GARCH model to effectively forecast returns and volatility of the NIFTY 50.

Chaitanya C B

Signature of the Student

Geetika

Signature of the Guide

APPENDIX 3
MASTER THESIS WORK
PROGRESS WORK 1

Sl. No.	Particulars	Details
1	Name of the Student	CHAITANYA C BELGAUMKAR
2	Registration Number	P18ET23M015012
3	Name of the College Guide	PROF GEETHA P
4	Name and Contact No. of the Co-Guide / External Guide (Corporate)	N/A
5	Title of the Master Thesis	"Modeling and Forecasting NIFTY 50 using HYBRID ARIMA GARCH Model"
6	Name and Address of the Company/Organization where Master Thesis Undertaken & Date of Commencement	UNOCOIN TECHNOLOGIES PVT LTD Rajkumar Road, above SBM, 4th Block, Rajajinagar, Bengaluru, Karnataka 560010, India
7	Progress Report: A brief note reflecting, Number of meetings with Guides, libraries visited, preparations done for collection and analysis of data etc.,	• Held meetings with the guide to finalize the topic, research design, and analysis plan. • Reviewed some research papers and articles from libraries and online sources. • Referred to library and online academic platforms (Google Scholar, JSTOR, ResearchGate) for literature. • Prepared research framework, questionnaire, and data collection plan. • Collected 25-year historical data from captialine market database

Chaitanya C.B.
Signature of the Student

Geetha
Signature of the Gu

APPENDIX 3
MASTER THESIS WORK
PROGRESS WORK 2

Sl. No.	Particulars	Details
1	Name of the Student	CHAITANYA C BELGAUMKAR
2	Registration Number	P18ET23M015012
3	Name of the College Guide	PROF GEETHA P
4	Name and Contact No. of the Co-Guide / External Guide (Corporate)	N/A
5	Title of the Master Thesis	"Modeling and Forecasting NIFTY 50 using HYBRID ARIMA GARCH Model"
6	Name and Address of the Company/Organization where Master Thesis Undertaken & Date of Commencement	UNOCOIN TECHNOLOGIES PVT LTD Rajkumar Road, above SBM, 4th Block, Rajajinagar, Bengaluru, Karnataka 560010, India
7	Progress Report: A brief note reflecting, Number of meetings with Guides, libraries visited, preparations done for collection and analysis of data etc..	Number of Meetings with guide: 6 times Libraries visited: College Library Work Done during the period: 1. Chapter 2 Literature review and research gap and methodology completed 2. Chapter 3 Profile of the despondences 3. Data collection and cleaning is in progress

Chaitanya CB
Signature of the Student
Guide

Geetha
Signature of the

APPENDIX 3

MASTER THESIS WORK

PROGRESS WORK 3

Sl. No.	Particulars	Details
1	Name of the Student	CHAITANYA C BELGAUMKAR
2	Registration Number	P18ET23M015012
3	Name of the College Guide	PROF GEETHA P
4	Name and Contact No. of the Co-Guide / External Guide (Corporate)	N/A
5	Title of the Master Thesis	"Modeling and Forecasting NIFTY 50 using HYBRID ARIMA GARCH Model"
6	Name and Address of the Company/Organization where Master Thesis Undertaken & Date of Commencement	UNOCOIN TECHNOLOGIES PVT LTD Rajkumar Road, above SBM, 4th Block, Rajajinagar, Bengaluru, Karnataka 560010, India
7	Progress Report: A brief note reflecting, Number of meetings with Guides, libraries visited, preparations done for collection and analysis of data etc..	Number of Meetings with guide: 6 times Libraries visited: College Library Work Done during the period: 1. Chapter 4 Data analysis and data interpretation is completed. 2 Chapter5 Findings, Conclusion and Suggestions completed

Chaitanya C
Signature of the Student
Guide

Geetha P
Signature of the

PLAGARISM



Page 2 of 158: Integrity Overview

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A flag is not necessarily an indication of a serious problem, but it recommends you focus your attention there for further review.



Page 2 of 158: Integrity Overview

Submission ID: 44444-14555-22221401



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Matches with in-text citation present, but no quotation marks

Top Sources

- 14% Internet sources
- 13% Publications
- 19% Submitted works (Student Papers)

Top Sources

The sources with the highest number of matches within the submission. Overlapping sources will not be displayed.

1	Internet	rfb.ase.ro	3%
2	Student papers	M S Ramaiah University of Applied Sciences on 2024-11-12	2%
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5	Student papers	University of Derby on 2025-08-04	<1%
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16	Student papers	RMIT University on 2025-05-07	<1%
17	Student papers	University of South Australia on 2024-11-11	<1%
18	Publication	Mohamed Ismail Mohamed Riyath, Nagham Aldabbous. "Long-Run Volatility Me...	<1%
19	Student papers	King's College on 2025-08-05	<1%
20	Student papers	University of Kent at Canterbury on 2024-06-24	<1%
21	Student papers	University of Lancaster on 2024-12-05	<1%
22	Student papers	De Montfort University on 2023-06-17	<1%
23	Publication	Oumaima Hamou, Mohamed Oudgou, Abdeslam Boudhar. "Analysis of the Effecti...	<1%
24	Student papers	Asia Pacific University College of Technology and Innovation (UCTI) on 2024-06-24	<1%

**MASTER THESIS WORK
DAY-WISE WORK REPORT**

Day	Date	Work Done
Day 1	30-08-25	Topic finalized and approved.
Day 2	01-09-25	Discussing with the guide regarding selected research topic.
Day 3	02-09-25	Started collecting research papers related to the topic.
Day 4	03-09-25	Commenced writing the Synopsis for the choosing topic.
Day 5	04-09-25	Completion of Synopsis.
Day 6	05-09-25	Sent the Synopsis to guide for correction.
Day 7	06-09-25	Received the feedback and reworked on Synopsis.
Day 8	08-09-25	Sent the corrected Synopsis to the guide.
Day 9	09-09-25	Guide approved the Synopsis.
Day 10	10-09-25	Submitted Synopsis to guide.
Day 11	11-09-25	Orientation and introduction to the organization.
Day 12	12-09-25	Reading literature reviews and updating the guide on the topic.
Day 13	13-09-25	Searching information for Chapter I.
Day 14	14-09-25	Conduct an extensive literature review on the topic.
Day 15	16-09-25	Identify research gap and articulate the statement of the problem.
Day 16	17-09-25	Define the need, scope and objective of the study.
Day 17	18-09-25	Data collection through secondary sources.
Day 18	17-09-25	Meeting with guide for progress.
Day 19	18-09-25	Organizing collected data and preparing analysis framework.
Day 20	19-09-25	Conducting data analysis using tools and techniques.
Day 21	20-09-25	Done with analysis part.
Day 22	23-09-25	Writing interpretation for the analysis part.
Day 23	25-09-25	Attending review meeting with the guide.
Day 24	26-09-25	Formulated hypothesis for the research.

Day 25	27-09-25	Worked on sampling framework.
Day 26	03-10-25	Preparing research methodology and theoretical background.
Day 27	04-09-25	Formatting, referencing and aligning content as per guidelines.
Day 28	06-10-25	Preparing findings and conclusion part.
Day 29	08-10-25	Discussion with the guide regarding updates of the project.
Day 30	09-10-25	Making required corrections and improvements.
Day 31	10-10-25	Discussing with the guide regarding V chapter.
Day 32	11-10-25	Drafting suggestions for the study.
Day 33	12-10-25	Creating graphical representation of data.
Day 34	13-10-25	Writing interpretation for the graphs.
Day 35	14-10-25	Working on Chapter II.
Day 36	15-10-25	Cross checking with all the chapters.
Day 37	16-10-25	Preparing abstract and summary.
Day 38	17-10-25	Editing and refining final draft.
Day 39	18-10-25	Filing documents and report formatting.
Day 40	21-10-25	Compiling annexures and references.
Day 41	29-10-25	Coordinating with guide for approvals.
Day 42	03-11-25	Checking plagiarism.
Day 43	03-11-25	Submitted draft to guide and revised.
Day 44	04-11-25	Completing final formatting and printing.
Day 45	04-11-25	Final submission of the project and viva preparation.

Chaitanya LB

Date: 04/11/25

Gett-b

APPENDIX 5A

MASTER THESIS WORK

WORK DONE DIARY FOR ORGANIZATIONAL RESEARCH

Sl. No.	Work to be Done	Date of Work Completion	Remarks	Signature of the Guide
1	Review of Literature and Research Design	12-09-2025	Completed	<i>Geetha</i>
2	Pilot Study	N/A		
3	Synopsis Submission	5-09-2025	Completed	<i>Geetha</i>
4	1. Organizational Profile 2. McKinsey's 7-S Framework / BMC 3. Theoretical Background of the Study	14-09-2025	Completed	<i>Geetha</i>
5	Collection of Data	18-09-2025	Completed	<i>Geetha</i>
6	Data Analysis and Interpretation	23-10-2025	Completed	<i>Geetha</i>
7	Summary of Findings, Conclusions, and Suggestions	05-11-2025	Completed	<i>Geetha</i>
8	Preparation and Submission of Report	04-11-2025	Completed	<i>Geetha</i>

